The Model

The Moon was built from the Earth while attached to it, and released where the Earth's rotation had already given it orbital speed

Zohar Duran · ORCID 0009-0005-3194-9387 · 15 October 2026

Abstract

This paper presents the LEF model — Lunar Elevation and Fission — as one continuous physical process. According to the model, the Moon was built from the Earth's material while attached to it, and released at the point where the Earth's rotation had already given it the speed of a free orbit. The paper computes nothing new: it tells the whole process as one sequence, presents each of the ten papers (2–11) that compute its stages, and gathers in one place what the series has measured and what the model requires.

Unlike a simulation, the model is a *process model*. It proposes a sequence of physical events, and for each event the series shows that standard physics permits it under the computed conditions, gives the number that sets it, and names the observation that serves to test it.

The process runs in stages 0–8. **The lifting mechanism:** the young Earth, hot and turning once every 5.5 hours, was at the stage in which its iron was still descending. At the source it held a pressure of 150 GPa — enough to lift a deep, foamed, volatile-rich melt lighter than 3,864 kg/m³ to the corotation line: the distance at which a body turning with the Earth is already orbiting it. **The body on the tip:** the pressure raised one bulge. On the tip of the bulge sat a body fed from within by the deep melt and covered from without by the Earth's outer shell, which rose with it as the skin of a dome: a squashed body over a 6,000-km contact, its upper dome reaching a height of about 3,650 km, holding the lunar mass at a density of 1,300 kg/m³. Material thrown from openings around the region returned at the speed it left, 5 km/s, and made craters on that shell while it still lay on the Earth. **The pear structure:** the bulge was drawn into a pear of the Earth's own material, about nine to twelve lunar masses by the width of its base (Paper 3). The pear is part of the Earth, held by the Earth's gravity like any other part of it, and is not thrown out, because its speed is not a throwing speed and it lies inside the corotation line. Its base, 9,000–11,000 km wide, stood near Kiritimati in the central Pacific, above the Pacific large low-velocity province at the bottom of the mantle, and its tip lagged to the west. **Rounding and flattening:** while attached, the body rounded outward, flattened — its equatorial diameter reaching 1.69 times its polar diameter, at 12,994 km — and rounded inward, crowding the craters of its polar regions (a clean ratio of 1.20 ± 0.05, rising with crater size to 1.53). **Heat transfer:** as long as the body was attached, its heat source was the Earth: the heat entered through Imbrium and Serenitatis, crossed the body and raised beneath the far side the dome that became South Pole–Aitken. Serenitatis peeled away first, while Imbrium stayed connected and was detached last. After release the heat is the Moon's

own, together with radiation from the nearby Earth, which faded as the Moon receded; the Moon cooled slowly — the maria first, and its centre later. **The release:** the lengthening pear slowed the Earth, and the gathering core sped it up: at release the day is 5.77 hours, and the corotation line lies at 16,399 km. The Moon, round and rigid, at 2,600 kg/m³, began to separate when its centre of mass reached that line, and was torn free slightly after, about 210 km past the line, 3.873 billion years ago. The system then held 1.007 times today's angular momentum, where classical fission needed 2.74 times. The Moon entered an orbit whose initial tilt was set by the latitude it left from, 22.5°, and this tilt decayed gradually to 5.145° at the present distance, the value observed. **The fate of the pear:** the release does not bring the pear down. The sorting process, which acts on it regardless, gathered it back into the Earth's interior over tens to hundreds of millions of years. Its heavy part sank to the base of the mantle beneath the base of the pear, its leading wall was folded into the stripes of the North Pacific, and its back was gathered into the arcs of the western Pacific.

Applying the same idea to Mars, by another route, gives its two small moons. The paper closes with the series' traffic light — the observations that were tested —, with the eighteen requirements of Paper 11 and their tests, with the two key observations of the model and their results (the geometry of the exposure zones, and the origin of the material of the crater record), and with the series of twelve papers and the work that remains open.

WHAT THIS PAPER PROPOSES

The Moon was built from the Earth, while attached to it, and released at the point where the Earth's rotation had already given it the speed of a free orbit. That is the whole of the model, and it is a process: a hot, fast-spinning Earth, with its iron still descending, concentrates a pressure beneath one region; the pressure raises one bulge, which is drawn into a pear; on the tip of the pear sits a body that grows from within and from without at once; the body rounds, flattens and rounds again while attached; and when its centre of mass reaches the corotation line it begins to separate, and slightly after, about 210 km past the line, it is torn free — round — into orbit. The pear is then gathered back into the Earth and leaves its folds in it.

Five numbers carry the model. Each of them is computed at length in the series.

- *The release radius.* The corotation line at the moment of release lies at **16,399 km** — about 2.6 Earth radii — with an Earth day of **5.77 hours** (Paper 2).
- *The angular-momentum budget.* The system at release holds **1.007** times today's angular momentum, where the classical route needs **2.74** times (Paper 2).
- *The source pressure.* The pressure available at the source is **150 GPa**; lifting a melt of 3,500 kg/m³ to the line costs 139.78 GPa and leaves a surplus of 10.49 GPa (Paper 3).

- *The density.* The density of the body on the tip of the pear is **1,300 kg/m³**, rising to **2,600** at release and to **3,344** today (Paper 3).
- *The age.* The separation took place **3.873 billion years ago**, the age at which the norite clocks of South Pole–Aitken were reset (Paper 8).

What the model requires. A parent still nearly uniform inside when the transfer begins (moment-of-inertia coefficient $k \gtrsim 0.397$); a deep melt lighter than 3,864 kg/m³ — which a volatile-rich foam satisfies and a dry melt does not; a shell stiff enough to keep its craters ($\gtrsim 10^{23}$ Pa s); a pear, held by the Earth's gravity like any other part of it, that stays inside the corotation line and carries the body to it; and a Moon round and rigid at release (axis ratio ≤ 1.04). Each requirement is stated with its test in Paper 11, and the requirements are gathered in Part E.

The requirements of the model, and the tests that will examine it, are collected in Paper 11 (The Requirements).

Introduction — a different kind of answer

The origin of the Moon is usually presented as settled: a Mars-sized body struck the young Earth, and the Moon accreted from the debris. That model has explained much and is supported by extensive computation, but explaining each of the Moon's most distinctive properties has required it to add a further mechanism. **Isotopic composition:** the Moon's material is isotopically identical to the Earth's, so the impact must have mixed the two bodies completely [2, 3]. **Orbital tilt:** the Moon's orbit is tilted five degrees from the plane in which a debris disk would have left it, so a resonance, a passing planetesimal or an Earth tilted past seventy degrees must have tilted it afterwards [4, 5]. **Crustal asymmetry:** the far-side crust is thicker than the near side, so a second moon must have fallen onto it [6]. These complications are not a failure of the collision model; they are the cost of a starting point in which the Moon is assembled from fragments in space.

The model presented here needs none of these mechanisms, because it does not assume assembly from fragments. In this model the Moon is lifted as one connected body out of the Earth, and only then released. Its material has the Earth's composition simply because it comes from the Earth. Its orbit is tilted because the body left from a particular latitude and not from an equatorial disk. Its far side is thick because it is built of many layers of compacted sial, which the forces of the spin carried up to the tip of the bulge, and of material that returned to it again and again; the near side was the base of the connection. What the model pays instead is the lifting, and that price is computed: the pressure, the angular momentum, the connection and the release.

This is a process model, and the term needs to be defined before the model is stated. A *simulation* solves the equations of motion for a chosen initial state and reports the results. A *process model* starts from a proposed sequence of events. For each step it shows that the physics permits the event, gives the number that sets it, checks that the numbers agree with one another, and names a

specific observation that tests it. The giant-impact hypothesis was a process model for many years before the first simulations ran [7]. The present model is at that stage, and the series is written so that it can be tested. No step asks the reader to accept it without its number.

How this paper is built. The paper tells the process first as a continuous story, and then presents the papers. Part A states what any model of lunar origin must explain, and why the lifting route was closed for a century. Part B tells the process in stages 0–8 as one continuous account, with the central number of each stage and the paper that computes it. Part C presents the ten papers that compute the model, one after the other: the claim of each, then a paragraph for each of its parts. Part D gathers the traffic light of the series — every observation tested, in green, orange or withdrawn. Part E collects the requirements and presents the two key observations of the model with their results. Part F lists the twelve papers and what remains open.

Part	Title	The question it settles
A	What has to be explained	Which properties must any model of origin account for, and why was the lifting thought impossible?
B	The process, in stages 0–8	What happened, in what order, and with what number at each stage?
C	The papers, one after the other	What does each of Papers 2–11 claim, and what does each of its parts establish?
D	The traffic light of the series	What has been measured, and how did each test come out?
E	Requirements and the key observations	What must hold if the model is right, and how did its two key observations come out?
F	The series	How are the twelve papers arranged, and what is left for future work?

THE ENVIRONMENT

The process took place in an ancient environment, on an Earth that no longer exists, and all the computations are stated in those conditions, not in the present ones. About 4.5 billion years ago the Earth was heavier by one Moon: its total mass, 6.0455×10^{24} kg, is today's Earth and today's Moon in one body. The body was hot throughout and expanded by its heat, with a volumetric radius of 6,517 km (about 6,721 km at the equator, because a fast-spinning body bulges there). The day was only 5.5 hours long. The iron was still descending to the centre; the organisation of the core had only begun, as the first stage of one long, continuous process. A steam atmosphere of hundreds of bars lay over a molten surface, and a cold, stiff, sialic skin floated on it.

Every number in the papers that follow is computed under these conditions. The conditions familiar to us today — a solid crust, a slow day, oceans, a Moon that recedes by a few

centimetres a year — do not apply to this period, and the differences are usually measured in orders of magnitude. The table below sets out the environment domain by domain.

Domain	The environment of the process	Why it matters to the model
The Sun	Luminosity about 30% lower than today, with far greater activity of the solar wind and of ultraviolet radiation [1]	Surfaces cool faster; volatiles are lost to space by hydrodynamic escape; the sky is not a heat source
The Earth's mass	6.0455×10^{24} kg — uniting the mass of the present Earth (5.972×10^{24}) and the mass of the present Moon (7.35×10^{22}) in one parent body	The Moon's material is all inside the parent; nothing has left it yet
The size	Volumetric radius 6,517 km, including a thermal expansion of 120 km. Because of the fast spin the body is a flattened spheroid, with an equatorial radius of 6,721 km and a polar radius of 6,128 km (Paper 2)	The thermal expansion is a necessary condition: with an expansion of less than 98 km the budget does not close (Paper 2)
The day	5.5 hours at the start of the process and 5.77 hours at release: the lengthening of the pear slows the spin, and the iron sinking to the centre speeds it up (Paper 2). The equatorial speed at the start of the process is about 2.1 km/s, and the sky turns four times faster than today	The corotation line lies at 15,818 km at the start of the process (5,020 m/s) and at 16,399 km at release — only about 2.5–2.6 Earth radii from the centre, where today it lies at the distance of geostationary satellites
Gravity	About 8.25 m/s² at the equator and about 9.5 at latitude 44°, on the flattened spinning parent	Gravity is not a fixed value; it changes with latitude, with the mass leaving and with the body rising above the surface
The interior	Hot through the whole section of the body; the iron in its first stages of descent, and the core only beginning to gather ($k \simeq 0.4$); a magma ocean beneath a thin chilled skin	Angular momentum is stored as in a nearly uniform body; the interior layers can flow; a pressure of 150 GPa is available at the source (Paper 3)
Water	No liquid water at the surface. The present oceans are held in a steam atmosphere of hundreds of bars over a molten surface; the volatiles are dissolved in the melt, and far more abundant than today	The volatiles lower the density of the rising melt to about 1,300 kg/m³ in the body on the tip; they do not raise the pressure above the lithostatic value; there is no ocean to fill a basin

Domain	The environment of the process	Why it matters to the model
The outer shell	A cold, stiff sialic skin, of 2,800 kg/m³, floating over a mobile base — the only rigid component in the system, and the only one able to keep a crater	Its sial, which the forces of the spin carried up to the tip of the pear and which gathered there in many layers, forms the Moon's far-side crust and carries its earliest record; to keep its craters it must be stiffer than $\sim 10^{23}$ Pa s (Paper 3)
The place	The base of the pear near Kiritimati (about 1.9°N 157.5°W, central Pacific), directly above the Pacific large low-velocity province at the bottom of the mantle; the pear leaned back westward and north over the north-central Pacific (the earlier reading of the source, about 25°N 165°E) (Papers 3, 9)	Every position is given in today's frame (see the box on the axes below)
Time	Start about 4.5 billion years ago. In the series' rough timeline, about 600 Myr from the start of the bulge to release at 3.873 Ga: about 400 Myr for the bulge and the pear, about 100 Myr for the flattening stage, and about 100 Myr more to release (Papers 2, 8, 11)	A working scenario, not a measured duration; how long "the Moon" was attached depends on what one calls the Moon (Part B)
Rates	The rates are not the present rates. The tidal recession rate of the Moon falls by about seven orders of magnitude from release to today, when it stands at 3.8 cm a year; a body released just inside the corotation line falls inward for billions of years, and then disrupts within tens of millions of years (Papers 4, 10)	Processes that take billions of years now took only thousands of years then; the order of events is set by rates that fell as the Moon receded

The data in the table draw the picture of a planet turning in five and a half hours, wrapped in a steam atmosphere, with a glowing surface on which a cold skin floats — and no Moon. Figure 1 shows the point about rates: a tidal recession of a few centimetres a year, as it is today, was close to the Earth a recession of an entirely different kind. The process described below is how the Moon separated from that ancient planet.

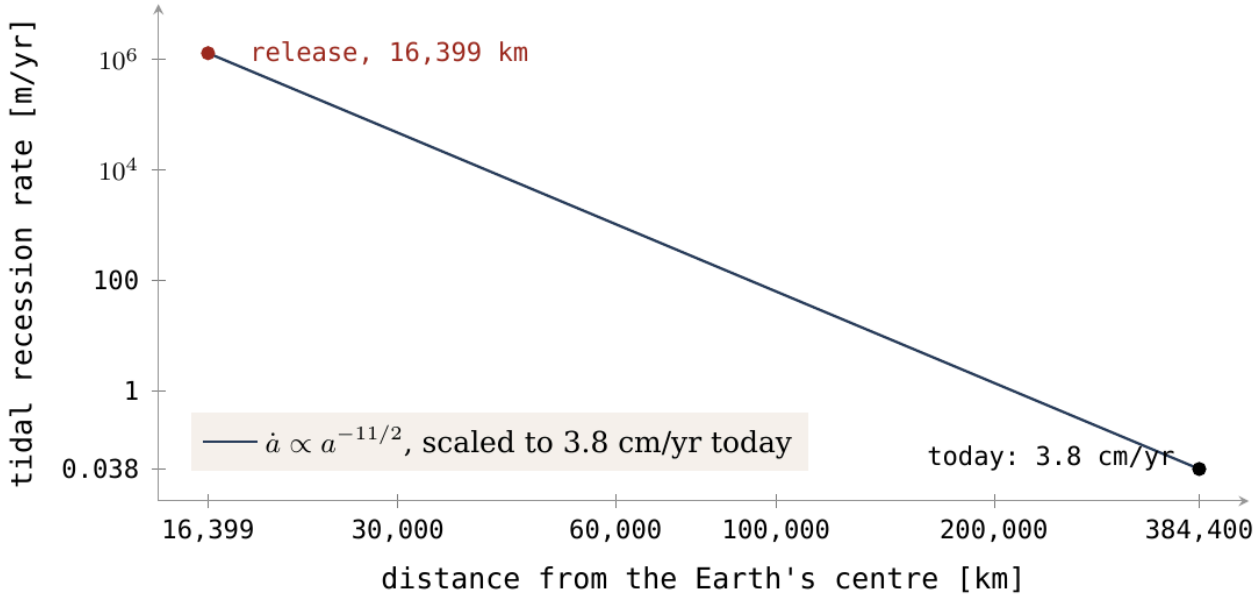


Figure 1. Present rates do not apply to the early Earth. The tidal recession rate, drawn for a constant dissipation and scaled to today's 3.8 cm/yr, falls by about seven orders of magnitude between the release line and the present distance. The curve shows scale only: a single constant dissipation cannot match both today's rate and the age of the system (Paper 4, §D.3).

WHAT IS NOT SOLVED — THE AXES

The system of axes of the Earth and the Moon, from release to today, is not solved in this series. It is a research problem of its own. It involves four planes: the Earth's equator, the plane of the Moon's orbit, the Moon's equator and the plane of the ecliptic. The motion of the axes is made of two separate mechanisms, acting at rates that differ by orders of magnitude: slow wandering of the planet's body relative to its spin axis (True Polar Wander), and precession of the spin axis itself in space. In addition, the frozen bulge of the Earth keeps an unknown memory of earlier spins, and the different components of the system feed back on one another.

For this reason the series gives every position in today's frame: the base of the pear near Kiritimati, the stripes of the North Pacific, the exits beneath South Pole–Aitken. Where a past orientation matters — the angle of 22.5° at release, or the equator of the stripes, inclined at least 10° to today's — this is stated explicitly in the paper that uses it. The solution of the axes is defined as future work and is not part of the series (Part F); that solution may change parts of the model, and the model is stated in a form that allows this.

PART A

What has to be explained

THE PROBLEM AND THE APPROACH

Which properties of the Moon must any model of its origin account for, which of them cost the assembled-from-fragments picture its additions, and why was the lifting route closed? **The approach:** list the seven properties that Paper 11 turns into requirements, set the additions of the collision model beside what a lifted Moon obtains from its geometry, and re-evaluate the angular-momentum objection at the place where this route actually releases the Moon.

Part A sets the target. Before the process is told in Part B, the reader should know what it has to deliver: a list of properties the Moon has, each of them measured, and each of them a property that any model of origin must produce. The list is short, and every item on it is examined again, with its test, in Paper 11.

A.1 The seven properties of the Moon

Identical isotopic composition. The Moon is made of the Earth's material: its oxygen, titanium, chromium and tungsten isotopes are indistinguishable from the Earth's, within a few parts per million [2, 8]. Isotopes are forms of one element that differ only in mass; their ratios differ from one body of the Solar System to another, and are hard to change once a body has formed, so they serve as a fingerprint of origin. For oxygen the Earth–Moon difference is only -1 ± 5 ppm (parts per million). **A small core.** The Moon has a small metallic core: its seismic radius is about 330 km, about 1.4% of the Moon's mass [9], against 32% for the Earth. **Asymmetry in crustal thickness.** The crust is thicker on the far side than on the near side [10] — by about 10 km between the centre of the near side and the far side, and by about 8 km in hemispheric means. **Synchronous rotation.** The Moon keeps one face permanently toward the Earth. **The order of preservation.** The far-side surface was closed and preserved before the fed near side quieted, and the far side is older than the near-side maria. **Water and volatiles.** The Moon's early melt carried water: melt inclusions — tiny drops of magma sealed inside crystals as they formed — hold 615–1,410 ppm [11]; the volatiles have since escaped to space, and the interior measured today is dry. **The centre-of-mass offset.** The Moon's centre of mass lies about 2 km toward the Earth from the centre of its figure.

Paper 11 turns these seven properties into seven basic requirements, R1–R7, each with the quantity that tests it; Table A.1 lists them. To the seven one more property is added here, because it is the property that a disk of debris cannot give without help: the Moon's orbit is tilted 5.145° to the ecliptic.

	Property	What is measured	What a lifted Moon must show
R1	The source of the material	$\Delta^{17}\mathrm{O} = -1 \pm 5$ ppm; full match in Ti, Cr, W isotopes	A composition reachable as a mixture of two terrestrial sources: the deep mantle feed and the rising sialic shell
R2	A small core	Seismic core about 330 km in radius (about 1.4% of the mass)	Little metal reached the Moon: the feed material was mainly silicate, with a limited terrestrial metal component (about 1.47% of the deep material, counting only the metal that reached the core); most core material sank back down the pear to the Earth
R3	Crustal asymmetry	Crust thicker by about 8–10 km in the far hemisphere (GRAIL)	The accumulation of many layers of compacted sial that migrated to the tip of the bulge, together with returning material that landed, on the far hemisphere
R4	One face permanently toward the Earth	Synchronous rotation	The side by which the Moon was attached to the parent is the side that faces the Earth after detachment
R5	The order of preservation	Far side older than the near-side maria	The far hemisphere closed and kept its record before the near hemisphere, which went on receiving feed from the pear, quieted
R6	Water	615–1,410 ppm in ancient melt inclusions	Volatiles in sufficient quantity, and in the right state, to lower the density of the feed and foam it, so that it can rise
R7	The centre of mass	An offset of about 2 km toward the near side	A crust of the modelled shape gives an offset of 1.24–3.11 km for density contrasts of 300–750 kg/m³
+	The tilt of the orbit	An inclination of 5.145° to the ecliptic	An orbit whose inclination is set by the latitude from which the body left the parent (Paper 4)

How to read the table. Each row joins a measured property to a step of the route: R1 to the source, R2 to the feed, R3 and R7 to the building of the far-side crust and the shape of the crust, R4 to the connection, R5 to the order in which the two sides were preserved, R6 to the conditions the feed needed. Paper 11 does not treat these as seven independent proofs: the thick crust on the far side (R3), the offset centre of mass (R7) and the placing of the maria are different expressions of one asymmetric distribution of material, which follows from the way the body grew on the tip of the bulge, and counting them separately would count one fact several times. In addition to the seven basic requirements, Paper 11 defines eleven further requirements, R8–R18, that follow from the

pear structure and from the later papers; the full list of requirements, all eighteen, is gathered in Part E.

A.2 What the collision model had to add

To evaluate the LEF model, a systematic comparison with the family of giant-impact models (Giant Impact) is useful. The table below sets the two approaches side by side, property by property. The middle column shows what the giant-impact family must add to obtain each property, usually a secondary working assumption or a dedicated dynamical addition; the last column shows what a Moon lifted from the Earth obtains from the geometric and mechanical structure of the route.

Geophysical / orbital property	The addition required in the giant-impact family	What follows from the lifting model (LEF)
Isotopic composition identical to the Earth's	Complete thermal and isotopic mixing of the impactor with the Earth, for example through an intermediate synestia state (synestia) or a collision of extreme energy [2, 3]	The material is the Earth's material, so no mixing mechanism is needed (Papers 3, 11)
Orbital inclination of about 5°	A later dynamical excitation mechanism: an evection resonance, close encounters with planetesimals, or a high initial obliquity of the Earth's spin axis [4, 5]	The inclination is set by the latitude of the release point, $i = \lambda$: 22.5° at release, decaying by the reduced decay law to the observed value, 5.145°, at the present distance (Paper 4)
Thicker crust in the far hemisphere	The assumption of a second, companion moon (companion moon), accreting at a late stage onto the far side of the main Moon [6]	Follows from the accumulation of many layers of compacted sial, carried up to the tip of the bulge by the forces of the spin, together with material that returned and landed again and again; the near side served as the base of the connection (Paper 3)
Small metallic core	The impactor's core sinks into the Earth and merges with its core, and only a small fraction of it is retained	All the Moon's metal is terrestrial. Much iron entered the pear, at a lower concentration toward its tip, and the feed that reached the Moon was mainly silicate with a little metal; at detachment the heavy core material lay in the last third of the body, did not cross the corotation line and sank back down the pear to the Earth (Papers 3, 11)

Geophysical / orbital property	The addition required in the giant-impact family	What follows from the lifting model (LEF)
Water in the early interior	Late delivery of water by the impactor or by secondary accretion, after the heat of the impact drove off the volatile components	The original melt was rich in dissolved volatiles, and they are what lowered its density and allowed it to rise (Paper 3)
Angular-momentum budget	Set by the geometry of the impact event, and requiring a later reduction by an evection resonance [4]	The system holds at release 1.007 times today's value, for a parent with an initial spin of 5.5 hours (Paper 2)

The comparison rests on the principle of scientific economy. Within the collision model each observed property requires a dedicated dynamical solution, added for it: complete mixing for the isotopic composition, a resonance or an encounter for the tilt, a second moon for the crustal asymmetry, and a loss of volatiles followed by a return of water. In the lifting model (LEF), by contrast, the same properties are derived as one whole from one cause — the mechanical route by which the material left the parent. The cost of this model is moved to the lifting itself, and the rest of the series computes what the lifting requires.

A.3 Why the lifting was thought impossible

The objection. Classical fission theory — the proposal that the early Earth shed the Moon because of a high spin rate — was first put forward by Darwin in 1879, but set aside within a few decades [12]. The main reason was arithmetic: a fission in which the material breaks away directly from the Earth's surface requires an extreme spin rate, and such a condition demands that the early Earth–Moon system hold a total angular momentum about four times the value measured today. *Angular momentum* is the total amount of rotation of a system — the spin of the Earth on its axis plus the Moon's motion around it. It stays fixed in an isolated system, in the absence of an external torque, and since no external agent could have removed so large an excess, the lifting route was thought impossible. The arithmetic itself is correct. Paper 2 shows the implicit basic assumption on which it rests: that the material leaves the parent directly from the surface, at the tangential speed of the surface.

Why the surface fails. At the surface of a parent turning once in every 5.5 hours, the rotation gives the material a tangential speed of about 2,133 m/s, whereas a free orbit at the height of the surface requires a speed of about 7,701 m/s. Because of the speed gap, about 5.6 km/s, any material released there falls back onto the parent. To close the gap at the surface the spin of the whole planet must be speeded up, and a faster spin inflates the angular-momentum budget. At the classical instability point of a spinning fluid body — the Jacobi bifurcation — the parent turns once in every 2.73 hours and holds an angular momentum of about 9.42×10^{34} kg m² s^{-1}, 2.74 times the

angular momentum of the Earth–Moon system today. The value of about four times (4.08) comes from the original reference calculation, made at a different point of the same sequence of spinning shapes; it is the source of the familiar statement.

Why the corotation line succeeds. The LEF model offers a different solution to the speed gap: not a faster spin of the Earth, but carrying the material outward while it remains attached to the Earth and turns with it as one body. A body turning with the Earth goes round once in every day, so its tangential speed rises linearly with distance, while the speed required for a free circular orbit falls with distance. The two speeds meet at only one distance, the *corotation radius*:

$$v_{\mathrm{rot}} = \Omega r, \qquad v_{\mathrm{orb}} = \sqrt{\frac{GM}{r}}, \qquad \Omega r_c = \sqrt{\frac{GM}{r_c}} \;\Rightarrow\; r_c = \left(\frac{GM}{\Omega^2}\right)^{1/3}.$$

Here Ω is the Earth's spin rate (2π divided by the length of the day), G is the gravitational constant and M is the Earth's mass. In words: the corotation radius is the one distance at which a circular orbit takes exactly as long as one turn of the Earth. For a 5.5-hour day it lies at 15,818 km, and both speeds there are 5,020 m/s; this value takes for M the mass of the Earth alone, as for a test particle, while with the full mass of the parent, 6.0455×10^{24} kg, the same formula gives 15,883 km (Paper 3, §A.9) — two definitions, and both are used in the series. Today's geostationary satellites sit on the same line, for the present day; with a day of a few hours the line lay only about two and a half Earth radii from the centre. A mass released there is already in orbit. An everyday picture: a child who walks from the centre of a turning roundabout to its rim moves faster at the rim, without anyone pushing the roundabout harder. Dynamically, the material rising along the pear gathers the tangential speed it needs gradually during the climb, and not in one throw from the surface; when it reaches the corotation radius and is released, it already holds the full speed required for a free orbit. Darwin computed what is needed to throw a moon off the surface; the LEF route grows a moon and carries it out.

The budget at the line. The spin rate the Earth needs to bring the corotation line this close is the rate it would have had if all of today's angular momentum of the Earth–Moon system were returned to the parent planet: a day of about 5.5 hours. A uniform parent of volumetric radius 6,517 km, flattened by its spin to an equatorial radius of 6,721 km and turning in 5.5 hours, holds an angular momentum of 3.466×10^{34} kg m² s^{-1}. During the mass transfer the lengthening of the pear acts to slow the spin, while the iron sinking to the centre acts to speed it up. At release the day is 5.77 hours, and the corotation line, with the Moon's own mass included in the calculation, lies at 16,399 km. At this point the Moon's orbit holds 5.9×10^{33} (17% of the system's budget; this is the orbital angular momentum of the pair about their common centre, with the reduced mass — counting the full lunar mass about the Earth's centre gives 6.0×10^{33}), and the Earth's spin holds 2.88×10^{34}. The total angular momentum of the system at release is **1.007** times today's (3.466 against 3.441×10^{34}), the same total it held at the start. This modest surplus, about 0.7% in the scalar sum and about 1.5–2.5% against the vector sum at the angles tested, has the sign required: since the release, solar tides have removed about 1% of the system's angular momentum. Table A.3 sets the routes side by side.

Detachment route / model	Where the speed match is made	Earth day	Angular-momentum ratio (to the present)
Classical fission (the original reference point)	the surface	—	about 4 (4.08)
Classical fission (Jacobi bifurcation)	the surface	2.73 h	2.74
LEF model (conserved budget)	corotation line: 15,818 km at the start, 16,399 km at release	5.5 $\rightarrow$ 5.77 h	**1.007**

Table A.3. Comparison of the detachment routes and their angular-momentum budgets: the angular-momentum price of the same product — a Moon in orbit — by route. The red fades as the price falls: the classical route evaluates the speed condition at the surface and pays two to four times today's budget; the LEF route evaluates it at the corotation line and pays what the system holds (Paper 2, §A.6).

How to read the table. The rows run from the most expensive route to the cheapest, and the colour fades with the price. The two red rows, of the classical models, require the Earth's surface to reach orbital speed — a condition that pushes the planet toward the threshold of mechanical break-up and demands an exceptional surplus of angular momentum. The LEF model makes this condition unnecessary: the material is carried out in corotation, and the speed match is made at the line, where a parent turning once in every 5.5 hours already supplies orbital speed. During the transfer the lengthening of the pear and the gathering of the core change the day and move the line, but they only move angular momentum between the Earth's spin and the Moon's orbit, and the total of the system is conserved: it is the same at the start and at release, 1.007 times today's (an earlier version gave 1.008 for the release state, from today's total rounded to 3.44×10^{34}). The objection that closed the route was evaluated at the wrong place.

What it costs, and what it requires. The mass-transfer process carries defined costs: about 15.6% of the initial angular-momentum budget is handed to the rising material in the base calculation, and about 16.5% in the coupled solution; at release, with the core included, the Moon's orbit holds about 17% of the total budget (17.0–17.1%; 17.7% is the share of the pear alone, without the core). These are transfers of angular momentum within the system, not losses from it. In addition, the mechanism sets a structural condition on the planet's interior, which Paper 2 states as a requirement: the parent must still be nearly uniform in its mass distribution at the start of the transfer, with $k \gtrsim 0.397$, where the moment-of-inertia coefficient k measures how the mass is spread between centre and surface (0.4 for a perfectly uniform sphere, about 0.33 for today's Earth, with its concentrated core). Had the gathering of the iron into the core already been advanced before the mass transfer, the planet would have stored less spin at the same day, and the angular-momentum budget would not have closed.

Summary of Part A. This Part presented seven measured properties (R1–R7) and an eighth — the tilt of the orbit — that a lifted Moon must produce from its geometry; it set against them the collision

model, which obtains each of them by a secondary addition; and it showed that the historical objection to lifting the Moon disappears when the speed condition is evaluated at the corotation radius instead of at the surface. Part B now describes the sequence of events by which the Moon got there.

PART B

The process, in stages 0–8

THE PROBLEM AND THE APPROACH

What happened, in what order, and with what numbers? **The answer:** one dynamical sequence made of nine successive stages, numbered 0 to 8; the sequence starts from an early Earth, hot and fast-spinning, and develops gradually to the Earth–Moon system familiar today. Each stage is computed or measured in one of the later papers of the series, and each stage is defined by one central number that the reader can hold on to. The stages run in the environment of the process set out at the opening of this paper: an Earth of 6.0455×10^{24} kg — today's Earth and Moon together — expanded by heat to a volumetric radius of 6,517 km, and turning once in every 5.5 hours, about 4.5 billion years ago.

This Part describes the model as one story, from beginning to end, without stopping to argue for each stage; the arguments are in the papers named at each stage, and Part C presents them paper by paper. The total duration of the process — from the start of the formation of the bulge to the release of the Moon into an independent orbit 3.873 billion years ago (3.873 Ga) — is estimated at about 600 million years. The division of this span is a rough scenario value, not a measured duration: roughly 400 million years (Myr) for the formation of the bulge, the rise of the dome and the lengthening of the pear, about 100 Myr of flattening of the body under the spin, and about 100 Myr from the end of the flattening to release (Paper 3, environment table; Papers 4, 6, 7). The one chronological anchor in the process is its end point: 3.873 Ga is the age at which the isotopic clocks of the norites of South Pole–Aitken were reset, and within the model this reset is the moment of separation itself (Paper 8). Figure B.1 lays the stages on this time axis.

A note on the definition of the body. How long the Moon counts as "attached" to the Earth depends on the definition of what one calls the Moon, and the definition should be stated, not assumed. If the Moon is defined by the material it is made of, it begins with the start of the gathering of material in the bulge, about 600 Myr before release. If it is defined as a distinct body sitting on the tip of the pear, it begins at stage 2. And if it is defined as a body held together by its own gravity, it exists only from the start of the inward rounding, at a distance of about 13,000 km from the Earth's centre (stage 4). All three are the same process seen from different points; the series uses the first when it speaks of the window of construction.

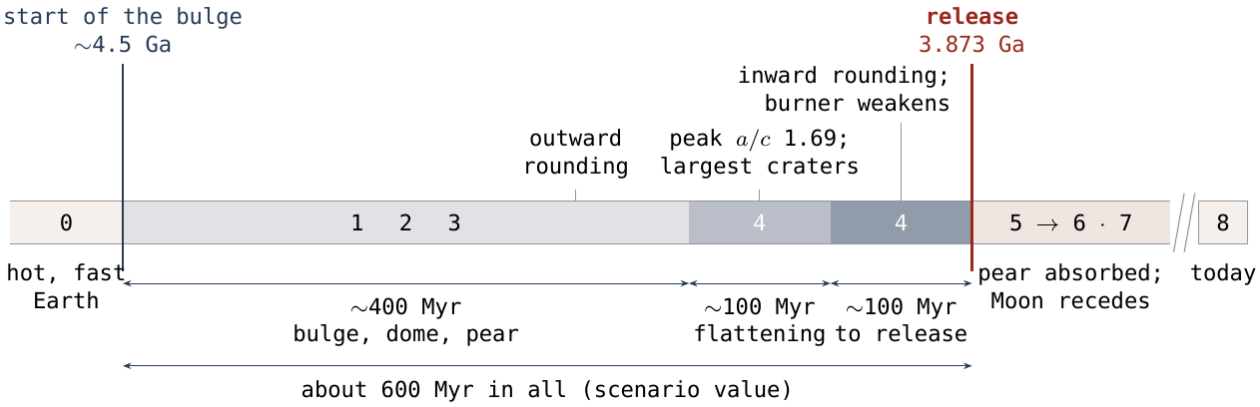


Figure B.1. The stages of Part B on a time axis. The axis runs from the start of the bulge, around 4.5 Ga, to the release at 3.873 Ga, the reset age of the norites of South Pole–Aitken (Paper 8), and on to today after the break. The shading deepens toward release, with the density of the body and the share of its history written into it; the three intervals, about 400, 100 and 100 Myr, are the rough scenario values of the series (Papers 3, 4, 6, 7). The positions of the events inside each interval are schematic.

The figure is read from left to right. The pale band before the start is stage 0, the hot Earth before the bulge. The longest interval, about 400 Myr, holds stages 1–3: the bulge, the dome and the pear. The two darker intervals hold stage 4: the body flattens with the spin and makes its largest craters, then rounds inward while the heat beneath its far side weakens. The red line is the release, stage 5; to its right the pear is absorbed into the Earth (stage 6) and the Moon recedes (stage 7), until the Moon and the Earth of today (stage 8).

A word on numbering. Paper 3 numbers its own stages from the outside of the body: its stage 1 is the flat shell, its stage 2 the body on the tip of the pear, its stage 3 the outward rounding. In this Part the stages are numbered for the whole process, from 0 to 8, and the flat shell belongs to stage 1, the bulge.

B.0 Before the bulge — the hot, fast Earth

The process starts from a point at which the Earth is not today's Earth. It holds the combined mass of the Earth and the Moon together, 6.0455×10^{24} kg, since the model does not assume that anything escapes: the whole of the Moon's material was originally part of the parent. In this period the planet is hot through the whole section of the body, and the dense iron is in its first stages of descent toward the centre to build the core. The heat caused an expansion of about 120 km, bringing the body to a volumetric radius of 6,517 km. Because of the fast spin — a day of only 5.5 hours — the body is flattened, with an equatorial radius of about 6,721 km and a polar radius of about 6,128 km (Paper 2). Its outer shell is sialic — made of *sial*, the light upper material rich in silicon and aluminium (SiAl), which lies over the layer of the denser *sima* (SiMa), rich in silicon and magnesium — and it is cold and stiff (Paper 3, §B.1). An atmosphere of vapour surrounds it.

The angular-momentum balance. Angular momentum is the quantity of rotation — mass times sideways speed times distance from the axis — and it is conserved when no outside torque acts.

The total quantity of rotation is stored in the parent and stands at $3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$, a little more than the combined angular momentum of the Earth and the Moon today: 0.7% more in the scalar sum, and 1.5–2.5% against the vector sum at the angles tested (Paper 2, §A.6). The sign matters: the tidal forces exerted by the Sun over four and a half billion years take angular momentum out of the system, so the parent must have had a little more than today, not less. Closing the budget under these conditions requires a structural condition: the planet's interior must be nearly uniform in its mass distribution. The moment-of-inertia coefficient k, the measure of how far the mass lies from the axis (0.4 for a uniform ball, smaller when mass gathers at the centre), must be at least about 0.397. The construction must therefore begin in the early stages of the organisation of the core, before much of the iron had gathered at the centre (Paper 2, §D.3). That organisation is one long, continuous process, without sharp limits: a little at the start, more while the pear grows, and a very long phase with the return of the pear, which continues to this day.

The corotation radius. A point on the surface of a spinning body moves sideways at a speed that grows with its distance from the axis, while the speed a free circular orbit requires falls with distance. The two are equal at one distance, the *corotation radius*: there a body that simply turns with the Earth already moves at the speed of a free orbit. At a spin of 5.5 hours it lies at

$$r_c = 15,818\ \mathrm{km}, \qquad v_c = 5,020\ \mathrm{m/s}$$

(Paper 2, §A.2.1). This number removes the historical limitation of the classical fission models. At the equatorial surface of the parent the tangential speed was only about 2,133 m/s, whereas a free circular orbit at the same height required a speed of about 7,701 m/s; this gap, about 5,568 m/s, prevents material from breaking away from the surface and makes it fall back. At the corotation radius, by contrast, the rotational speed of the body equals the required orbital speed: the material is not thrown, but climbs gradually along the attached structure, gathering tangential speed from the planet's spin, until it reaches that line. The whole model is the answer to one question — how the Moon's material got from the surface to that line while still turning with the Earth.

The interior asymmetry and the source pressure. The descent of the iron to the core does not take place evenly and symmetrically. An asymmetry in the structure of the interior focuses the rise of mantle material into one preferred feed region, located north of the equator, which Paper 3 calls Lobe A (§A.4). At this deep point the weight of all the material above it produces an available pressure:

$$p_s = 150\ \mathrm{GPa}\ \ (\ 150.26\ )$$

One gigapascal (GPa) is about ten thousand times the air pressure at sea level. The upwelling is a pressure, not a force: opening a flow path from the deep interior lets the squeezed material press upward along it, as toothpaste rises when the tube is pressed. Lifting a deep melt with a density of 3,500 kg/m³ to the corotation radius costs 139.78 GPa, and leaves a pressure surplus of 10.49 GPa that drives the rise. The condition holds for any deep melt with a density below about 3,864 kg/m³, and fails for higher values (Paper 3, §A.9–A.10). A melt rich in water and in dissolved volatiles meets this requirement, whereas a dry, dense melt cannot climb to the required height. The volatiles are what make the Moon possible.

What we now have: a hot, expanded, fast-spinning Earth, still nearly uniform inside, with a corotation radius at 15,818 km and a source pressure of 150 GPa that can lift a light, volatile-rich melt to that radius.

B.1 The bulge — the pressure from inside

The pressure available in the mantle acts asymmetrically and lifts one defined region of the Earth's surface, producing one bulge (Paper 3, §C.7). Before the bulge rises to a significant height, the sialic crust over the active feed region still lies flat ("the flat stage"), and this stage is the starting point of the record of the Moon. Deep mantle material reaches the shell through feed openings and is thrown out through them at speeds of kilometres per second; part of it returns and lands on the shell (Paper 3, §B.1; Paper 6).

Out and back at the same speed. A ball thrown straight up, with no air to slow it, returns to the hand at the speed at which it was thrown. Material thrown upward from the parent likewise returns to the surface at the speed at which it left: a particle thrown at 5 km/s lands back at 5 km/s. The spin adds one component to this: when thrown, the material keeps the sideways speed it had on the surface, but as it rises this speed carries it through a smaller angle per second than the surface beneath it turns — like a child who walks outward on a turning roundabout and falls behind the boards beneath her. During the flight the ground beneath moves on eastward, and so the material lands to the west of the opening from which it was thrown. For a throw speed of 5 km/s the flight rises to a height of 1,932 km, lasts about 28 minutes, and the material lands about 1,032 km west of the opening — about a thousand kilometres (Paper 6, §K.1).

A landing of 5 km/s is a shock. When two bodies of the same material meet head-on at 5 km/s, the pressure behind the shock front is the product of the density of the material, the speed of the front and the speed of the material behind it. For basalt: 2,700 kg/m³ $\times$ 6,750 m/s $\times$ 2,500 m/s, which is

$$p \simeq 46\ \mathrm{GPa\ at\ the\ contact}$$

(45.6 GPa; Paper 6, §J.2). This value exceeds the pressures at which the minerals of basalt begin to record shock. Even at a landing speed of 2 km/s the same calculation still gives about 13 GPa. As a result, the returning material makes craters of the kind the literature reads as the mark of an impact by a body from space. Gravity does not soften the landing: it gives back on the way down exactly the speed it took on the way up.

The oldest craters of the Moon are made at this stage. They are made on the Earth, in a shell that still belongs to the Earth, before the Moon has formed as a body of its own. Nearly all the Moon's craters, some 99%, are the product of terrestrial material that returned and fell; foreign material, if it is found at all, will appear in very small amounts, space dust apart (Paper 6). This matters for everything that follows: the Moon's oldest surface did not form on the Moon. It formed here, and was carried up.

The mechanism of gathering the shell. The shell that will become the Moon's skin is drawn and gathered inward from a region the size of the Pacific Ocean — about 1.0×10^{14} m² without its marginal seas, and 1.65×10^{14} m² with them, 1.4 to 2.4 times the surface of the upper dome the shell will cover — as a cloth laid flat on a table gathers inward when its middle is lifted (Paper 3, §B.1). The shell does not slide far: the inward gathering belongs to the flat stage, before the dome has developed.

What we now have: one bulge rising under a stiff sial shell; openings in that shell throwing material that returns at the speed it left, 5 km/s out and back, 28 minutes in the air, about a thousand kilometres to the west, landing at 46 GPa; and a shell already carrying craters before the body exists.

B.2 The dome — a body on a broad tip

The bulge takes shape as a body seated on the tip of the Earth's rising extension. The tip is broad: the contact between the body and the tip is 6,000 km across. The body is a squashed ball on that tip, made of two parts. Its *lower dome* lies on the curved tip and follows its outline — the tip, curved with a radius of about 4,000 km, enters the body about 1,350 km under the middle of the contact, as a head pushes up into a cap pulled down over it. Its *upper dome* rises above the lower structure and reaches a height of

$$3,650\ \text{km above the rim of the contact}$$

(Paper 3, §B.2). The upper dome makes up most of the surface, about two-thirds (7.0×10^{13} m², 67%), and the lower dome on the tip about one-third (3.4×10^{13} m², 33%).

The density follows from the shape. The mass equals the Moon's mass; the shape fixes the volume, 5.65×10^{19} m³, about two and a half times the Moon's volume today; and mass and volume together fix the density: 1,300 kg/m³. The taller the dome over the same contact, the lower the density — a dome as high as a hemisphere gives about 2,000 kg/m³, and the lowest possible dome, 2,450 km high, gives today's density, 3,344 kg/m³ (Paper 3, Figure 7). The adopted tall dome is not a free variable. 1,300 kg/m³ is the foamed state that the pressure test of stage 0 requires — a melt full of gas bubbles, lighter than the material it is made of, as rising dough is lighter than its flour and water — and it is the density at which the flattening of stage 4 begins (Paper 7). The earlier version of this paper described the stage as a single hemisphere on a base 6,000 km across, with a surface of 56.5 million km² (5.65×10^{13} m²), three times the Moon's far hemisphere (Paper 2, E12; Paper 11, §B.3). Paper 3 has replaced this description with the two domes on a curved tip; the density, 1,300 kg/m³, and the order of magnitude of the surface are kept, and in both pictures neither number is chosen: both follow from the base.

The same lunar mass	The body on the tip (stage 2)	The spherical body at release	The Moon today
Total surface	1.04×10^{14} m²	4.49×10^{13} m²	3.79×10^{13} m²
Total volume	5.65×10^{19} m³	2.83×10^{19} m³	2.20×10^{19} m³
Radius	— (double-dome structure)	1,890 km	1,737 km
Mean density	**1,300 kg/m³**	**2,600 kg/m³**	**3,344 kg/m³**

Table B.1. One mass in three states (Paper 3, §B.2). Read from the first column to the last: the body grows smaller in size and denser.

The table presents the history of the Moon's density in three columns: the same mass, large and light on the tip, a sphere of radius 1,890 km at release, and today a sphere of radius 1,737 km at a

density of 3,344 kg/m³. The shading deepens down the rows toward the density, the quantity that drives the whole sequence.

The body is still held by the Earth. At this stage the body's own gravity — the pull of its own mass, acting to hold it together — is small against the tidal stretch of the Earth, that is, the difference between the Earth's pull on the near side of the body and its pull on the far side. Their ratio is about 0.11: the body's own pull is only about a ninth of the stretch (Paper 3, §B.2). This means that the body is held by the Earth; it is a dome on a tip, and not yet a body of its own.

The shell rises with it and slides to the tip. The stiff sial shell of stage 1 rises together with the body of the pear, as the skin of a balloon rises with the balloon. As it rises it gradually separates from the base beneath it, slides up the pear to its tip and gathers into the body, and there it becomes the surface of the upper dome, and in time the Moon's crust (Paper 3, §B.3). This is why the tip separates as a body of its own. The condition that makes it a shell is stiffness: a viscosity of the order of 10^{23} Pa·s or more — viscosity is the resistance to slow flow; honey has more than water, and a cold shell vastly more than either — which keeps a crater 100 km wide from sagging over ten million years under the gravity of the parent, 8.25 m/s² (Paper 3, §B.6). A shell that holds its craters carries them with it: a crater made on it before it was covered or carried goes with it and keeps its size. The thick crust of the far side is built of many layers of this sial, which migrated gradually to the tip during the bulge under the forces of the spin and were compacted there, together with material that returned and landed on it again and again (Papers 3, 11). The depth this crust reaches is not fixed here; it may be much deeper than has been assumed.

What we now have: a light body of the Moon's mass, 1,300 kg/m³, seated as an upper dome over a lower dome on a tip 6,000 km wide, the upper dome 3,650 km high; still held by the Earth; and wearing the Earth's sial shell, with its craters, as its skin.

B.3 The pear — where it stood, and how it leaned

Beneath the body, the Earth itself is drawn outward. The pressure draws the one bulge into a *pear*-like structure, made of the Earth's own material: broad at its base on the Earth's surface, narrowing outward, with the body on its tip (Paper 3, §C.7). The pear is not a pipe through which material is pumped, but a body of the Earth, standing.

The structure is held by the Earth. A narrow passage cannot do this work: Paper 3 shows that a cylindrical passage of no mass fails the balance of loads, and this balance is reached only above a radius of about 1,000 km (1,017 km), where the passage already holds about half a lunar mass (§A.13). The connection must be broad, hot and massive, and it carries its sideways load by its shape. The pear meets this requirement: it is a single continuous arc, without a waist. It is part of the Earth, and it is held by the Earth's gravity like any other part of it; it is not thrown out, because its speed is not a throwing speed and it has not crossed the corotation line. Its width is bounded from above by the energy needed to lift it:

Width of the pear's base	Mass of the pear	Lifting work	
9,000 km	8.8 lunar masses	8.1×10^{30} J	
10,000 km	10.4 lunar masses	9.3×10^{30} J	
11,000 km	12.1 lunar masses	1.05×10^{31} J	the order of the core's formation energy

Table B.2. Properties of the pear at release: width, mass and lifting work (Paper 3, §C.7). The orange deepens with the cost.

A wider base increases the mass held, and lifting more mass costs more work; at a width of 11,000 km the work reaches the order of the energy released by the descent of the iron into the core, of order 10^{31} J. The series takes the base as 9,000–11,000 km. From this base width Paper 3 derives the mass of the pear: about nine to twelve lunar masses — about nine, some 11% of the Earth's mass, for a 9,000-km base (Paper 3, §C.9; Paper 11, R10); Paper 9 allows seven to twelve. The lifting work is estimated at about 8×10^{30} J, more than five times (about 5.4 times) the energy needed to lift the Moon itself (an earlier version said about six times) (about 1.5×10^{30} J). The descent of the iron, which drives the upwelling, is the same evolutionary process that pays for the pear. While the pear stood, the Earth's core was displaced about 270–370 km toward it (Paper 3, §E.5).

The pear lengthens and its tip narrows. Between the start of the process and release the pear roughly doubles its length, from about 4,000 to about 8,700 km above the Earth's surface, while the width of its tip narrows by half, from 6,000 to 3,000 km, in step with the outward rounding of the body on it (stage 4). The whole pear lies inside the corotation line. Along it the effective gravity points toward the Earth everywhere, so the pear is compressed along its length, it has no waist that could pinch, and once the Moon has gone — nothing of its material can stay in orbit (Paper 3, §C.7).

Where it stood. The base of the pear stood on the Earth's surface near Kiritimati, in the Line Islands of the central Pacific,

$$\text{base: } 1.9^{\circ}\text{N},\ 157.5^{\circ}\text{W}$$

above the Pacific LLVP (large low-velocity province) — a broad region at the bottom of the mantle, seen in seismic imaging, through which earthquake waves travel more slowly than in the mantle around it. The point near Kiritimati falls inside the slow region in all five cluster models tested, and within the -1% contour of two tomographic models, SMEAN (-2.25%) and SMEAN2 (-2.36%); the centre of the province lies 1,476–1,657 km south-south-west of it (Paper 3, §E.4).

How it leaned. The pear did not stand upright. Its outer part, carried far from the spin axis, had to move faster sideways to keep up with the Earth's turn, and lagged — as the arms of a skater lag when she spreads them — so the whole pear tilted back, westward, and its tip and the Moon on it trailed behind (Paper 3, §C.8). It leaned north as well, from a base near the equator to a tip at a latitude of about 22.5°N, the release latitude of Paper 4; the rotation does not produce this lean,

since its sideways push is always directed toward the equator; the northward lean is produced by the oblique feed from the northern source (Paper 3, §B.1). The pear also slowed the Earth: by carrying material outward it took angular momentum from the Earth's spin (stage 5).

The footprint of the tip is seen today. When the pear was gathered back into the Earth, the tip came down to the west of the base and left its outline on the floor of the North Pacific: two bends where the outermost stripes leave their parallels and turn toward each other, at 23.8°N 164.9°E and 5.3°S 173.2°E (Paper 9, §C.1). The two bends are

$$3,360\ \text{km apart}\ (3{,}040\text{–}3{,}700),\ \text{centre}\ 9.3^{\circ}\text{N},\ 169.3^{\circ}\text{E}$$

and their midpoint lies about 3,760 km west of the base (3,762 km by the coordinates; an earlier version gave 3,770), on the trailing side. The base and the southern edge of the tip lie on the equator of the stripe system, whereas the centre of the tip is offset about 15° north of it (Paper 9, §C.2). This footprint is not the 22.5° of release: the release latitude is the direction of the Moon's centre as seen from the Earth's centre while the pear stood, whereas the footprint is the place where the tip came down when the pear was gathered back, in the frame of the stripes — two different quantities in different frames. The width of the footprint falls between the diameter of the contact at release, 3,000 km, and the dome, 3,650 km: the region the stripes turn toward is as wide as the tip. Inside it, a mirror image of the near side at the size of the Moon on the tip — larger than today's Moon by the factor 1.37, the cube root of the ratio of 3,344 to 1,300 — places Serenitatis, Imbrium and the contact centre within the bounds of the footprint (Paper 9, §C.3). All these positions are given in today's frame; how the Earth's axis has moved since is not solved in the series and is left to future work.

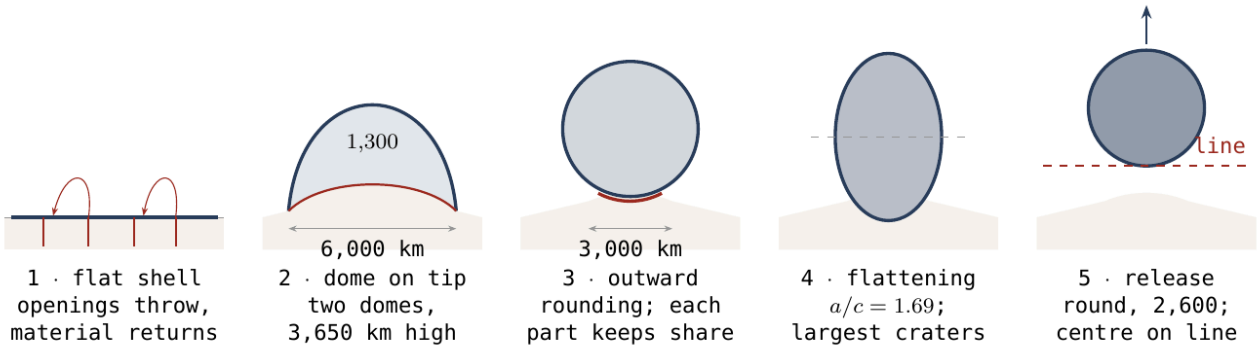


Figure B.2. The body on the tip of the pear, from the flat shell to release (stages 1–5; after Paper 3, Figure 5, and Paper 7). Navy: the shell, which rises with the pear and slides up to its tip, as the skin of the body. Red: the openings on the flat shell (1), the lower dome on the curved tip (2) and the contact (3). The shading deepens with the density, 1,300 to 2,600 kg/m³. The flattening (4) is drawn about the spin axis, which lies horizontal in the picture. Schematic, not to scale.

Figure B.2 shows stages 1–5 in five pictures. Follow the navy line: it lies flat in the first picture, becomes the skin of the upper dome in the second, and stays the skin of the body through the rounding, the flattening and the release. Follow the red line: the contact narrows from the 6,000-km lower dome to the 3,000-km disc, and lets go at the corotation line.

What we now have: a pear of nine to twelve lunar masses, its base 9,000–11,000 km wide near Kiritimati above the Pacific LLVP, held by the Earth's gravity as part of it, inside the corotation line, leaning back to the west and north, with the body on its tip; and on the floor of the Pacific, between two bends of the stripes, the footprint of that tip.

B.4 The rounding — outward, flattened, inward

While still attached to the Earth and turning with it, the body changes its shape three times: it rounds outward, it flattens with the spin, and it rounds inward. Each change of shape leaves on the Moon a signature read to this day: the outward rounding writes the near side and the far side, the flattening the largest craters, and the inward rounding the crater density at the poles (Papers 3, 7).

The outward rounding: each part keeps its share. As the body's own gravity grows, the squashed body on the tip becomes a ball, at constant volume. The contact narrows from 6,000 to 3,000 km and stays attached. The ring of the lower dome between the two — the peeled annulus — peels away from the tip, from the rim inward, together with the sial it sat on: the sial slides up the pear toward the tip — the direction of the outward sorting — and goes with the body; the slide is eased under shear by the "lubricated" layer separating the sial from the sima. The everyday picture is a half-deflated ball pressed onto a rounded stand and then inflated: its wall peels from the stand from the edge inward (Paper 3, §C.4a). This amounts to about 1,500 km of inward motion of the contact's edge, on a weak interface. The peeling from 6,000 to 3,000 km, and the rounding that went with it, took millions of years: slow at the start and faster toward the end — hundreds of millions of years in the early part, tens of millions in the late part. The final position of each part of the surface is fixed by one rule: **in the outward rounding each part keeps its relative share of the surface** (Paper 3, §B.4).

The original structure on the tip	Share of the surface	From the sub-Earth point	On the Moon
The contact, 3,000 km	7%	0°–31°	a near-side cap about 1,900 km across, exposed only at release
The peeled annulus	26%	31°–70°	the belt, exposed gradually during the outward rounding
The upper dome	67%	70°–180°	the far side, exposed to space from the start of the process, its craters carried up

Table B.3. The surface translation rule (Paper 3, §B.4). Red, orange, blue: from the part that touched the Earth to the end, to the part open to space from the start.

A share of a sphere's surface, measured from one point, corresponds to a fixed angle from that point: a cap holding 7% of the surface reaches 31° from its centre, and a cap holding 33% reaches 70°. The point is the sub-Earth point, the point of the body that faced the Earth directly. The surface of the ball is smaller than that of the squashed body — 7.1×10^{13} against 1.04×10^{14} m², a factor of

0.68 in area and 0.83 in length — but since every part keeps its share, the change is the same everywhere and moves no zone and no crater relative to another, as a photograph reduced in size keeps every figure in its place. The rule fixes the size of the zones; the map fixes where they lie today. The contact is centred on the tip of the pear, at a latitude of about 22–24°N; the tests of the series measure the three zones about 24.1°N 335.2°E, the centre of the contact as the author outlined it, 34.0° from today's sub-Earth point (Paper 11). Read on the map, the zones nest one inside another: the base of the bulge, 6,000 km, centred at 16.6°N 13.1°E, holds 99% of the area of the dark basalt of the near-side maria; the last contact, 3,000 km, holds two-thirds of that area (66%) — shares of area, not counts of maria (Paper 5); and Mare Imbrium, the opening through which the feed entered, lies within the bounds of the contact (Paper 3, §B.4). The earlier version of this paper placed the edge of the contact cap at about 41°; the share rule places it at 31°.

The flattening. A soft body spinning fast takes the *Maclaurin figure*, the exact equilibrium shape of a spinning uniform fluid: wider at the equator and lower at the poles. The light body, turning with the Earth once in every 5.5 hours at a density of 1,300 kg/m³, flattens strongly, because its own gravity is weak against the spin:

$$a/c = 1.69 \text{ at the start of the rounding, at } 12,994 \text{ km}$$

where a/c is the ratio of the equatorial radius to the polar radius (1.00 for a sphere; Paper 7, C.1). The distance 12,994 km is the rigid Roche limit for a body of 1,300 kg/m³ — the distance outside which a body's own gravity can hold it together against the Earth's tidal stretch — so the body begins to hold itself, and to round inward, at the same distance at which it becomes able to survive (Paper 7; "about 13,000 km" in Paper 3). During the flattening most of the craters form, and among them the largest: the sorting of the deep material has advanced, the material thrown out is light and rich in water, and the poles, which took their outline in the outward rounding, receive their share of the impact record (Paper 3, §B.6). Large outflows belong to the flattening stage, and in it the far-side crust is built layer upon layer, from the sial and from the returning material.

The inward rounding. As the body climbs along the pear its density rises; its own gravity grows accordingly; the flattening relaxes and it rounds inward. The shell, being stiff, does not shrink with the surface: the craters on it keep their size, and the surface separating them closes, most of all at the poles — as the spaces between buttons sewn on a cloth close when the cloth is gathered, while the buttons stay the same size (Paper 3, §B.6; Paper 7).

The stage on the tip of the pear	Mean density	Earth day	a/c	Crater-density ratio it leaves (poles/equator)
Start of rounding, ∼ 13,000 km	1,300	5.50 h	**1.69**	1.59
On the pear, rising	1,600	5.55 h	1.44	1.38
On the pear, densifying	2,000	5.62 h	1.30	—
Far side reaches the line	2,300	5.70 h	1.24	—
Late attached stage	2,600	5.77 h	1.20	1.17

The stage on the tip of the pear	Mean density	Earth day	a/c	Crater-density ratio it leaves (poles/equator)
Centre of mass on the line — release	2,600	5.77 h	≤ 1.04, round	1.0 for later craters
Today	3,344	—	1.00	—

Table B.4. The figure of the body along the climb on the pear (Paper 3, §C.7; Paper 7). The navy fades as the flattening disappears; the last column is the pole-to-equator crater-density ratio that rounding from that figure leaves (Paper 7).

Reading the table from top to bottom shows that the density rises, the Earth's day lengthens slightly, and the axis ratio of the body falls from 1.69 toward 1. A crater made at the peak of the flattening is carried through the whole rounding process, and ends 1.59 times denser in the polar caps than in the equatorial belt; a later crater, made on a figure of 1.20, ends 1.17 times denser; and a crater made after release, on a round body, is not denser at all. The prediction is therefore not a single value but a range that rises with the age of the craters, and so also with their size, since most large craters formed in the early stages. This range is measured. In a clean count — the maria and the South Pole–Aitken basin removed on both sides, so that old ground is compared with old ground, caps of 30° around the poles against the belt within 15° of the equator — the pole-to-equator crater-density ratio is

$$1.20 \pm 0.05 \text{ for craters} \geq 20\,\text{km}, \text{ rising to } 1.53 \text{ for } 60\text{–}100\,\text{km}$$

— 1.09 for 20–30 km, 1.15 for 30–40, 1.26 for 40–60, 1.53 for 60–100, and 1.84 $\pm$ 0.38 for the few craters of 100–300 km (Paper 7, D.1–D.2). The order and the range are what the model predicts. The craters themselves did not shrink with the surface: the diameter at which simple bowls give way to complex craters with central peaks lies at 16.5–17.1 km on the far side and at 14.3–15.1 km on the near side, with no difference of scale between them, and the polar craters are, if anything, the larger (Paper 3, §B.6; Paper 7, §D.5).

The burner. As long as the body stayed attached, the openings on its near side were fed from the Earth through the contact, and opposite them, beneath the far side, heat gathered: the *burner* of Paper 8 — heat from the Earth's interior that crossed the attached body along the axis of the openings and warmed the far side from below, as a flame warms the base of a pot. The burner had two exits, named after the near-side openings they face: **Imbrium, the main one, and Serenitatis, the secondary**. The heat reached the surface where the crust was thin, and led to a series of changes: the region rose into a dome; the softened crust slid sideways to the margins and piled up there into a thickened ring; volatile material was driven off; the layers of returned material were cooked and compacted; and iron-rich material from the Earth's interior rose to the base of the region. The craters in the region were erased or relaxed: within 10° of the centre of South Pole–Aitken, on ground cleaned of the lava plains, the maria and a halo of 200 km around each of them, the density of craters of 20 km and more is only 0.51 of that just outside the rim of the basin (Paper 8). The melt that solidified above the burner is dated in the Chang'e-6 norites: $4,247 \pm 5$ million years.

The dome of SPA lies south of both exits. The position of the pear and the forces of the Earth's rotation — the horizontal push inside the rotating pear — carried the heat south of the point opposite the exits. On the flattened figure the centre of the dome lay at an angular distance of about 17–21° from them (17.0° from Serenitatis and 21.2° from Imbrium; the exits are the points opposite Imbrium, 37.6°N 340.0°E, and opposite Serenitatis, taken here at 27.3°N 18.4°E); the change from the flattened figure to the sphere widened this gap to the 24–29° of today (24.1° and 29.2°) (Paper 8, §C.3–C.4). The burner weakened gradually as the peeling narrowed the passages: Serenitatis peeled away first and was cut from the Earth, while Imbrium stayed connected and was detached last. After release the heat source was the Moon's own heat, together with the radiation of the nearby Earth, which faded as the Moon receded; with time everything cooled slowly, the maria first and the Moon's centre later, and the dome cooled, contracted and collapsed into the deep basin seen today — as a soufflé, raised by heat, sinks below its rim when it leaves the oven.

The far side seals; the near side stays open. The far side, exposed to the cold of space, lost its heat and sealed its old openings early, before release; the pressure beneath it was trapped (Paper 5). The near side, by contrast, faced the hot Earth: radiation from an Earth of about 2,000 K, with a view factor of 0.205, held the near-side surface temperature near 1,345 K, above the solidus — the temperature at which the material begins to melt — so the near side could not freeze while attached, and its openings stayed open (Paper 5, §H.4).

What we now have: a body that rounded outward and so wrote its near side, its belt and its far side; flattened to an axis ratio of 1.69 and made its largest craters; rounded inward, and raised the crater density at the poles to 1.20 $\pm$ 0.05 and up to 1.53 for the larger craters; and, beneath its far side, a burner with two exits that raised and then dropped the dome of SPA, south of both exits, and weakened as the passages narrowed toward release.

B.5 The release — the centre of mass on the corotation line

While the pear stands, two opposing processes act on the Earth's spin (Paper 2, §A.4). **The lengthening of the pear (slowing):** the lengthening pear carries material outward and slows the Earth, as a skater slows when she spreads her arms; acting alone, it would lengthen the day by 0.94 h, from 5.50 to 6.44 h, and move the corotation line from 15,818 to 17,569 km. **The gathering of the core (speeding up):** the iron gathering into the core brings mass toward the axis and speeds the Earth up, as the skater speeds up when she pulls her arms in; with the core organised as Paper 2 computes it (concentration $D = 2$), it shortens the day by 0.67 h. The combination of the two mechanisms gives:

$$T_{\oplus} = 5.77\,\mathrm{h}, \qquad r_c = 16,399\,\mathrm{km\ on\ the\ equator}, \qquad 17,287\,\mathrm{km\ at}\ 22.5^{\circ}$$

(16,331 km for a body of negligible mass; 16,399 km when the Moon's own mass is included in the two-body orbit; Paper 2, §B.1; Paper 3, §C.8). 16,399 km is about 2.6 Earth radii. Off the equator, a body turning about the axis moves on a smaller circle, and so must be carried a little farther out before its speed matches orbital speed: the line lies at $r_c / \cos^{2/3}\varphi$, 888 km farther out at 22.5° than on the equator. The earlier version of this paper placed the end state at 17,570 km with a 6.44-hour day — the effect of the pear alone; with the core taken into account, the release state of the series is 5.77 h and 16,399 km.

The body is round. At release the density of the body is 2,600 kg/m³ and its radius 1,890 km. It is round, and its outer shell is rigid: a shell of rigidity 10 GPa or more holds its axis ratio to at most 1.04 (Paper 2, §B.7). The flattening and the rounding are over; they happened while the body was attached. Its near point lies about 14,509 km from the Earth's centre, and its far side about 18,289 km; the Earth itself, flattened by its spin, has an equatorial radius of 6,701 km (a little less than the 6,721 km of the 5.5-hour parent, since it now turns more slowly), so about 7,800 km separate the Earth's surface from the Moon — the length that the pear bridges (Paper 2, §B.1).

Why the centre of mass, and why the line. A body moving as one piece responds to the sum of the forces acting on it as if all its mass were concentrated at its centre of mass. Inside the line, the Earth's pull is larger than circular motion needs, so the body presses on the contact: the contact is compressed and holds. The far side reaches the line first — at a density of 2,300 kg/m³ and with a 5.70-hour day (Table B.4) — but as long as the centre of mass is inside the line, the sum stays directed inward. Once the centre of mass crosses the line the sum reverses, and the body begins to pull on the contact (Paper 3, §C.5).

The body is pulled off. The contact is not a brittle fracture surface but a hot, viscous layer. It resists, but only moderately: the body begins to peel from it when its centre of mass reaches the corotation line, and detaches slightly after, about 210 km past the line. It is not released gently; it is pulled outward with force — and this is also what drew up the spatula ridges on the floor beneath the tip (stage 6). The internal pressure beneath the body assists. At a latitude of 22.5° the radial component of the load vanishes on the line, and a sideways load of 4.06×10^{22} N remains, directed toward the equator; the broad contact carries this load, not a narrow neck (Paper 3, §C.2).

Carried, not thrown. The pear lifts the body and carries it round the axis. On the equator, at a distance of 16,399 km, the sideways speed the rotation has given the body is 4.96 km/s, and the speed of a circular orbit there is the same 4.96 km/s: that is what "on the line" means (Paper 3, §C.8). A body thrown from low on the Earth has height, but not the sideways speed an orbit requires: a body thrown at 7 km/s from 8,991 km, the top of a body sitting low on a 5.5-hour Earth, rises to 24,433 km and comes back to 854 km from the centre, deep inside the Earth. Only the carried body arrives with the speed it needs.

The angular-momentum balance closes. The pear passes the Moon about a sixth of the system's angular momentum, taken from the Earth's spin. At release the system holds

$$1.007 \text{ times today's angular momentum}$$

— a surplus of 0.7% in the scalar sum, with the sign that the loss of angular momentum to solar tides requires — whereas the classical fission route, in which the Earth must spin itself to break-up, needs 2.74 times today's angular momentum (Paper 2, §B.6, §A.6.5). The objection that closed the fission route — a budget about four times larger than required — was computed at the Earth's surface, where release would need speeds so high that they would break the Earth apart. In the route proposed here the pear raises itself to the region of corotation and releases the material at its tip, here the Moon: the pear grows until the material at its tip detaches, and then the sorting draws the pear back in. The classical objection was computed at the Earth's surface; this route is computed at the tip of the pear.

The body survives. The Moon's centre lies far beyond the rigid Roche limit of a body of 2,600 kg/m³, 10,314 km — about 7,200 km outside it, at 17,496 km (Paper 4). The fluid limit, which applies to a body with no strength, lies farther out, at 19,971 km; the rigid, round body is held by its shell, and the series requires that its interior be stiff enough to pass the 10^1–10^3 years it spends inside that distance (Paper 2, §D.3).

What the release does to the Moon. The contact, 3,000 km across, peels from the tip. Over it the heat had already melted the sial away and smoothed the surface; the body leaves with the shell over its far side, the peeled sial over its belt, and over the contact a surface smoothed by heat and gravity (Paper 3, §C.4a). The near-side maria were until this moment open passages of the pear's interior: through them the Earth's pressure passed from the Earth, through the pear, into the Moon, and throughout the whole period of attachment they stayed hot, carrying magma from the pear into the Moon. At release this passage is cut. The maria do not freeze at once: at the moment of detachment they are hot and of low viscosity, which is why their floors are smooth, like a frozen lake; they freeze only after detachment, slowly and from the outside inward, over hundreds of millions of years and up to one or two billion, kept warm by the Moon's own heat — and not by the Earth's pressure — and in the first stages also by the radiation of the nearby Earth. Their floors therefore froze only after nearly all the returning material had already landed — Mare Imbrium holds 11 craters of 20 km and more per million km², against 205 on the highlands (Paper 5). The contact maria, which froze last, are the cleanest.

When. The norites of South Pole–Aitken crystallised above the burner $4,247 \pm 5$ million years ago; a later event reset their clocks at

$$3.873 \pm 0.008\ \mathrm{Ga}$$

— an event strong enough to restart the radioactive clocks, but not to make the material anew. Within the model this event is the separation itself, the end of the attached stage (Paper 8, B14–B15). It falls at the end of the window of construction, as the model requires.

What we now have: a round, rigid Moon at a density of 2,600 kg/m³ and with a radius of 1,890 km, pulled off after its centre had crossed the corotation line — 16,399 km on the equator, 17,287 km at 22.5° — on a 5.77-hour Earth, with the speed of a circular orbit already given to it; the angular momentum of the system 1.007 times today's; the near-side maria cut from the Earth's pressure and still hot; and the date 3.873 Ga.

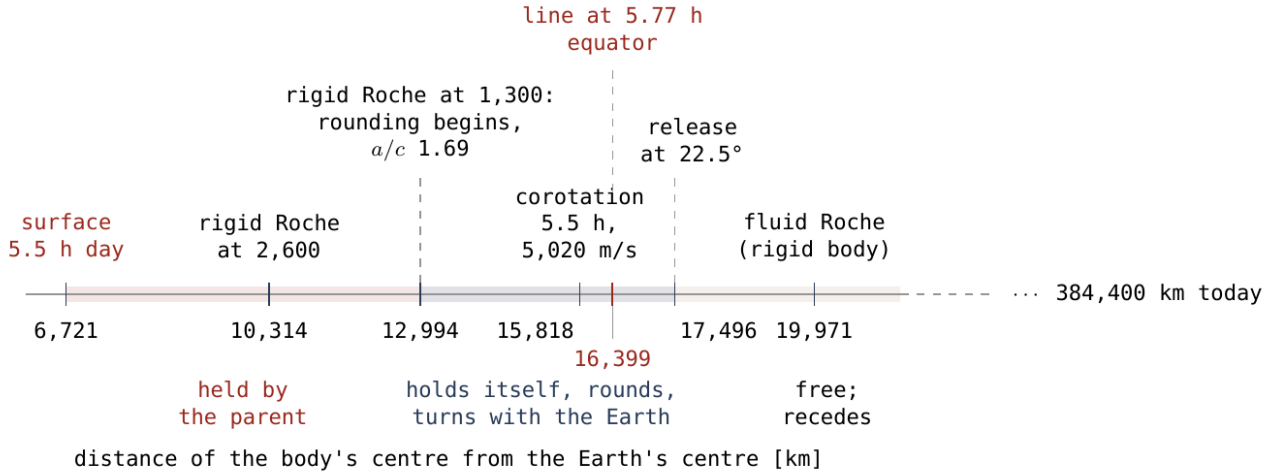


Figure B.3. The process on one axis of distance, with the reference values of the series (Papers 2, 3, 4, 7). The three bands are the three dynamical regimes: the body held by the Earth, the body holding itself and rounding while it turns with the Earth, and the free Moon. Not to scale beyond the break.

Figure B.3 tells the same story as Figure B.1, this time in distance rather than in time. Look first at the three bands: the red band, where the body is held by the Earth; the navy band, which begins at 12,994 km, where a body of 1,300 kg/m³ can first hold itself and the inward rounding begins; and the pale band beyond the release at 17,496 km, where the Moon is free. Then look at the two lines near the middle: the corotation line of the 5.5-hour parent, 15,818 km, and the line of the 5.77-hour Earth at release, 16,399 km. The fluid Roche limit lies beyond the release; the Moon does not obey it, because at release it is rigid.

B.6 The fate of the pear — absorbed into the Earth

The detachment of the Moon does not make the pear collapse or return. The sorting acts on the pear whether the Moon is on it or not, and after release it gradually gathers its material back into the Earth over tens to hundreds of millions of years. The pear does not leave the Earth: it is gathered back into it and absorbed. Since the whole pear lay inside the corotation line, nothing of its material can stay in orbit; no remnant of the connection is left attached to the Moon or circling the Earth (Paper 3, §C.9; Paper 11, R10). The mass returns from high outside the Earth to deep inside it, and the potential energy it gives up enters the Earth as heat. This absorption leaves three marks, each in its own place.

The heavy part sank to the base of the mantle. The pear worked as a settling column: along it the effective gravity pointed toward the Earth, so heavy, iron-rich material sank back while light material rose to the tip — which is why the Moon kept only a small core. After release the heavy part of the pear went down to the base of the mantle beneath the base of the pear, at the place where the Pacific LLVP is identified today; the position test passes in every model examined (Paper 3, Part E). We propose — with reservation, since the nature of the deep provinces is still open, and the Pacific province may in part be core-related material that did not finish sorting — that this province is largely the returned pear. The model predicts that the Pacific province is denser and richer in iron

than the African one, lower and broader. The Earth's core, displaced about 270–370 km toward the pear while it stood, returned to its place as the pear was gathered back (Paper 3, §E.5).

The leading wall was folded into the stripes. The Earth turns from west to east, so the eastern wall of the pear was its leading side, pushing into the direction of the rotation and stretched in that one direction for as long as the pear stood. When the pear was gathered back, this wall was smeared along the floor in the direction of the rotation and folded — hence the long, straight, parallel stripes of the North Pacific: Mendocino, Pioneer, Murray, Molokai, Clarion and Clipperton (Paper 9). The stripes are parallel to one equator, and the base of the pear lies on that equator, 1.6° from it. They pass beneath the volcanic ridges that cross them, unbroken: they were made by folding from the sides and from below, and not cut from above. And they are as long as the pear was tall:

$$\text{stripes } 9,360\text{ km} \quad \text{against a pear of } 9,680\text{–}9,800\text{ km}$$

— from the centre of the tip to the eastern end of Clipperton, 4% shorter than the pear, whose length runs from the Earth's surface to the corotation line at release (Paper 9, §C.4). A wall laid down along the floor covers ground about as long as it was tall, as a felled tree lies across a length equal to its height.

The back was crumpled into the arcs. The western side, the back of the pear, was gathered back in no order: the crumpled arcs of the western Pacific and of Indonesia — the Banda arc 8,100 km from the base, the Philippines and Sulawesi 9,040–9,070 km, and Java 10,300 km, again where the length of the pear reaches (Paper 9, §B.8). East of the base the marks are straight and parallel; west of the tip they are curved, crowded and turned in many directions. As the tip itself was pulled away, the viscous layer between it and the floor was drawn up into ridges, like the ridges a spatula leaves on a wall: in the region of the detachment all the mountains were pulled up from above — these are the spatula ridges, the Marshall–Gilbert–Tuvalu chains in the west and the Line Islands in the east (Paper 9, Part D). Mountains pushed up from below are to be sought elsewhere, not in this region.

The equator of the stripes. Fitted together geometrically, the stripes are parallel to an equator inclined by at least 10° to today's equator (the first fit, 11°, crosses today's equator at 159°W, near the base; values up to about 20° fit almost equally) (Paper 9, §E.2). The stripes were not laid down while the pear stood but as it was gathered back, after release; they record the equator of that time. Until then a mass of seven to twelve lunar masses had stood high above one side of the Earth; when it was gathered into the Earth, the Earth's distribution of mass changed, and the Earth could begin to turn toward a new orientation. How far it turned, how fast and about which line — these are the subject of the study of the axes, which this series leaves to future work.

The Earth goes on throwing. The release did not end the Earth's throwing activity. Its openings went on throwing material; most of it fell back to the Earth, and part of it reached the Moon, then a separate body receding from a distance of a few Earth radii. To reach it the material had to leave at ever higher speeds. All the material was very hot: the large pieces were thrown first, and after the magma storm calmed, small particles, still hot, went on being thrown (Paper 3, §C.11; Paper 6). The Moon kept the record. On the highlands alone, with all the maria excluded on both sides so that the comparison is like for like, the near side — which always faces the Earth — carries 1.19 times as

many fresh craters of 1–2 km as the far side (1.12–1.27), and the craters on the maria grade gradually from large, melted structures to small, sharp ones (Paper 6, Study 9; Papers 3, 5). A flux of bodies from an outside source predicts no near-side excess. The Earth was the last scatterer of the small craters on the Moon.

What we now have: a pear absorbed into the Earth — its heavy part in the Pacific LLVP beneath its base, its leading wall in the stripes, 9,360 km against a pear of 9,680–9,800 km, and its back in the arcs of the western Pacific and of Indonesia — an equator of the stripes inclined at least 10° to today's, and an Earth that went on throwing material at the Moon after the release.

B.7 The recession — the orbit and its tilt

The tilt is the latitude. The *inclination* of an orbit is the angle between the plane in which the body goes round and the Earth's equator. A body released while turning with the Earth at latitude λ moves eastward along its circle of latitude; the orbit it enters must pass through the Earth's centre, because that is where gravity pulls; and a plane through the Earth's centre and through a point at latitude λ, running eastward there, is tilted by λ to the equator:

$$i_0 = |\lambda|$$

— three lines of algebra from $\mathbf{h} = \mathbf{r} \times \mathbf{v}$, with no mass, no distance and no spin rate, and nothing to adjust (Paper 4, §A.2). Released at a latitude of 22.5°, the Moon begins its path at an inclination of 22.5°. A disk cannot do this: a disk of debris settles onto the plane of the equator and starts with almost no tilt, which is why the lunar inclination has been an open problem, and every remedy for it in the impact picture adds a further component to that picture.

The latitude. The adopted latitude, 22.5°, is that of the centre of the contact, about 3,000 km across, on which the Moon sat; the upper end of the latitude of the opening, 25°, is a sensitivity case (Paper 4). The series does not claim that the body moved in latitude on its contact before release. The base of the pear stood roughly on the equator of that time; whether that equator is today's equator is a question for the study of the axes, left to future work.

Nearly circular, and outward from the start. The corotation radius, corrected for a latitude of 22.5°, lies at a distance of 17,287 km from the Earth's centre. The separation begins when the Moon's centre of mass reaches this point, and the final release takes place at a distance of 17,496 km, about 210 km beyond the corotation line, when a little more than half of the Moon's body has crossed the line. At this distance the tangential speed of the body is slightly higher than a circular orbit needs, and so the release point is fixed as the perigee of a nearly circular orbit ($e_0 \simeq 0.037$, apogee 18,830 km). Had the body been released on the equatorial line at the same latitude, its speed would have been too low for a circular orbit: it would have entered an eccentric orbit ($e_0 = \sin^2 \lambda = 0.1464$) in which the point of detachment is the apogee, and the Earth's tide would at first have drawn it inward. Beyond the corotation line, the Earth's own spin is faster than the Moon's orbital motion; as a result, the tidal bulge the Moon raises on the Earth is carried ahead of it, and its pull draws the Moon forward along its orbit, so the orbit grows: the tide pushes the Moon outward from the first moment, and it recedes from its first orbit (Paper 4, §B.2). After release, the Moon's own spin follows its orbit: about 6.2 hours at release, 14 hours at a distance of 30,000 km, and 41

hours at a distance of 60,000 km; the small bulge that the spin and the tide raise on the rigid body falls from an axis ratio of 1.03 at 30,000 km to a value of 1.00 at 60,000 km (Paper 7, environment table, after Paper 4).

The tilt decays. As the tidal forces push the Moon to a greater distance, the inclination of its orbit falls gradually. The reduced law of Paper 4, $\tan i \propto a^{-1/2}$, takes the inclination at release, 22.5°, to

$$5.145^\circ \text{ at today's distance} = 5.145^\circ \text{ observed}$$

(Paper 4, §D.1). Putting a latitude of 25° into the same law gives 5.788°; the latitude that returns the observed angle is 22.5° itself. The reduced law does not fully integrate the orbit through the Laplace-plane transition — the distance at which the slow precession of the orbit passes from being set by the Earth's bulge to being set by the Sun — or through the Cassini state, and these are defined in Paper 4 as requirements.

The rate. The rate of recession falls very steeply with distance, as $a^{-11/2}$, because the tidal bulge weakens rapidly as the Moon moves away; as a result, nearly all the time of the journey is spent in its slow, late part. Today the Moon recedes by about 3.8 cm a year, at a distance of 384,400 km. A single constant dissipation coefficient cannot give at the same time today's rate of recession and the age of the system: the value that reproduces 3.8 cm a year brings the Moon to its present distance in only about 1.6 billion years; the value that fills the 3.87 billion years since release ($Q \approx 29$) gives a present rate of about 1.5 cm a year, two and a half times too small, and the value that fills 4.5 billion years ($Q = 33.8$) gives a rate three times too small. Hence the Earth's dissipation coefficient has varied over time, a finding consistent with detailed models of the tides (Paper 4, §D.3).

The picture at this point: an orbit whose inclination at release equals the latitude from which the Moon left, 22.5°, nearly circular, receding from the start, and brought by the reduced law to 5.145° at the present distance, the value observed.

B.8 Today — what is seen on the Moon and on the Earth

The process leaves a Moon and an Earth with distinct signatures, measurable today on both bodies — and these signatures are indeed measured (Paper 11).

On the Moon.

- *A terrestrial isotopic composition.* The Moon's oxygen isotope ratios match those of the Earth within -1 ± 5 ppm (parts per million): the Moon's mantle is the Earth's mantle, lifted by the pressure and carried along the pear structure (Paper 11, R1; Paper 3, §C.10).
- *A small core.* The seismic core reaches a radius of only about 330 km, about 1.4% of the lunar mass: the feed material was mainly silicate and poor in metal, since the iron was still descending, and the pear sent most of what rose into it back down (Paper 11, R2).
- *Asymmetry in crustal thickness.* The far-side crust is thicker than that of the near side by about 8–10 km — about 10 km between the centre of the near side and the far side, and about 8 km in hemispheric means. The far side is the upper dome, built layer upon layer from the Earth's shell and from the returning material; the near side is the base, which carried only the sial it sat on,

and over the contact lost even that to heat. For a far-side crust 55 km thick, the far hemisphere holds 2.6×10^{21} kg of crust, against 2.85×10^{21} kg of shell on the dome: the two values agree to about 10%, with no parameter adjusted between them (Paper 11, §B.3, corrected 15.10.2026). The model further predicts that the far-side crust will show enrichment in iron with depth, since its deeper layers formed earlier and were less sorted (Paper 3, §B.7).

- *The centre-of-mass offset.* The Moon's centre of mass lies about 2 km toward the Earth from the centre of its figure; a light crust thicker on the far side moves it toward the near side, and in the example of Paper 11 the offset is 1.24–3.11 km for the density contrasts tested (Paper 11, R7).
- *One face to the Earth*: the near side is the side that served as the base of the body on the tip (Paper 11, R4).
- *The maria.* A mare is an opening over a feed dome. Of the 34 maria catalogued in Paper 5 (Appendix A), 33 sit on a dome of the gravity field — a result that no random placement of 34 points reached in 20,000 trials — and of 73 such domes, 44 lie under bright highland, 36 of them on the far side: buried openings, which worked and were later covered, with the craters of their time, by the layers of material that built the land around them (Paper 5, Part D). The near-side cap, which served as the contact, holds the great maria; the openings through which the feed passed are the dense, gravity-high basins, and Paper 5 examines which of them threw material out and which only carried it under cover. The crust test of Paper 5 used the 36 openings first marked by the author, of which the catalogue keeps 34 (with point 35, Copernicus H, and point 36, Sinus Aestuum); the two counts are kept apart there.
- *Polar crater density.* 1.20 ± 0.05 in a clean count, rising with crater diameter to 1.53: the written record of the inward rounding (Paper 7).
- *The South Pole–Aitken basin.* A crater-poor centre — about 0.51 of the density just outside the rim — a reduced mantle beneath it, a thickened ring of crust that slid off the dome, and the centre of the dome 24–29° south of the two exits, Imbrium and Serenitatis (Paper 8).
- *The origin of the crater record.* Nearly all, some 99%, are the product of returned terrestrial material; the oldest were made on the Earth before the Moon was a body; the small late ones were made of material the Earth threw after the release, and their near-side excess is 1.19 times (Paper 6).

On the Earth.

- *The Pacific LLVP* lies beneath the base of the pear near Kiritimati, and is identified as a slow region in all five models tested (Paper 3, §E.4).
- *The stripes of the North Pacific* represent the leading wall of the pear, folded: their length is 9,360 km against a pear of 9,680–9,800 km, and their equator is inclined at least 10° to today's (Paper 9).
- *The footprint of the tip*, 3,360 km wide, bounded between two bends of the stripes, centred at 9.3°N 169.3°E, about 3,760 km west of the base; on either side of it the two chains of spatula

ridges (Paper 9).

- *The arcs of the western Pacific and Indonesia* represent the back of the pear, at 8,100–10,300 km from the base (Paper 9).

In the sky. A lunar orbit at a distance of 384,400 km, inclined 5.145° to the ecliptic, and a recession rate of about 3.8 cm a year (Paper 4, target state).

Each of these signatures is a requirement of Paper 11 or a measurement of Papers 5–9, and each is gathered with its status in Part D and with the requirements in Part E.

B.9 The stages in one table

The table below gathers Part B in one place: for each stage it shows the central process, the numerical parameter that defines it, and the paper in which the calculation or the measurement is made. The colour deepens from the hot Earth to the release, as the body gathers, densifies and rounds, and turns to red after the release, when the story passes back to the Earth and to the free Moon.

Stage	Description of the process	Leading numerical parameter	Reference paper
0 Before the bulge	A hot, expanded Earth, with a nearly uniform mass distribution, its iron descending; one feed region under one region of the surface	5.5 h; $r_c = 15,818$ km, 5,020 m/s; 150 GPa at the source; melt $< 3,864$ kg/m³	2, 3
1 The bulge	One bulge raised by the pressure (the flat stage); material thrown through openings in the shell lands as returning material and makes craters on the Earth's shell	5 km/s out and back; 28 min; ∼1,000 km west; 46 GPa	3, 6
2 The dome	A body of lunar mass on a broad tip, upper dome over lower dome; the stiff sial shell rises with it as its outer skin, carrying craters	contact 6,000 km; dome 3,650 km; 1,300 kg/m³; self-gravity/tide 0.11	3, 11
3 The pear	A pear held by the Earth's gravity as part of the Earth, its base near Kiritimati above the Pacific LLVP; leaning back west and north; the tip footprint between the stripe bends	base 9,000–11,000 km; 9–12 lunar masses; tip 6,000 $\rightarrow$ 3,000 km; footprint 3,360 km, about 3,760 km west	3, 9

Stage	Description of the process	Leading numerical parameter	Reference paper
4 The rounding	Outward rounding by the share rule (contact, belt, dome); flattening and the largest craters; inward rounding crowding the poles; the burner beneath the far side, exits Imbrium and Serenitatis, weakening as the passages narrowed	0–31°, 31–70°, 70–180°; a/c 1.69 at 12,994 km; poles 1.20 $\pm$ 0.05, to 1.53; SPA dome 17–21° $\rightarrow$ 24–29°	3, 7, 8
5 The release	The centre of mass reaches the line and the body is released about 210 km beyond it; the round, rigid Moon leaves on a nearly circular orbit; the 3,000-km contact peels; the near-side maria are cut from the Earth's pressure and freeze slowly afterwards	5.77 h; 16,399 km (17,287 at 22.5°; release at 17,496); 2,600 kg/m³, 1,890 km; 1.007 $\times$ today's angular momentum; 3.873 Ga	2, 3, 4, 5, 8
6 The fate of the pear	Absorbed back into the Earth: the heavy part to the LLVP, the leading wall to the stripes, the back to the arcs; the Earth goes on throwing material	stripes 9,360 km vs pear 9,680–9,800 km; equator of the stripes $\geq$ 10°; near-side fresh craters 1.19	3, 6, 9
7 The recession	Initial inclination equal to the latitude of detachment; a nearly circular orbit; the tide pushes outward from the start; the tilt decays	$i_0 = 22.5^\circ \rightarrow$ 5.145° (= observed); $\dot{a} \propto a^{-11/2}$	4
8 The modern era	The Moon: the Earth's material, small core, thick far side, 34 maria, crowded poles, SPA. The Earth: LLVP, stripes, arcs	-1 ± 5 ppm; core $\sim$330 km; far side +8–10 km; centre of mass $\sim$2 km; 3,344 kg/m³	5–9, 11

Table B.5. Summary of the stages of the model (0–8), the numerical parameters and the papers of the series. Navy deepens through the construction as the body gathers, densifies and rounds; red marks what follows the release.

Reading the table from top to bottom tells the model again in nine lines. The first half of the table traces the course of events; the second half supplies the quantitative validation that makes the description a model: a numerical parameter for each stage, computed or measured, and the paper in which the reader can check it. Part C now presents those papers, one after the other.

PART C

The papers, one by one

Whereas Part B presented the process as one sequence, this Part sets out the ten papers that form the foundation of the model, in the order of the series. Each paper is presented in the same structure: first its central claim is given in a box of two or three sentences, and then its parts are set out — every Part and every appendix — paragraph by paragraph. Each paragraph carries the title of the original part and its key numerical values. The reader who wants only the structure of the series can read the claims and stop there; the reader who wants to know what a paper actually computes, and where in it to look, reads on through its parts. The numbers are those of the papers themselves, and each paragraph names the section (§) in which they are found.

C.2 Paper 2 — LEF Mechanism in Principle

> **THE CLAIM**
>
> The Moon's material left the Earth while keeping full rotational synchrony (corotation), and was released at the point where the tangential speed it already had equalled the speed a free orbit requires: the corotation radius, 15,818 km for a 5.5-hour parent, and about 16,400 km at the 5.77-hour day of release. Evaluating the condition of detachment at this line, and not at the surface, shows that the angular-momentum budget closes at 1.007 times today's (an earlier version gave 1.008), while the classical fission route needs 2.74 times; the objection that closed the door on a terrestrial origin since Darwin does not apply to this route.

Front matter — the environment, the idea and the introduction. Before any calculation, the paper fixes the state to which each value belongs, so that no number is carried from one moment to another. The parent unites today's Earth and Moon in one body, 6.0455×10^{24} kg, where the Moon's mass, 7.35×10^{22} kg, is counted once — inside the parent before the transfer, and in the Moon after it — leaving the Earth 5.972×10^{24} kg. The parent is hot and expanded (volumetric radius 6,517 km, an adopted expansion of 120 km), and its mass distribution is nearly uniform, since the organisation of the core is at its beginning; its day is 5.5 hours, and as a Maclaurin spheroid — the flattened equilibrium shape of a uniform spinning fluid — its equatorial radius is about 6,721 km. The introduction separates two questions that were bound together for a century and a half: the *speed* the material needs at the moment of release, and the *angular-momentum budget* needed to bring the system to that state. The 5.5-hour day is not chosen: it is derived by returning today's angular momentum, 3.44×10^{34} kg m^2 s^{-1}, to the spin of a hot, undifferentiated Earth (5.54 h), plus the roughly 1% that the solar tides have removed since.

Part A — The problems before us. Part A states, one by one, the conditions that a body built from the Earth's own material must meet to reach orbit whole. The first is the velocity gap that ended classical fission: at the equator of a 5.5-hour parent, the surface speed is 2,133 m/s, whereas a free orbit at the same place requires 7,701 m/s, and material released from the surface enters an ellipse

with a semi-major axis of 3,495 km, which runs back into the parent (§A.1). The gap closes with distance, since the speed carried by the rotation grows outward while the orbital speed falls, and the two are equal at the corotation radius, $R_{sync} = \left(GM/\omega^2 \right)^{1/3}$ — the synchronous-orbit relation of standard mechanics, used to place geostationary satellites, applied here at a new place (§A.2, A.5). The pear structure bridges the Earth to this line: a hill with a base about 9,000 km wide and a 3,000-km contact at the Moon, bridging about 7,800 km from the Earth's surface to the Moon on the equatorial line (about 8,700 km for the release at 22.5°, Paper 3), and lying wholly inside the line (§A.3). Two mechanisms set the length of the day at release: the lengthening of the pear carries rotation outward, and alone would lengthen the day from 5.50 to 6.44 h and push the line to 17,569 km; on the other hand, the iron sinking into the core speeds the Earth up, like a skater drawing in her arms, and for a core twice as dense as the mantle ($D = 2$) this mechanism shortens the day by 0.67 h (§A.4). The budget is built next: a uniform parent holds $L_0 \simeq 3.466 \times 10^{34}$ kg m^2 s^{-1}, a value 0.7% above the sum of today's spin and orbit (1.5–2.5% with vector addition at the angles tested; an earlier version gave 1.5–2.6%), and the transfer process hands about one sixth of it to the material, 15.6% in the base calculation (§A.6). The surplus requires a parent essentially uniform at the start of the process, $k \gtrsim 0.397$, against about 0.33 for today's Earth with its core. For comparison, at the classical Jacobi bifurcation the parent would need a day of 2.73 hours and a budget 2.74 times today's angular momentum; the familiar "four times" (4.08) belongs to another point along the same sequence (§A.6.5). The Part closes with the Roche limits for a body of 2,600 kg/m^3 — 19,971 km for a fluid body and 10,314 km for a body that keeps its shape (§A.7; an earlier version gave 10,313 km, the same within rounding) — and with the tidal watershed: outside the corotation line the tide pushes a satellite outward, and inside it draws it inward, as Deimos and Phobos show today at Mars (§A.8).

Part B — How the problems are solved. One unified calculation meets all the problems of Part A with one consistent set of numbers (§B.1). From the 5.5-hour starting point, the lengthening of the pear and the gathering of the core together give a day of 5.77 hours at release; under these conditions the corotation line lies at 16,331 km for a test particle, and at 16,399 km, about 16,400 km, when the Moon's own mass is included — about 2.6 times today's Earth radius. The Moon begins to separate when its centre of mass reaches this line, and is torn free slightly after, about 210 km past the line; with a radius of 1,890 km its near side lies at 14,509 km, and the Earth's equator, flattened to 6,701 km at the slower spin, lies about 7,800 km below it. At this point the rotation gives the Moon exactly the orbital speed, about 4,960 m/s, so no gap remains and no push is needed (§B.2). At release the Moon's orbit holds 5.9×10^{33} (17%) — the orbital angular momentum of the pair about their common centre of mass, with the reduced mass; 6.0×10^{33} with the full lunar mass about the Earth's centre — and the Earth's spin holds 2.88×10^{34}; the system holds 1.007 times today's angular momentum, and the Earth's moment-of-inertia coefficient implied by this, $k = 0.355$, agrees with the value 0.360 of a two-layer Earth with $q = 0.55$ and $D = 2$ (§B.6). No giant impact and no later removal of angular momentum are needed. The Moon's centre at release lies well beyond the rigid Roche limit (9,484–12,994 km for 1,300–3,344 kg/m^3) and inside the fluid one; it is held by its rigid outer shell, which at a rigidity of 10 GPa or more holds its axis ratio to at most 1.04 — the body is round, its flattening and rounding stages having been completed during the attachment (§B.7; Paper 7). With separation, the centre of the body lies a little beyond

the line, where the tidal bulge raised on the faster-turning Earth runs ahead of it and draws it outward from the start (§B.8).

Part C — Summary. A single table pairs each problem of Part A with its solution in Part B: the speed comes from the rotation, and no gap remains on the line; the pear bridges to the line, and the Moon is released as a whole body; the day at release, 5.77 hours, comes from the two mechanisms together; the centre of mass is released at about 16,400 km; the angular-momentum budget closes at 1.007 times today's and agrees with the core structure; the body is rigid and lies well beyond the rigid Roche limit; and it recedes from the moment of release.

Part D — Claims, tests and novelty. The paper gathers its requirements and claims in one place. Three requirements head the list: a core centralising during the mass transfer ($D \approx 2$); a rigid body at release ($\mu \geq 10$ GPa); and a continuous pear structure from the Earth to a 3,000-km contact, with no neck remnant carried away by the Moon. Five central claims follow, each with its status and its test — corotation (an application of standard physics), closure of the budget (a consistency test), sensitivity to interior structure (a computed result), the non-equatorial correction (a geometric correction) and survival (an open requirement handed to the next papers). The adopted values are presented within their sensitivity windows: a 5.5-hour day within a range of 4.90–5.54 h for an expansion of 120 km; an expansion of 98–520 km, and a volumetric radius of at least 6,495 km; a parent with $k \geq 0.397$ at the start of the transfer, against about 0.33 for today's Earth; and a release beyond the line, since a body released inside it spirals inward. The novelty is stated plainly: the physical laws are classical and standard; the new component is the place where the speed condition is evaluated, and the joint computation of mass, shape and spin in one connected transfer.

Part E — Conclusions. A body of the Moon's mass can reach orbit from the Earth's own material without a collision from outside. The process is explained as a single chain: the Earth's rotation carries the material outward; the line where carried speed equals orbital speed sets where the Moon is released; the balance between the lengthening of the pear and the gathering of the core sets where that line lies; and the angular-momentum budget the Earth started with is enough, with a surplus of about 1% that the Sun will remove later. The earlier conclusions are kept with their reference values: 15,818 km is the position of the line before the transfer, and about 16,400 km its position after the pear and the core have acted; the material gains 15.6% of the budget in the base calculation and 16.5% in the coupled solution; about 17.7% is the final orbital share of the coupled state with the pear alone ($D = 1$), whereas at release with $D = 2$ the Moon's orbit holds about 17.0% (an earlier version gave the 17.7% as the share at the end of the process). The conclusions chapter outlines six future calculations for other groups: a stratified Maclaurin solution of the centralising core, the dissipation coefficient of the early Earth from the 5.77-hour day, an equation-of-state value for the thermal expansion, a tidal integration of the onset of recession, a visco-elastic survival calculation for the 10^1–10^3 years spent inside the fluid Roche limit, and a modern computation of the classical Jacobi sequence.

Appendix A — Derivation of corotation, definitions and constants. The appendix fixes the definitions on which every number rests. It keeps apart the total initial mass of the parent and the final mass of the Earth, and a radius measured from the Earth's centre from one measured from the common centre of mass of the pair; in the two-body case the gravitational parameter is

$G\,(\,M_{\oplus}+m\,)$. The value 15,818 km is derived from one stated definition (the final Earth mass, in the test-particle approximation); a different definition gives a different number, and not the same value under a new label.

Appendix B — The rotating geometry and the coupled angular-momentum budget. The appendix sets out the construction of the parent step by step: the mass, 6.0455×10^{24} kg; the size, a cold-density radius of about 6,397 km, expanded by 120 km to a volumetric radius of 6,517 km at a mean density of about 5,214 kg/m^3; the shape, from the Maclaurin equation, $e_0 \simeq 0.4107$, $a_0 \simeq 6,721$ km, $c_0 \simeq 6,128$ km; and the budget, $I_0 \simeq 1.092 \times 10^{38}$ kg m^2 and $L_0 \simeq 3.466 \times 10^{34}$. The expansion scan traces the limits of the admissible window: with no expansion, the budget falls 3.3% short of today's; at an expansion of 98 km it equals it; at the adopted expansion of 120 km it exceeds it by 0.7%; and at an expansion of 520 km by 15%, the adopted limit of what solar tides can remove at later stages.

Appendix C — The differentiation calculation and the sensitivity tests. The two-layer calculation behind the offset compares end states, not a function of time: an inner and an outer layer turning at a common angular speed, each layer boundary a surface of equal effective potential, and the moment of inertia derived from the density and shape profiles. For $q = 0.55$, the final day falls from 6.438 to 5.274 h, and the line moves inward from 17,569 to 15,382 km, as D rises from 1 to 3; $D = 2$ gives 5.77 h and 16,331 km, and at a value of $D = 2.5$ the speeding-up by the core almost exactly cancels the slowing by the pear. Freeing the constraints on the boundary shapes changes the radius by at most about 2.5 km; the results are reliable to the kilometre.

Appendix D — The vector correction to the budget and the hand-off to Paper 4. Off the equatorial plane, the Moon's orbital angular momentum and the Earth's spin angular momentum no longer point along a common axis, and they add as vectors: $L_0^2 = S^2 + H^2 + 2SH\cos\lambda$. In the sensitivity test for a latitude of 25°, the corrected release radius lies at about 17,177 km (test-particle approximation) or 17,240 km (two masses), and the Earth's axis is tilted by 3.85° or 3.84° from the total vector; the earlier value 4.48° was set aside, since it came from mixing vectors expressed in different frames. The adopted latitude, 22.5°, is recomputed in Paper 4.

Appendix E — Roche calculations and the survival assumptions. The appendix gives the background to the two Roche limits and supplies a first measure of the tidal load. At a distance of 15,818 km, the tidal stretch across a body of radius 1,890 km is about 0.381 m/s^2, against a surface self-gravity of 1.373 m/s^2 — a ratio of 0.277: self-gravity wins. When these figures are turned into a stress, 1.87×10^9 Pa, and into a time, the load would change by half the shape of a body of viscosity 10^{19} Pa s in about 85 years; this figure is kept as a parametric dependence and not as a disruption time, and the deformation of the attached and released body is defined as a stated survival requirement.

Appendix F — Prior work, novelty and record of tests. Component by component, the appendix separates standard physics (circular velocity and corotation, Maclaurin and Jacobi figures, the sign of tidal transfer, the Roche limits) from the paper's own application: the combination of mass, shape and budget in a connected-transfer scenario. The appendix names the bibliographic checks still

owed before submission, and records that the absence of a survival solution is marked as an open issue, and is not counted as a test passed.

Appendix G — The physical environment table binding on the calculations. The appendix gathers the full environment, in rows E01–E70, and records quantity by quantity what is adopted, computed, reproduced, open or a present constraint, in each of three states: before the division of mass, during the construction of the body, and close to release. Two later updates by the author add the balance of volatiles — an ancient inventory larger than today's, found dissolved, in bubbles or as vapour, according to pressure and temperature — and the cooling cycle, in which volatile-rich material rises, cools and sinks back, like a pot of soup on a stove. A binding timeline puts the stages in order without assuming dates: as the body peels, the passages that carry the Earth's pressure into it narrow; at the moment of detachment that pressure stops reaching the Moon, and the outward ejections stop with it; the rounding is complete before release.

C.3 Paper 3 — LEF Mechanism in Practice

> **THE CLAIM**
>
> The Moon was fed from within and rose from without — one body, built by two mechanisms at once. From within, the parent's own pressure, 150.26 GPa at its source, lifted a foamed melt to the corotation line with a pressure surplus of 10.49 GPa; from without, the Earth's outer shell rose with the body as the skin of its upper dome, and became its crust. Beneath the body the Earth was drawn into a pear structure, which is part of the Earth and held by its gravity, and the body on its tip was released, round, slightly after its centre of mass reached the corotation line, about 210 km beyond it; the sorting process then gathered the pear back into the Earth.

Front matter — the environment, the idea and the introduction. The paper analyses the construction of the Moon as one process acting on one body, in the environment defined in Paper 2: a parent of 6.0455×10^{24} kg and 6,517 km, a day of 5.5 hours at the start and 5.77 hours at release, and the corotation line at 16,399 km in the equatorial plane and at 17,287 km at a latitude of 22.5°. The time window of the scenario is about 600 Myr from the start of the bulge to release at 3.873 Ga — about 400 Myr for the bulge and the pear, about 100 Myr for the flattening stage, and about 100 Myr more to release — with an interval of 100 Myr kept as a sensitivity case. The picture is a balloon filled from a pipe beneath it, while its skin is the stretched surface of the ground on which it lay: the filling is the feed of deep, hot material; the skin is the Earth's sialic shell — sial being the light upper layer, rich in silicon and aluminium, lying over the denser sima, rich in magnesium. Neither of the two sources alone builds the Moon. The corotation mechanism links spin speed to orbital speed, but it neither lifts material nor fixes the moment of detachment; the paper therefore asks what pressure, what mass and energy balance, what flow structure and what connection the construction requires.

Part A — From within: the transport problem and the material balance. The first step is a mass balance in which every kilogram has one place: the parent's mass is divided at every moment

among the Earth, the growing body, the material in transit and what has escaped, and the mass of material that is ejected and lands back is subtracted and added again (§A.1). Rotation alone cannot do the lifting: for a body of the lunar mass touching the parent, 8,991 km from its centre, the Earth's pull is 4.93 m/s^2, whereas the outward push of the 5.5-hour spin is only 0.905 m/s^2; every kilogram still weighs about two fifths of its weight on today's Earth (§A.2). The mechanism examined is therefore continuous feed under pressure, and Part A defines the balance it must satisfy: along the path, the pressure must carry the weight of the column of material in the combined field of gravity and spin — its hydrostatic head, proportional to the density of the material — and the friction of the flow (§A.3).

Part A (continued) — A deep source, a developing body and different thermal boundaries. A wandering feed path would build many heaps and not one Moon; the source is a preferred region of the reorganising interior, Lobe A, north of the equator, which feeds the same place for a long time, as a pot heated off-centre keeps boiling at one spot (§A.4). The passage is kept open by the heat of the feed and of the hot bodies on both sides; the heat keeps the material fluid, but does not support the walls of the passage, whose stability must come from its shape (§A.5). The volatiles, held in an inventory larger than today's, are dissolved at depth and come out as the pressure falls, foaming the melt; their role is to lower the density of the rising material, and not to raise the source pressure above the weight of the material above it (§A.6). A solid skin forms over the hot body quickly — one metre in about 1.9 days when the surface cools hard, and 50 km in about 13.2 Myr (158 Myr with a warm surface) — so the outer envelope can stiffen while the interior and the feed stay mobile (§A.7). The order of the stages is fixed: the feed, the rise of the dome and the ejections overlap. As the peeling advances, the passages that carry the Earth's pressure through the pear into the body narrow: Serenitatis peels first and is cut off from that pressure, and afterwards its activity continues only in weakened form, through Imbrium, which stays connected. At the moment of detachment the Earth's pressure stops reaching the Moon, and with it the Moon's internal pressure and the ejections stop; the crater era therefore ends only after release (§A.8). After release, small and medium drops thrown out by the Earth are drawn to the nearby Moon and make small craters; the closing of the outer cap plays no part in this, and the activity of the far-side maria stops when cooling ends the internal pressure.

Part A (continued) — Are the pressure, the rate and the energy sufficient? The push is computed exactly. At the centre of the parent, the weight of the overlying material, slightly relieved by the spin, gives a source pressure of 150.26 GPa (§A.9). A column of melt from the centre of the parent to the corotation point of the whole parent, 15,882.9 km, costs 139.78 GPa for a deep melt of 3,500 kg/m^3 foamed to 1,500 kg/m^3 above the surface, and leaves a surplus of 10.49 GPa; at a density of 4,000 kg/m^3 the calculation fails by 3.92 GPa, and the threshold is a deep melt with a density below about 3,864 kg/m^3 — a condition that a volatile-rich melt meets and a dry melt does not (§A.10). Delivering the whole lunar mass needs 2.33×10^7 kg/s over 100 Myr, or 3.88×10^6 kg/s over 600 Myr, and the flow through a passage grows as the fourth power of its radius (§A.11). Lifting the Moon from the Earth's surface to 16,400 km costs about 1.47×10^{30} J, less than one per cent of the parent's binding energy; this cost is paid by the same evolution that drives the upwelling, the descent of the iron (§A.12). The joint test of pressure, flow and load in one state — 22.5°, 5.77 h, the body's far side on the line — gives the sharp negative result of the paper: a thin, massless pipe

cannot build the Moon. The balance of the radial forces is reached only from a passage radius of about 1,017 km, where the passage itself already holds 0.46 lunar masses, and a sideways load of order 10^{22} N remains, which a fluid at rest cannot carry. The connection must therefore be broad, hot and massive (§A.13).

Part B — From without: the body on the tip of the pear. Part B follows the Earth's outer skin through four stages, and maps each piece of it onto today's Moon. At the flat stage, the sial lies over an active feed and already carries craters from material that was ejected through its openings, returned and landed; the Pacific basin, whose area is 1.4 to 2.4 times the surface of the later upper dome, marks the region from which the shell was gathered (§B.1). The base of the pear stood near Kiritimati (1.9°N, 157.5°W); the region at 25°N, 165°E, identified earlier as the source, is its back, which leaned west as the rotation made the outer part lag, and north, following a feed that rose obliquely from a source north of the equator. At the intermediate stage the body sits on the wide tip of the pear, as a squashed ball on a rounded support: a contact 6,000 km across, a lower dome hugging the tip (3.4×10^{13} m^2, 33%) and an upper dome standing about 3,650 km above the rim (7.0×10^{13} m^2, 67%) — the height at which the full lunar mass has the foamed density of 1,300 kg/m ^{3} that Part A requires: the body then encloses 5.65×10^{19} m^3, twice the 2.83×10^{19} m^3 of the sphere at release, and its upper dome is about three times the far hemisphere of that sphere (§B.2). The shell does travel: it rises with the body of the pear, separates from it gradually and slides up to the tip, into the Moon — which is why the tip separates as a body of its own; the sial slides upward, in the direction of outward sorting, a process eased under shear by the lubricated layer between the sial and the sima. The exposed ring of the underside of the body contracts about 1,500 km inward over millions of years, on a weak interface beneath a stiff shell with a viscosity of 10^{23} Pa s or more — the viscosity that keeps a 100-km crater from relaxing over 10^7 years (§B.3). The translation rule follows: in the outward rounding each part keeps its relative share of the surface, so the 3,000-km contact (7%) becomes a near-side cap within about 31°, the peeled annulus (26%) becomes a belt from 31° to 70°, and the upper dome is translated to the far side (§B.4). The hill and the contact can be drawn on today's near side: the hill holds 99% of the area of the dark basalt of the near side, and the contact 66% (shares of area, Paper 5), and Mare Imbrium, the opening through which the feed entered the body, lies inside the contact. The openings are of two kinds — vents on the crown and on the flanks that threw material outward, and covered conduits beneath the lower dome that only carried it (§B.5). The craters made on the exposed shell travelled with it and kept their size; most of them, and the largest, formed in the flattening stage, and in the inward rounding the poles gathered them (§B.6). The far-side crust is made of many layers of compacted sial, which migrated up to the tip under the forces of the spin, and of material that returned again and again, and its iron content rises with depth; for a thickness of 55 km and a density of 2,550 kg/m^3 over the far hemisphere, its mass is about 2.6×10^{21} kg, 3.5% of the lunar mass, and together with the near-side crust, about 1.5×10^{21} kg, the mass of the crust reaches 4.1×10^{21} kg, about 5.6% — within the range of 4–6% that GRAIL gives for the whole crust; the remaining 94–96% is the deep material of Part A (§B.7). The far side is thicker, by about 8–10 km, because the two sides were built differently; the near side is thinnest beneath the maria, because they are openings.

Part C — The connection and the release. The deep source pressure does not act directly on the body's cap: it is spent on pushing the column, and the body's own weight gives only about 3.37 GPa

at its centre (§C.1). At the corotation line, at a latitude of 22.5° and a distance of 17,287 km, the radial component of the requirement on the connection vanishes, but a sideways load of 0.559 m/s^2 remains, 4.06×10^{22} N — a shear stress of 5.74 GPa over a disc 1,500 km in radius — which a narrow, hot neck cannot carry (§C.2). The corotation line is where the radial pull turns over: beyond it the rotation pulls the body outward (§C.3). Along the ring between the 6,000-km rim and the 3,000-km disc, the body peels away gradually together with the sial it sat on, as an inflating ball peels from a rounded stand; over the central disc the heat melted the sial and smoothed the surface (§C.4, C.4a). The hot layer over the disc resists the separation, but only moderately: the body begins to peel when its centre of mass reaches the corotation line, and detaches slightly after, about 210 km past the line — pulled outward with force and not let go gently; this is the pull from above that drew up the spatula ridges (§C.5). The base of the pear stood roughly on the equator of that time; whether that equator is today's belongs to the axes, future work, and the release latitude is the 22.5° of Paper 4. The release state handed to Paper 4 is a corotation line at 16,399 km in the equatorial plane or at 17,287 km at a latitude of 22.5°, and release about 210 km beyond it, at a 5.77-hour day (§C.6); an earlier release at 20,595 km rested on the lengthening of the pear alone, and with the centralising core the line lies at about 16,400 km. The release distance is set by the search of §C.5a, in which two reduced tests — the present inclination and the present eccentricity for a hot young Moon — meet near 210 km beyond the line, when a little more than half the Moon has crossed it. Beneath the contact lies one bulge drawn into a pear structure — wide at its base, narrowing in a single arc without a waist to the contact, wholly inside the corotation line. The pear is part of the Earth and is held by its gravity like any other part of it; it is not thrown out, since its speed is not a throwing speed and it has not crossed the corotation line. At release it is about 8,700 km long and its base 9,000–11,000 km wide; from the width of the base the paper derives 8.8–12.1 lunar masses, lifted at a cost of 8.1×10^{30} to 1.05×10^{31} J, the order of the energy released by the formation of the Earth's core (§C.7, C.9). Along the climb the body on the tip flattens to an axis ratio of 1.69 near 13,000 km, and rounds inward to a ratio of 1.20 in the late attached stages; at release it is round. The pear lifts the Moon radially, and because the Moon is attached, it gives it the sideways speed of an orbit, 4.96 km/s in the equatorial plane; a body thrown from low on the Earth, even at 7 km/s, falls back, since a push adds height but not sideways speed (§C.8). The pear structure sorted what it carried: the sial drained to the tip and became the crust, the Earth's mantle became the Moon's mantle, and the iron sank back down the pear, which is why the Moon's core is one to two per cent of its mass (§C.10). After release the Earth went on throwing material at the Moon — all of it very hot: first the large pieces, and, as the magma storm calmed, small pieces, still hot — and the near-side highlands, with all the maria excluded on both sides of the comparison, carry 1.19 times as many fresh craters of 1–2 km as the far side: the Earth was the last scatterer of small craters on the Moon (§C.11).

Part D — Predictions, tests and conclusions. The requirements are collected in thirteen rows that one Earth, one feed and one body must meet at once, and they are not counted as independent votes: feed under pressure, construction rate, energy balance, a stable attached body, an outer envelope stiffening over a mobile base, a connection that carries the load and then fails, and the order of stages; to these are added the share rule (zones at about 31° and 70°), the carried record (the oldest craters densest on the far side, less in the belt, and least of all in the cap, with no difference of scale), the layered crust of the far side (iron content rising with depth, tested in central

peaks), the release on the line, the northward lean of the pear, and its gathering back whole into the Earth, with no residue in orbit (§D.1). The paper closes with one negative result — a massless cylindrical passage, transmitting pressure alone, cannot build the Moon — and with one geometric result that answers it: the broad, hot, massive connection that the negative result demands is the wide tip of the pear (§D.3–D.4).

Part E — What the pear left in the Earth. The pear did not follow the Moon. The release did not bring it down: the sorting process, acting on the pear whether the Moon was there or not, gathered it back into the Earth over tens to hundreds of millions of years. Its heavy, iron-bearing part sank to the base of the mantle beneath the base of the pear; its leading wall was folded into the stripes of the North Pacific; and its back was gathered into the arcs of the western Pacific (§E.1–E.2). The model proposes a distinction between the two large low-velocity provinces at the base of the mantle: beneath the Pacific — largely the returned pear, part of the sorted material that went back into the Earth, on a drained lobe; beneath Africa — a lobe never disturbed, in which the reading of an impact model sees remnants of the colliding body, and which requires no relation to the Moon's source (§E.3). The position test has been made and passes: the base near Kiritimati lies inside the Pacific province in every model tested — five of five slow votes, inside the -1% contour of SMEAN (-2.25%) and SMEAN2 (-2.36%) at a depth of 2,800 km, 1,229–1,584 km from the edge — about 1,500 km north of the centre of the province. Composition, shape and isotopes are the further tests (§E.4). While the pear stood, its gravity drew the Earth's core a few hundred kilometres toward it — 270–370 km for bases of 9,000–11,000 km — and as the pear was gathered back, the core returned; the east–west asymmetry of the inner core is the first record to compare (§E.5). The further work lists nine calculations, from the motion of the body on the contact layer to a three-dimensional simulation of the connected body and the equation of state of the deep melt.

Appendix A — Sources, assumptions and novelty. The appendix sorts the components of the paper by their standing: standard physics (gravity, pressure and momentum balance; Poiseuille flow in a passage; Stefan's law for the separation of two surfaces joined by a viscous layer), prior observations used as constraints (water in lunar melt inclusions, the GRAIL crust, the lunar core), calculations made here (the joint test of pressure, flow, load and the mass of the passage; the body on the tip and the share rule), and elements of the model proposed for testing (the connected feed and Lobe A). No priority is claimed beyond what a dedicated review would support.

Appendix B — Constants and limits of reproduction. The appendix gathers the constants and the input data that are test values and not measurements — viscosity, segment densities, passage width, feed duration, and the thermal inputs of the cooling estimate. The day of 5.77 hours and the release distance are taken from Paper 2; everything the paper computes from them can be checked from its own equations. A value such as 10.49 GPa is printed to four digits to let the reader repeat the calculation exactly, and not because the margin is known to that precision.

C.4 Paper 4 — The Orbit

> **THE CLAIM**

> The inclination of the Moon's orbit is the latitude it left from: a body released while moving eastward in rotational synchrony with the Earth at latitude λ enters an orbit inclined by exactly λ to the Earth's equator, with no parameter adjusted. Released at a latitude of 22.5°, about 210 km beyond the latitude-corrected corotation line (17,287 km), at a distance of 17,496 km, the Moon enters a nearly circular orbit and recedes from the start; the reduced tidal law carries its initial inclination, 22.5°, to 5.145° at the present distance, the value observed.

Front matter — the environment, the idea and what the paper proposes. The paper analyses the orbit from the moment of release onward, in the environment set in Papers 2 and 3: masses of 5.972×10^{24} and 7.35×10^{22} kg after the division, a gravitational parameter of the pair $K = 4.0347667 \times 10^{14}$ m^3 s^{-2}, a day of 5.77 hours at release, a released body at a density of 2,600 kg/m^3 and with a radius of 1,890 km, and a broad viscous contact 3,000 km across. The release latitude is the latitude of the *centre* of that contact, 22.5°: a body has one velocity, and the width of its base does not spread the inclination. The paper keeps apart three places: the base of the pear near Kiritimati, its back at 25°N, 165°E, and the release latitude of the Moon's centre. The opening from which the body rose is identified in the range between about 10° and 25°N. The base of the pear stood roughly on the equator of that time; whether that equator is today's is left to the axes, future work. 25° is kept as the demanding sensitivity case. The destination is today's orbit: a distance of 384,400 km, an eccentricity of 0.0549, an inclination of 5.145° to the ecliptic, and a recession rate of about 3.8 cm a year.

Part A — Which inclination results at the moment of release? The inclination is defined as the angle between the plane of the orbit and a reference plane, represented here by the Earth's equatorial plane at release. A body carried round on a fixed circle of latitude moves parallel to the plane of the equator; at the moment of separation it must enter an orbit lying in the one plane that contains both its velocity vector and the Earth's centre, and that plane is tilted by the angle of the latitude — like a stone let go from a raised ring of a merry-go-round, leaving in a plane that passes through the centre. In three simple steps of algebra, $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ gives $\cos i_0 = \cos \lambda$, $i_0 = |\lambda|$; mass, distance and spin rate cancel and do not affect the angle (§A.2). This distinction sets the mechanism apart from the disc model, a disc that settles over the equatorial plane and starts with almost no tilt, so that the tilt must be added to it at a later stage; here the tilt follows from the place on the turning Earth where the body was attached (§A.1). A radial velocity component changes the shape of the ellipse but does not affect the plane of the orbit; a northward or southward component would only make the plane steeper. At release the Earth's spin angular momentum holds 2.876×10^{34} and the Moon's orbital angular momentum 5.90×10^{33} kg m^2 s^{-1}; the angle of 22.5° between them divides into 3.78° between the Earth's spin axis and the total vector, and 18.72° between the plane of the orbit and the total vector. These values do not represent the inclination of the orbit to the ecliptic or the present obliquity of the Earth's axis, 23.44°; the joint evolution of the axes of the Earth and the Moon after release is future work, not solved within the series (§A.3).

Part B — Does the body enter an orbit that survives and recedes? A body at latitude λ circles the spin axis at a radius of $r \cos \lambda$, not at r, and so moves more slowly than an equatorial body at the same distance. On the equatorial line, 16,399 km at the 5.77-hour day, it is too slow: its orbit has

eccentricity $\sin^2\lambda = 0.1464$, the point of detachment is the apogee, the perigee falls to 12,209 km, and the tide the body raises on the Earth draws it inward. When the body is carried a little farther out, to $r_c \cos^{-2/3}\lambda = 17,287$ km ($x = 1.0542$), the corotation radius corrected for its latitude, its speed equals exactly the circular speed (§B.1). The Moon begins to separate when its centre of mass reaches this line, and the release takes place about 210 km beyond it, at $r_0 = 17,496$ km ($x = 1.0669$). At this distance its speed is slightly higher than the circular speed, and the release point is the perigee of a nearly circular orbit ($e_0 \simeq 0.037$, apogee 18,830 km); the tide pushes the Moon outward from the first moment and goes on pushing as it recedes (§B.2). The centre of the body lies about 7,200 km outside the rigid Roche limit of 10,314 km; the rigid limit applies because the Moon is released as a round body, with a rigid outer shell and an axis ratio of at most 1.04 (Paper 2, §B.7). The starting orbit is therefore $a_0 \approx 18,163$ km, $e_0 \simeq 0.037$, with the perigee at the release point, and $i_0 = 22.5^\circ$ to the equatorial plane of that time (§B.3). An earlier version placed the release farther out, at 20,595 km; with the core centralising during the mass transfer, the day at release is 5.77 h, and the release just beyond the latitude-corrected line replaces it.

Part C — How does the orbital plane change on the way out? The initial value of the inclination, 22.5°, is defined relative to the Earth's equatorial plane, whereas today's value, 5.145°, is measured relative to the ecliptic, the plane of the Earth's orbit round the Sun. Close to the Earth the equatorial bulge makes the plane of the orbit precess about the equator; at large distances the Sun makes it precess about the ecliptic; the Laplace surface bends from one to the other (§C.1). An order-of-magnitude scale for this transition, with the bulge scaled to the day of 5.77 hours, is about 17.07 Earth radii; the release point lies at about 16.1% of it, deep inside the region where the equator governs the plane of the orbit. Whether the orbit can follow the turning plane smoothly depends on the tilt of the Earth's axis, the line of nodes, the eccentricity and the recession rate, and requires an integration of the axes system (§C.2). The Moon's fossil figure and its Cassini state are further constraints, and not stages fixed by the release latitude; a published thermal-orbital model finds that the Moon's figure froze before it reached about 16 Earth radii (§C.3).

Part D — How much of the inclination survives, and in what time? The simplest law of tidal damping, $di/da = -\frac{1+\epsilon}{2a}\sin i\cos i$, integrates to an exact solution, $\tan i \propto a^{-(1+\epsilon)/2}$: a law in $\tan i$, and not in $\sin i$, a distinction that matters at a starting angle of 22.5°. From $a_0 \approx 18,163$ km to 384,400 km the distance grows about 21 times, and $\tan i$ falls about 4.6 times: 22.5° becomes 5.145°, the value observed today, so the latitude that returns the observed angle is 22.5° itself. A release 8% beyond the line would give 5.282°, and a starting point at 25° would give 5.788°. Hence the final angle is governed by the release latitude, and not by the starting distance (§D.1). Since the same rate sets both the recession and the damping, the Earth's tidal quality factor Q cancels from this law — but not from the thermal history, which sets how strongly the Moon itself dissipates at each distance (§D.2). The clock test supplies the dimension of time: with $Q = 12$ constant, today's recession rate comes out at about 3.70 cm a year, but the Moon reaches its present distance in only about 1.60 billion years; Q of 33.8 fills 4.5 billion years, but gives a present rate about three times too small. The time available after release at 3.873 Ga is about 3.9 billion years, not 4.5; against 3.87 billion years the clock fits for $Q \approx 29$, where today's recession rate falls to about 1.5 cm a year, and the conclusion is the same. No constant value of Q fits both constraints; the Earth's dissipation must have varied with time, as models of ocean resonances find (§D.3).

Part E — What has been proved and what will be tested? The Part sets out seven claims by status, from what is fixed to what is still to be computed: the source of the inclination — a geometric identity that holds for a release without a north–south velocity component or a torque at separation; the initial ellipse — two-body mechanics; the initial recession and survival — computed, with the deformation check left open; and the present inclination, the distance with its clock and the geographic link of the release latitude to a site on the Earth — each of them requiring an integration or an independent reconstruction. The identity $i_0 = |\lambda|$ stands independently of the geographic link, which bears on the source site alone. The contribution is stated explicitly: the identity is standard mechanics; what the model adds is that the starting orbit is fixed by the construction of Papers 2 and 3, and not chosen to fit. The further work begins with a coupled thermal-orbital integration from the release state, through the Laplace-plane transition and the Cassini state, with a time-dependent dissipation; and continues with the reconstruction of the Earth's dissipation history from the 5.77-hour day, an analysis of the obliquity channel opened by the tilt of 3.8° at release, the solidification of a magma ocean under a one-sided lid, and the fossil figure.

Appendix A — Formulas for reproducing the tidal test. In the constant-time-lag framework, the tide the Moon raises on the Earth pushes it outward as long as $\omega \cos i/n$ exceeds a factor that depends only on the eccentricity and equals 1 for a circular orbit: the Earth must turn, in the plane of the orbit, faster than the Moon goes round. The test gives $0.7526 < 1.1962$ on the equatorial line (contraction), equality on the corotation line, $1.077 > 1.013$ at the adopted release point, about 210 km beyond it (expansion), and $1.1660 > 1.0532$ at $x = 1.08$ (expansion); the sign was also checked by direct integration of the tidal forces along one orbit. A short code reproduces the ellipses, the tidal sign and the angles 5.145° and 5.282°.

Appendix B — Scope of verification and availability. The appendix states what was checked and what was not. An independent Newtonian integration of the reference orbit returned to its starting point within about 1.6×10^{-10} of its distance, confirming the orbital calculation in the monopole approximation; it does not test the mechanics of the connection or the rheology. The full coupled thermal-orbital integration is not included in the paper, and is not reported as if performed.

C.5 Paper 5 — The Maria

> **THE CLAIM**
>
> A mare is an opening in the Moon's shell lying over a *feed dome*: a body pushed up from a focus beneath the surface, through which the material rose, was thrown out and came back as the returning material that made the craters. The dark basalt plain is only the last layer that solidified in the opening. When this definition is applied, the positions of the maria do not coincide with the spread of the basalt: 33 of the 34 maria sit on a dome in the gravity field, and 44 of its 73 domes lie under bright highland, 36 of them on the far side.

Paper 5 asks what a mare is, if it is not a basin dug by an outside body and flooded by lava at a much later stage. It answers with a definition, and then tests it in five independent data sets: the

gravity of the domes, the height of the floors, the craters on the floors and around them, the iron and titanium of the ground, and the thorium. The paper consists of fifteen Parts and one appendix.

Part A — Terms and definitions. The literature uses one word, *mare*, for a dark surface, the hole beneath it and the story of how it was filled. Part A gives each its own name. The *feed dome* is the dense body lying beneath; it is not an intrusion that came to rest, but the shape the flow built as it passed upward, as a strong spring raises a mound over its outlet while it runs. The *opening* is the site where the feed reached the surface; it is not a pipe with walls, but a region of runny melt inside stiffer material, through which pressure passes. An *exit* threw material outward; a *mouth* carried material while still covered and was exposed only later, so that a single structure can begin as a mouth and end as an exit. The *mare* is the opening together with its last surface, and the *basin* is the margin built of the opening's own returning material. A further rule separates the place from which a crater's material came from its landing site: the craters seen on a given surface are the background, plus the craters made by a nearby opening, minus those the opening melted or covered while it was active. A crater count therefore reflects a sum of three processes, and is not a simple clock.

Part B — What the series has fixed. The analysis rests on two inputs fixed in the earlier papers and not derived again. From Paper 3 comes the body on the tip of the pear: the tip 6,000 km across at the start of the rounding and 3,000 km at release; the *hill*, the 6,000-km region that rested on the tip, centred today at 16.6°N 13.1°E and holding 99% of the area of the near-side dark basalt; the *contact*, the last 3,000 km that still touched the Earth at release, centred at 24.1°N 335.2°E and holding 66% of it, with Imbrium inside it (14.8° from its centre); the zones of exposure, a cap of 31° and a belt to 70°, carried onto today's Moon by the rule of proportional share; and the light, foamed body of 1,300 kg/m³. From Paper 6 comes the origin of the craters: some 99% of them are attributed to returned terrestrial material. The thin crust of the near side is a result *of* the openings and of the heat that flowed beneath them, which removed its light upper layer, and not the other way round.

Part C — What a mare is, how many maria there are, and the mare as the source. The literature counts surfaces of dark basalt: 646 mapped units under 97 names, 189 of them of 1,000 km² and more, and 20 basins of 500 km and more. The model counts openings lying over feed domes. The author identified 34 maria on the gradients of the GRAIL gravity field (14 on the near side and 20 on the far side); in parallel and independently, an automatic search took as a feed dome every round high of 100 km and more standing 50 mGal and more above its surrounding ring, and found 73 domes. The two catalogues were built independently, and 33 of the 34 maria lie on a dome (Study 1). In 20,000 random sets of 34 points not one set reached 33; by chance, about 8 of 34 points fall on a dome. The one exception, Ingenii, lies over South Pole–Aitken, whose heat cooked its dome until it vanished. The dark plains with no dome beneath them — Procellarum, Tranquillitatis, Nubium, Frigoris and others — are not maria but *plains*: crust melted in place by the heat of the Earth, with no channel to depth. The second half of the Part reads the ground. Around six maria and three plains, 11,574 craters of 5–60 km were chosen, and the sample was frozen before any chemistry data were read, together with 5,922 control craters more than 600 km from any basalt (Study 2). Around the maria the ground carries 5.96 wt% FeO within 50 km and still 3.40

at 300–600 km, against 2.30 in the control areas; around the plains the value falls to 2.60, close to the controls. A mare scattered its material over great distances; a plain only spilled over its edge. Each crater is a little poorer than the ground around it (-0.26 wt% within 50 km, -0.01 in the control areas): it threw aside the thin, iron-rich *veneer* the mare had laid, and exposed the poorer base beneath, as Dionysius on the margin of Tranquillitatis shows.

Part D — The openings, visible and buried. If the opening, and not the basalt, defines the mare, openings must exist also in regions where no basalt is seen. The gravity field allows them to be identified through their cover. An opening is told from a crater by the crust beneath it: a dome pushed from below thins the crust over its centre, whereas a crater made from above does not. Of the 36 marked openings, 26 show this *crust jump* of 8 km and more (Moscoviense $+29$ km, Orientale $+28$), against only 4 of 169 ordinary craters (Study 3); the six openings that do not show it, Tsiolkovskiy among them, are craters that landed on a lava area. Sorted by cover, 36 of the 40 far-side domes are buried, and 22 of the 33 near-side domes lie under a mare; counted by domes, the far side has more openings than the near side, 40 against 33. Over a buried dome the iron and thorium concentrations equal those of the surrounding land (-0.19 wt% FeO, $+0.05$ ppm), the crust is thicker by 8–13 km, and no crowd of craters gathers on it (Study 4): it is an opening with a blanket laid over it, buried together with the craters of its time. Two of the near-side buried domes, Balmer–Kapteyn and Schiller–Schickard, coincide with the cryptomaria known in the literature. Procellarum is the contact disc itself, and holds several domes. The generations of openings, and the classification of each as an exit or a mouth, are recorded twice, by appearance and by position, and both readings are kept.

Part E — The feed dome. When the feed stopped, the material left in the dome froze into a body richer in iron and titanium, and denser than the crust around it; its strength is defined as the excess of gravity at its centre over its surrounding ring. Among the open maria Orientale reaches 740 mGal, Moscoviense 689 and Nectaris 538; the buried far-side domes are equally strong (Freundlich–Sharonov 519), so a buried mare is not a failed opening. The floor above a dome should be lowest over its centre, and the more so the stronger the dome. The candidate mechanisms are the weight of the dome, suction from beneath, and thermal expansion and contraction with heating and cooling — in the model, the most significant factor — whose effect grows the larger the mass in diameter and depth; which of them acts is a test still to be made. On the open maria the centre lies low relative to the margin in 10 of 12 (median -267 m), and the rank correlation coefficient between dome strength and the height of the centre is $\rho = -0.59$ (Study 5). The two exceptions are Serenitatis and Imbrium, the large maria of the contact, whose centres stand slightly above the margins ($+60$ and $+43$ m). Throughout the attachment they were hot, and carried magma from the pear into the Moon. Serenitatis froze first, very close to the Earth, and the Earth’s gravity field drew it outward more strongly; Imbrium froze later, after detachment, as the Moon receded, and the pull acted on it less — hence $+60$ against $+43$ m, with Imbrium the lower of the two. The literature attributes a similar sag to the load of the mascon; the model accepts the shape, but defines the dense body differently — the frozen feed dome, found also beneath the 44 openings that have no basalt.

Part F — The floor, the floors and the last scatterer. A liquid at rest cannot hold a slope, so a floor that froze as one body of melt shows one level. All 16 large floors show one level, and so do 188 of 189 smaller units; all 16 lie below their rim, by about 3 km (median 3.4 km), and below the surrounding land (median 1.6 km), a result that by chance would occur once in 65,000 (Study 6). Wrinkle ridges lie at 0.71–0.77 of the radius and graben at 0.76–1.22: the margin was stretched and the interior squeezed as the centre sank. In the contact, Imbrium, Copernicus H and Sinus Aestuum lie at the level of the neighbouring plains (-0.25, $+0.13$ and $+0.03$ km), with no step at the edge; Serenitatis lies 0.84 km lower, and its margins are the layers of craters that came out of it. Beyond the contact, 9 of the 13 floors with a measurable plain lie below it (median -0.52 km): the six large basins — Nectaris, Crisium, Humboldtianum, Smythii, Orientale and Moscoviense, the last from only 21 pixels of plain — lie one to two kilometres lower (1.0–2.2 km), Humorum and Apollo about half a kilometre, Grimaldi (-0.13 km), Poincaré, Crüger–Sirsalis and Schiller–Zucchius within 0.2 km, whereas Ingenii, lying over South Pole–Aitken, stands 2.1 km above; the depth below the plain belongs to the large basins, and not to the small maria (an earlier version stated that every mare beyond the contact lies one to two kilometres below its plain). Around the far-side maria the surface rises in a *collar* built of the maria's own returning material, $+742$ m around the buried maria against $+122$ m in the control area (Study 16). Every floor is far cleaner than the surrounding land: for craters of 2–5 km the density on the floor is about a twelfth of the density of its land (Study 7). The few craters on the floors order them by the time they froze, and the contact solidified last: the floors beyond the contact carry a density of craters of 10 km and more 83 times that of the contact floors, and the hill floors 33 times, with the catalogue entries of the basins themselves left out (Study 12; an earlier version, which included them, gave 58 and 24). The source of these late craters is the *last scatterer*, the Earth, which went on throwing material in all directions after release.

Part G — Size and state: the maria and the craters in time. This Part is shared with Paper 6 (its Part E), and its main study is numbered jointly: Study 8 here, Study 5 there. Within Imbrium three craters record three states — Sinus Iridum half melted, Plato smeared and Kirch sharp. Every crater of 3 km and more on the near-side basalt (3,478 in all) was measured against highland craters of the same size. Within the contact the floors carry about a third of the small-crater density of the neighbouring plains (263 against 836 per million km² at 2–5 km): the rest of the craters *vanished* into the active maria. On the floors the share of sharp craters falls from 71% at 3–5 km to none at 20–40 km, and the share of melted craters rises from 17% to 85%. This trend comes from two clocks running together: the rate at which the maria solidified (a frozen lid thickens as the square root of time, so a mare ten times deeper stays liquid a hundred times longer), and the material that arrived from the Earth, which grew smaller in size, though it stayed hot, since reaching the Moon took 8.8 km/s at a distance of 17,000 km and 10.6 km/s at 60,000 km. Therefore a large crater is an early crater made on a soft floor. On the plains of the contact, heated from below by the Earth, the rims of the large craters were lost on the side facing the plain ($+24\%$ of directions) and the surviving crests were blunted, whereas on the plains beyond the contact they were only flooded (Study 15).

Part H — The engine. An opening stays active as long as its melt is fed and can flow. The force is carried by a liquid: a solid would have to bear 13–16 GPa against a strength of about 0.2 GPa, whereas a liquid passes the pressure without resistance, as brake fluid passes the push of the pedal. At that time enormous amounts of volatiles were in and around the material: they lowered the

density of the melt, expanded it and drove the ejection of the material that built the margins. They have since escaped to space, and the dry interior measured today is the Moon after they left. The flux through an opening divides between what stays in the body and what is thrown out, and only the thrown part made craters. An opening seals when its flux falls below the *critical flux*, at which the heat it brings no longer balances its cooling, as a tap left running keeps a pipe from freezing; computing the value of the critical flux is a stated requirement of the model. During the attachment, the near side faced an Earth of about 2,000 K that filled about a fifth of its sky (view factor 0.205), and held its surface temperature near 1,345 K, above the solidus: throughout the attachment the near-side openings stayed hot and open and carried magma from the pear into the Moon, whereas the far side, which faced space, froze. At release the near-side maria were still hot and of low viscosity, which is why their floors are smooth like a frozen lake; they solidified only afterwards, slowly and from the outside inward, over hundreds of millions of years and up to one or two billion.

Part I — The geometry of exposure, and the generations. As the tip narrowed from 6,000 to 3,000 km, the edge of the contact moved inward across the shell like a front, and the distance of an opening from the contact centre tells when it was exposed — as a deflated ball inflated on the floor lifts its faces off one by one, the centre last. A covered opening worked as a mouth, and became an exit when the front passed it, provided it was still active; this condition is boxed in the paper as $t_{shutdown} \geq t_{exp}$. Three generations follow from this: generation I on the far side, open from the start of the process; generation II in the belt, mouths first and exits later; generation III in the contact, the last mouths, exposed toward release. The maria threw out most of their material during the flattening; the far side sealed in the inward rounding stage; the near side was still hot at release and froze only after it, which is why Imbrium carries 11 craters of 20 km and more per million km², against 205 on the highlands. Several openings on the limbs of the disc and on the far side stayed active a little longer. The size of an opening keeps the memory of its feeding: the six largest feed domes, 450–909 km across, all lie on the near side, and the median diameter is 375 km there against 227 km on the far side, where the largest is Orientale, 435 km (an earlier version stated that the openings of 200–1,200 km lie almost all on the near side).

Part J — Chemistry. Iron and titanium concentrations fall in rings around every mare: pooled over six maria, from 6.9 wt% FeO at 0–25 km to 3.1 at 100–150 km, holding at 3.3–3.4 out to 600 km, against 2.3 in the control areas; around Procellarum they fall to the control level (Study 9). The falling trend records the sealing as well: light, volatile-rich material flew far in the early stages, and heavy, iron-rich material landed near the opening in the late stages. Thorium, which stays in the last liquid of a cooling melt, is concentrated in the contact: the contact disc, 17.5% of the surface, holds 95% of the thorium above 3 ppm. A cap of the same size moved over 16,200 centres places the contact centre, fixed before the thorium data were read, in third place, only 47 km from the leading position (Study 10). Within the contact the plains and the highland carry nearly identical thorium concentrations (5.62 and 5.48 ppm), and the maria the lowest (3.99): the last thorium-bearing liquid sank back down their channels. Every mare of the contact is poorer than its surrounding ring (Study 13). Reading the maria as basins of external impacting bodies requires about 1×10^{19} kg of foreign mass, yet the Imbrium melts match no chondrite group, and the late basins added only little (Study 11).

Part K — Dating. An opening is characterised by five clocks: when it formed, when it was exposed, when it began to throw, when it sealed, and the age of its surface today — as a well is dug, uncovered, flows, is capped, and leaves a last puddle at its mouth. The radiometric ages (Apollo 3.1–3.9 billion years, Chang'e-5 about 2.0 billion years, Chang'e-6 about 2.8 billion years) record the point at which the sample cooled enough to hold its daughter atoms, and not when the opening formed. Crater counts are used only to order surfaces relative to one another, and are never converted to years, because the constant rate of arrival that the conversion assumes is exactly what Paper 6 replaces. The order of sealing derived from the floors is: the buried far-side openings first, the open maria of the belt next, and the maria of the contact last.

Part L — Statistics. Every comparison of the ground is like for like: the maria are removed from both sides, and the comparison says so explicitly. Areas are computed on the sphere, craters are assigned by their centres, and every dome is checked on two separate filterings of the gravity field. One unified table gathers every measured result together with its sample and its uncertainty, among them the sign test of 16 of 16 ($p = 1.5 \times 10^{-5}$), the collar of $+471$ m against $+122$ m (Mann–Whitney $p = 0.009$) and the sharp craters of the contact floors, 82% against 56% (Fisher $p \approx 5 \times 10^{-6}$).

Part M — Predictions and tests. Four working assumptions that guided the research were changed by the measurements: a peak of craters on every rim (found only as a property of the volatile-rich maria), a moat along the walls (the lowest point lies at 0.57 of the radius), a lava flow building the far margins (basalt covers only 0–7% of the ring), and mare area as a measure of the volume of ejection. Of the fifteen studies, thirteen achieved full validation (green) and two are borderline cases (amber): in Study 13 the thorium part held, whereas the titanium part did not separate the zones; in Study 16 the collar does not separate from the control when the open far maria are counted alone. The studies are numbered from 1 to 16; Study 14 was cancelled and the other numbers kept. Then twelve predictions are presented, each with its test — among them the presence of an opening above every feed dome at some stage, a dense body beneath every floor that sank, ground near a mare richer in its signature than ground far away, and basin melts that carry in their isotopic signatures no meteoritic mass of a body of the size the literature assumes.

Part N — Discussion. The five data sets come together into one sequence. The openings were exposed in the order set by the narrowing of the contact; each opening scattered material over the land around it as long as it stayed open; when its feed stopped, its floor froze as one level and sank over the dome; the far openings were buried under the layers they themselves built, while the near ones stayed hot and open as long as the body was attached, and froze slowly after release; and the contact holds the thorium. The readings accepted in the literature explain parts of this picture — the round basins, the thorium of Procellarum, basalt in low places — but they do not explain the 44 domes without basins, the thorium shared by basalt and highland within the contact, or the fact that floors sank only over strong domes. The deciding tests are the buried domes, the floors over them, the iron rings and the foreign mass.

Part O — Conclusions. Eight conclusions, in order: a mare is an opening lying over a feed dome; counted this way, there are 73 domes, 44 of them buried, and 33 of 34 maria lie on a dome; the buried domes carry the cover of the land and 8–13 km of extra crust; every floor is one level below

its rim, which sank over its dome; the maria are the source of the iron around them; the floors are clean, and the contact floors the cleanest of all; the contact holds the thorium; and the basins, when read as the product of external bodies, require a foreign mass that their melts do not carry. The further research defines the calculation of the critical flux, the rim ring against measured volatile concentrations, the last scatterer as a function of distance, the ruthenium and osmium isotopes of each basin, the thorium at a depth of 26 km, and the youngest basalt against a centre fixed in advance.

Appendix A — The catalogue of openings. The appendix lists the 34 maria and gives for each its position, its distance from the sub-Earth point, its zone and generation, the feed dome beneath it and its cover: 4 in the contact (generation III), 11 in the hill (II) and 19 beyond it (I); three of them — Ingenii, Apollo and Bailly — lie over South Pole–Aitken. The author first marked 36 openings by eye, numbered 1 to 36, point 35 being Copernicus H and point 36 Sinus Aestuum; these 36 marks are the sample of the crust test. Six marks did not meet the definition of a mare and four were added later (37–40), so the catalogue holds 34 maria while keeping the original numbering. Where Paper 1 refers to the 34 maria it means this catalogue; where it refers to the crust test, it means the 36 marks.

C.6 Paper 6 — The Craters

> **THE CLAIM**
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> The oldest craters of the Moon were made on it in the period when it was still attached to the Earth: material thrown from the openings returned to the shell at the speed it left — 5 km/s out, 5 km/s back, 46 GPa on basalt — and the craters were carried upward together with the shell. Some 99% of the Moon’s craters are of returned terrestrial material thrown from the maria. The turning of the attached body sorted the landings: the trailing half collected the most craters, while the leading half collected the hardest.

The paper focuses on two questions: can returning material land fast enough to make a crater and leave the marks of shock, and do the craters of today’s Moon read as the record such landings would leave? It answers the first by means of the physics of the flight, and the second by means of measurements over the whole Moon. The paper consists of seventeen Parts and an appendix, and its studies are numbered jointly with Paper 5.

Part A — Terms and definitions. The words *crater*, *impact* and *depth* carry the reading of the literature, so they are defined first. *Returning material* is material thrown from an opening that came back to the surface; a *landing* is its arrival, the word impact being kept for bodies from an external source; a *crater* is the site where returning material landed. The term depth hides three heights, each with its own reference plane: the height of the rim above the land, the depth of the floor below the land, and the distance from the rim to the floor, which is their sum — like a bowl set into a table, which has a lip above the tabletop and a bottom below it. Two craters can share the same third value and divide it differently between the first two, and this division is what the longitudes record.

The rims are not equally preserved: 43% of the craters at 90°W fall in the better-preserved half, against 57% at 180°, so rims and floors are compared within the deeper, better-preserved half of each size bin.

Part B — What the series has fixed. The craters were made through the three stages of Paper 3. In the outward rounding the craters on the shell changed together with it, in the same way everywhere; in the flattening stage most of the craters formed, and among them the largest, from light, water-rich material; and in the inward rounding the craters kept their size, and the poles gathered them (Paper 7). A crater made on the shell moved together with it, and is not to be recomputed as if it had formed in today's lunar gravity. Craters of 5 km and less, which mix primary landings, lumps of ejecta and spray, are not included in the comparison.

Part C — The craters, how many there are, and the crater as a record of the returning material. The source is the maria of Paper 5 together with the buried feed domes; on the parent their material landed at the speeds of the flight, and at a later stage, from the body itself, at the body's escape speed or less — about 2.0 km/s at a density of 1,300 kg/m³, and 2.3 km/s at release. Most of what a mare throws falls back into it, as most of the drops of a pot boiling over fall back into the pot; the drops that fall outside it are those that make the craters. The craters of 20 km and more were counted in four equal-area caps, after removing the maria and South Pole–Aitken: 249 per million km² at 90°E, the trailing half, against 139 at 90°W, the leading half; the ratio, 1.79 (1.56–2.05), stays far from one, and the excess is concentrated in craters smaller than about 70 km (Study 1). The literature expects the opposite trend — an excess on the leading half. The source test (Study 2) finds the mare's iron in the ground and in the craters, falling with distance, as in Paper 5, and the thorium follows it, from 2.16 ppm near the edge to 1.02 in the control areas. A last section asks how heavy a returning body had to be: on the parent, at 8.25 m/s² and 5 km/s, the same crater needs a mass 7 to 23 times that of a body at 15 km/s on today's Moon — a sensitivity of the scaling law that a calibration of the target must turn into a mass.

Part D — The craters, visible and buried. Craters were lost by two mechanisms, and the paper keeps a separate term for each. *Buried* is used only on the far side: a source that worked early was covered, together with its craters, by the layers of the far side. Over the 44 buried domes the crater density is 0.48–0.95 of that in the ring around them, and never exceeds it, and the large craters are denser far from the visible maria (1,040 per million km² beyond 1,000 km, against 467 within 150 km): they come from the buried domes and the visible maria alike (Study 3). *Melted* is used for the floors of the maria: a crater that landed on an active floor sank into it and closed. Of 79 craters of 10 km and more on the floors, 49 are melted (62%); over 1.2 million km² of floor only one sharp crater of 20 km and more remains, 0.8 per million km² against 207 on the highlands (205 in Paper 5, on a slightly different highland area); and on the 703,000 km² of the contact floors not one crater of 20 km and more survived (Study 4). Beneath the melted craters the base of the crust is raised $+0.8$ km above their floor in 13 of 16, with no sharp edge in the gravity field: the material sank into the liquid mare and spread, and left no body in the shape of the crater (Study 14).

Part E — Size and state: the maria and the craters in time. This is the Part shared by Papers 5 and 6, in the same text as Part G of Paper 5 (above): its main study is Study 5 here and Study 8 there, and the study of the plains of the contact is Study 15 in both. The Part shows that the share

of melted craters on the floors rises with size, and that the small craters of the contact are sharp, and reads both through two clocks: the cooling of the maria, and the material from the Earth, which arrived after release smaller in size, though still hot.

Part F — The rim. The rims of 1,457 craters of 20–100 km were traced along 72 azimuths in four caps, after removing the maria and South Pole–Aitken. The rim records the forces that acted at the moment of landing. The signs of a hard landing point to the leading half: in craters of 20–30 km its rims are a quarter more irregular than those of the trailing half (9.2 against 7.4 thousandths of the diameter), jump more between neighbouring azimuths (3.9 against 3.5), and are more often elongated (17% against 12%). The height of the rim above the land is greatest on the far side, a fifth to two fifths above the trailing half in craters above 30 km, and least on the leading half among the craters of 20–30 km. The model reads this through four forces: on the far side the outward push of the shared rotation lifted the thrown-up material; on the leading half the ground ran into the landing material; on the trailing half the landing was ordinary; and on the near side, facing the Earth, the same push pressed the material down. Between the two halves it is the rim that differs, and not the floor.

Part G — The bottom of the crater below the ground. The floor records the ground on which the material landed. In craters of 20–30 km the floor lies 54 thousandths of the diameter below the land on the far side, against 27 on the near side: deep in the crushed and melted layers of the far side, and near the surface on the near side, where the maria filled them from below. On the near-side basalt, the floors of the large melted and smeared craters stand at the level of the basalt, and not at the depth of a crater, and a third of them stand above the basalt outside — a rise that froze before it could settle, which a floor filled from outside cannot do (Study 10). The high floors are poorer in iron and titanium than the low ones (13.3 against 14.4 wt% FeO, and 1.5 against 2.5 wt% TiO_2): they froze with the light, volatile-rich material, whereas the low ones settled with the heavy material (Study 11).

Part H — The floor of the crater. The surface of the floor records what shook it and what filled it. In the better-preserved craters of 20–30 km, the floors of the leading half are 1.5 times rougher than those of the trailing half, and crater by crater a rough floor goes with a jagged rim, most strongly at 90°W (correlation 0.40) and least at 0° (0.02): the shock of the landing. The near side has the largest share of flat floors in every size bin: the maria filled them from below. A third reading, that the leading half met more space dust, predicts a larger excess on the shallower craters, and is not supported by the measurements.

Part I — The central peak. The peak records the energy of the landing. In craters smaller than 50 km the leading half carries the most peaks, most clearly near the threshold size (60% of the craters of 20–30 km, against 49% on the trailing half); at 50–100 km nearly every crater has a peak; the height of the peaks follows the rims — highest on the far side and lowest on the near side — the mark of the rotation the shell shared with the parent. Rim, floor, roughness and peak come together into one study (Study 6): the leading half carries the marks of a hard landing, the far side the highest rims and the lowest floors, and the near side the lowest rims and the highest floors.

Part J — The engine. To be thrown at speed u each kilogram needs $u^2/2$ at the opening: 2, 4.5, 12.5 and 32 MJ/kg for 2, 3, 5 and 8 km/s, respectively. The pressure that lifted the deep material (the lift cost of 139.78 GPa in Paper 3) was spent on the lift itself, as the push of a pump is spent in raising water to a roof tank; the ejection energy must come from the expansion of the volatiles at the opening. A landing of material on identical material at 5 km/s drives the material behind the shock front at 2.5 km/s; on basalt (a front speed of 3.5 km/s plus 1.3 times the material speed, a density of 2,700 kg/m³) the contact pressure is the product of density, front speed and material speed, 45.56 GPa, and 12.96 GPa at 2 km/s — above the thresholds at which basalt minerals record shock. On the way back, a body a kilometre across crossing a vapour column of 100 bar keeps about half its speed. A solid lump and a cloud of particles of the same mass do not deliver the same shock, and the receiving surface must be stiff enough, to a depth comparable with the size of the crater, to keep it. All material thrown at less than about 10.748 km/s stays bound to the parent and returns to it.

Part K — The geometry. Material thrown vertically from the turning parent keeps its energy and its turning motion; when it returns to the same radius, gravity gives back the speed it took, and the ground beneath turns at the same rate, so it lands at exactly its ejection speed. Higher up it turns more slowly than the ground, as a skater slows her turn by spreading her arms, and so lands to the west of its starting point: at 5 km/s it rises to a height of 1,932 km, stays in flight 28 minutes and lands 1,032 km to the west. On a higher target the landing speed falls with height up to the corotation radius, about 15,883 km here, and rises beyond it; hence a trend of shock strength in time belongs to one path, and is not a property of the mechanism. Earlier versions of the series gave shock ceilings of 113 and 97 GPa and a trend of 102 to 86 GPa; reading the flights path by path showed that these hold only for one family of paths, and they are no longer claimed. The last section turns to the body's own ejecta: material thrown from the trailing half falls back while the ground moves away from it, and lands softly; on the leading half the ground keeps moving into it and the landing is hard, as a runner meets the rain on the face and not on the back; from the far side the material falls back toward 170°E to 100°E. The trailing half therefore holds the most craters, and the leading half the hardest.

Part L — The chemistry. Around every mare its iron and titanium concentrations are highest next to it and fall outward, with an excess out to 600 km — ten times and more the reach of sideways mixing by later craters (Study 2, presented here in full). The crater is poorer than the ground around it because it threw the veneer aside. Thorium also forms a halo around the maria, but the thorium map is too coarse to analyse single craters. Inside the crater, from the centre to the rim, one material lies at every longitude, the same on the halves that were landed on hard and soft (Study 7): the crater is filled with the returning material that made it. Reading the visible craters as the product of external bodies requires 4.4–5.0×10^{19} kg of foreign mass, two thirds of it for South Pole–Aitken alone; at the retention rate accepted in the literature, this leaves 2–6% in the upper 5 km of the crust (an earlier version gave 2–5%), which in fact holds 0.9–2% at most — up to 6.7 times less than required; the whole bombardment the literature assumes requires 3.7×10^{20} kg, against 2.1–2.7×10^{19} in crust and mantle together, fourteen to eighteen times more, and the ruthenium isotopes of the crater melts, where they have been measured, are the Earth's (Study 8). The ejecta deposits of the craters on the basalt are poorer in iron than the basalt, as the lighter material that the Earth could throw after release requires (Study 12). The small fresh craters, of 1–2 km, are 1.19 times

more numerous on the near side than on the far side, while the older craters, of 10–20 km, are only about half as numerous there: the Earth, always in the near side's sky, was the last scatterer (Study 9).

Part M — Dating. The paper orders the crater populations without converting counts to years: first the craters buried with the far-side sources; then the craters made while the body was attached, which rose together with the shell; then the craters that landed on the active mare floors and melted; and last, the sharp craters on the frozen floors and the small fresh craters on the land. The last population is the material captured after release: with release the Earth's pressure no longer reached the Moon, and the Moon's own ejections stopped, while the Earth went on throwing small and medium drops, some of which were drawn to the Moon. Material from the Earth arrived at a few km/s plus the Moon's own escape speed, 2.4 km/s at today's radius (2.3 km/s at release), and not at the 15–20 km/s of an asteroid, and it is made of the Earth's material. Hence the predictions for the maria: craters small relative to the mass that made them, shock below the asteroidal range, and no contrast of composition between the returning body and the ground it landed on.

Part N — Statistics. The shapes were measured on 1,457 craters of 20–100 km on the SLDEM2015 elevation model at a resolution of about 60 m, and compared only within the same size bin and the same cap. Peaks and crests are identified by fixed rules set before the comparison and applied alike to every region. The intervals are bootstrap intervals; the counts of Studies 1 and 9 were also resampled by whole cells of 5° by 5°, so that craters clustered around one source are not counted as independent, and the ratios stayed clear of one. The chemical data are analysed against the calibration scatter of the maps, 0.81 wt% FeO and 0.43 wt% TiO_2.

Part O — Predictions and tests. The Part sets out fourteen focused tests, each with its claim and its measurement: among them ejection and flight, the trend in the series of events, the terrestrial source, the maria as the source, the spray ring of the active openings, the buried feed domes, the longitudes, the missing foreign mass, and the record carried on the shell. Of the fourteen studies, eleven achieved full validation (green) and three are borderline cases (amber). In Study 6 some of the intervals of the rims of the larger craters include zero; in Study 11 the iron difference, measured against the basalt just outside each floor, is not beyond chance; in Study 12 the craters that stayed within the basalt do not yet separate the model's reading from that of the literature. The numbering is joint with Paper 5: Study 5 here is Study 8 there, Study 15 is the same study in both papers, and Study 13 of this paper (like Study 14 of Paper 5) was cancelled, the other numbers kept.

Part P — Discussion. The findings of the paper come together into one process recorded in three records: where the material landed — set by the turning of the attached body; how hard it landed — written in the centre and the rim; and on what ground it landed — written in the floor. The trailing half carries 1.4 to 1.8 times the large craters of the leading half; a flux from outside would give an excess of some tens of per cent on the leading half, and would give no reason for the gentler craters of the trailing half, while the body's own ejecta give both findings at once. The size distribution of the craters remains a constraint to be met, by carrying the sizes and speeds of the thrown material through the stages of break-up, landing, saturation and burial. What is new is the distinction between the return speed and what the source can throw, the speed relative to a turning surface, and the parent's gravity and vapour as part of the conditions of crater formation.

Part Q — Conclusions. Nine conclusions, five on the physics and four on the record: a returning body lands at the same speed at which it was thrown; at 2–5 km/s the landing makes 13–46 GPa; the source must throw fast, and the material must cross the vapour and arrive whole; a shock trend or a shock ceiling belongs to a path, not to the source; a crater made on the parent must be computed at the parent's gravity, about 8.25 m/s², five times the Moon's today; the craters carry no foreign mass; the maria are the source, and craters were lost by burial on the far side and by melting on the mare floors; and the two halves record the body's own ejection. The tests of Part O decide the attribution measurement by measurement.

Appendix — Reproduction and the novelty map. The flight table is recomputed step by step as a segment of a Kepler ellipse leaving from an ejection radius of 6,721 km: the semi-major axis, the eccentricity and the mean motion, the exit angle, the flight time, and the westward shift of the landing site as the difference between the angle the surface turns and the angle the material sweeps. There is no general flight ceiling of 44 hours; near the escape energy the flight time lengthens without limit. The novelty map separates knowledge existing in the literature (secondary craters, transfer of material from the Earth to the Moon, shock and crater scaling laws) from the model's hypothesis (internal ejection onto the shell of the body attached to the pear) and from the calculations of the paper itself, and places the attribution of the observed craters with the tests of Part O.

C.7 Paper 7 — The Poles

> **THE CLAIM**
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> The polar regions of the Moon carry a higher crater density per unit area than its equatorial region, because the young Moon was flattened by its fast spin while it stood on the pear, received its craters evenly on that flattened figure, and then rounded at constant volume while still attached, so that its polar surface closed up and the craters on it were gathered toward the two ends of the spin axis. The expected crowding is not a single value but a range that rises with the age of the craters — about 1.6 for craters formed at the peak of the flattening, about 1.2 for those formed late in the attached period, and 1.0 for those formed after release — and the clean count, in which the maria and South Pole–Aitken are removed on both sides, gives 1.20 ± 0.05, rising with crater diameter from 1.09 to 1.53.

Part A — The observation. This part is devoted to measuring where the craters stand, before any theoretical explanation. The data rest on the Robbins (2019) catalogue and examine craters of 20 km and more (6,973) and of 40 km and more (2,329); the data are always expressed as a density — a number of craters per unit area — so that regions of very different size can be compared. In the raw count the polar caps (of radius 30° about the north and south poles) hold 219 craters per million km² against 166 in the equatorial band $|\phi| < 15°$, a ratio of 1.32; for craters of 40 km and more the ratio rises to 1.51, and the band adjoining the north pole reaches 300 (1.80 times the equator). Cut by longitude the data show an even sharper unevenness (values from 84 to 251, a factor of 2.97),

but this dimension reflects a separate east–west pattern: a latitude band contains every longitude, so the two sets of profiles are analysed separately. Removing the basins and maria narrows the longitudinal range to a ratio of 1.68, but leaves the polar density in place.

The filtering uses a mask that removes the 20 basins of 500 km and more and the LROC mare polygons; their area is removed from the denominator, and their craters from the count — as one counts trees per hectare on dry land, not on the pond. After the filtering the two bands adjoining the poles show densities of 305 (south) and 300 (north). Caps of decreasing radius reveal how the excess is arranged: about the north pole, where no basin or mare reaches within 20° of it, the density rises steadily from 189 in the cap of 60° to 300 in the cap of 15°, and levels at 298 in the cap of 10°. The southern hemisphere looks flat in the raw count only because South Pole–Aitken occupies it; once the basin is removed it too shows a rise toward its pole (305 in the cap of 15°, and 348 in the cap of 10°, on 36 craters). At both thresholds the excess is larger for the larger craters.

A count in six equal caps of 30° about the six ends of the Moon's three axes, after the mask, shows a clear order: the sub-Earth point is the sparsest (148), the leading point next (174), the poles and the far side at a higher level (221–246), and the trailing point shows the highest value (252). This is the order of a body flattened by its spin and stretched by the tide along the line facing the Earth. The trailing excess (1.45 in the caps, about five counting standard deviations, but only about one in the pole-to-pole bands) lies about 90°E, on the side that carries more maria of its own; the model reads it as returning material from the openings landing about them, $N_{\mathrm{obs}} = N_{\mathrm{background}} + N_{\mathrm{return}} - N_{\mathrm{erased}}$, and the link between mare area and crater density is set as a requirement. Inside South Pole–Aitken the ring within 10° of its centre holds 132 craters per million km², 0.59 of the density just outside its rim; the finding is measured here and passed on for further analysis in Paper 8.

The basic comparison on which the paper rests is like for like: the maria (22% of the equatorial band and only 1% of the caps) lower the equatorial density and inflate the ratio, while South Pole–Aitken (24% of the caps) lowers the polar density. When both are removed on both sides, the ratio goes from 1.32 to 1.11 $\pm$ 0.04 (maria removed) and settles at **1.20 $\pm$ 0.05** (maria and South Pole–Aitken removed). The figure after $\pm$ is one counting standard deviation; with 95% confidence the clean ratio lies between 1.10 and 1.30 (an earlier version read the $\pm$ as the 95% interval).

Part B — What we tried before the answer. Six alternative readings were examined and set aside. Each was held to four questions, in this order — the sign of the change, the size of the effect, the shape along the spin axis, and its survival under the mask — and each left something to the account. Preservation on old highlands is a real factor, ordering the crater distribution at global scale (219, 164 and 99 craters per million km² on the far side, the limb belt and the contact disc, respectively); yet the polar excess survives the mask, the crown of the far side (221) stands below the poles, and the north pole has a centre. Rotational flattening of a surface already cratered gives the opposite sign to the findings: it gives $\rho_{\mathrm{pole}}/\rho_{\mathrm{equator}} = c/a$, 0.83 and 0.59 for the figures of the attached stage. Behind the scenes, this is where the method changed: the analysis moved from looking for a mechanism that makes a flattening to looking for one that undoes it. The tidal stretch into the Roche figure ($1 : 0.637 : 0.601$) likewise thins the poles and crowds the sub-Earth point, the reverse of the observations; the model keeps this figure as the shape to be undone. Stronger gravity at the pole moves nothing sideways: on a spinning body the pole is a hilltop of the effective

potential, about 1.6×10^5 J/kg or 120 km above the equator, and it becomes the engine only when the densifying body can no longer carry its flattening. A sink at the pole would require a density peak at a point, but the northern density levels off within 15° of the pole, and every force of a rotating body is symmetric about the axis and so makes a ring, not a point. A uniform change of scale leaves every ratio at 1.00. What remains is one change of shape: a flattened body returning to a sphere. It answers all four questions, and it alone makes the excess depend on crater size.

Part C — The mechanism. A soft spinning body takes the Maclaurin equilibrium figure, whose flatness is set only by the spin and the density: the relation between $\Omega^2/\pi G\rho$ and the eccentricity rises to a maximum of 0.4493 ($a/c = 2.72$), so at a spin of 5.5 hours no fluid body lighter than 1,070 kg/m³ can hold together. When the body first held itself, at the rigid Roche limit for 1,300 kg/m³ (12,994 km; 10,314 km at 2,600), it turned with the Earth once in 5.5 hours, and its axis ratio was **1.69**. As the body densified on the pear, the same spin supported less flattening (1.44, 1.30, and about 1.20 late in the attached stage, when 71% of the flattening was already gone), and at release the body was round and its shell rigid (axis ratio at most 1.04, Paper 2 §B.7) — a requirement of the model, not computed in this paper. The free fluid tidal figure at 2,600 kg/m³ ($1 : 0.637 : 0.601$ at 20,595 km, just beyond the fluid Roche limit of about 20,000 km for that density) is used only to illustrate the tide's share; at release, at 16,400 km, no fluid figure exists, and the body is held by its rigid shell.

When a flattened figure rounds at constant volume, the polar surface shrinks by $(c/a)^{2/3}$ and the equatorial surface opens by $(a/c)^{1/3}$, so craters formed evenly end the process crowded at the pole point, against the equator, by exactly a/c; averaged over caps of 30° against the equatorial belt, the crowding factors are 1.59, 1.38, 1.26 and 1.17 for figures of 1.69, 1.44, 1.30 and 1.20, respectively. The poles are the standing points of the change because the spin axis passes through them: as the flattening relaxes every meridian converges back to them, as the spokes of a wheel meet at the hub, and the whole ring 90° from the sub-Earth point is crowded, the poles most of all. Compaction from 1,300 to 3,344 kg/m³ shrinks every element alike (0.53) and changes no ratio. The engine of the rounding is the body's own gravity, growing as it compacts — $\Omega^2/\pi G\rho$ falls from 0.37 to 0.17 at nearly constant spin — and the rounding releases $2\text{–}3 \times 10^{27}$ J, about 2.4% of the binding energy, enough to warm the interior by 30–40 K. Had the rigid shell, strained by ten to twenty per cent, carried the change of shape, it would have faced stresses of 2–12 GPa against a strength of about 0.1 GPa; the pressure escapes through the existing vents of the maria, and the material thrown out returns and lands as craters — strongly while the figure is flattest, and weakly in the later stages. Large craters formed early and small ones late, so crater size serves as a clock.

Along the time line, the tide measure $k = M_\oplus/\pi\rho r^3$ falls from 2.01 at the hemisphere stage and 0.67 at the start of rounding to 0.17 at release, and at no stage drops below the fluid threshold of 0.09. The base, narrowing from 6,000 to 3,000 km, carried what the body's own gravity could not, and at the start of rounding the effective gravity at the two ends of the tidal axis was negative. The detachment from the base was not gentle: the body was pulled off with force slightly after its centre of mass crossed the corotation line, about 210 km beyond it, where the month is born equal to the day (5.77 hours). In the series' rough time line about 600 Myr separate the start of the bulge from release at 3.873 Ga — some 400 Myr for the bulge and the pear, about 100 Myr for the flattening

stage, and about 100 Myr for the inward rounding — so the pace of the change of shape was set by densification, not by relaxation. The figure of the body was fixed well before it reached 16 Earth radii (Paper 4 §C.3), and the fossil flattening of today, $f \approx 0.001$, is four hundred times smaller than the 0.4 of the peak. The material demands of the two layers meet in one requirement: an interior of 10^{21}–10^{22} Pa s that follows its figure within 10^{3}–10^{4} years, under a shell of 10^{23} Pa s or more (the shell Paper 3 requires to keep the craters made on the Earth). On this shell the craters ride whole, like tiles on a floor pushed together at the joints: it shortens toward the poles and fractures toward the equator.

Part D — Tests. Five tests were set, together with their pass lines, before the counts were made; every ratio is given with its 95% interval (exact binomial for counts, and a bootstrap of 20,000 resamplings for medians). *The density ratio:* predicted 1.1–1.6, and the clean count gives 1.20 $\pm$ 0.05 (95% interval 1.10–1.30); craters below 20 km are not included in the comparison, since secondaries, spray and saturation govern there. *The size clock:* the ratio rises steadily with diameter — 1.09 (20–30 km), 1.15 (30–40), 1.26 (40–60), 1.53 (60–100), and 1.84 $\pm$ 0.38 (100–300) — and read backward through the geometry of Part C these values point to figures of 1.10, 1.17, 1.31, 1.63 and 2.00; the craters of 60–100 km recover the peak flattening of 1.69, computed in Part C from spin and density alone. Because the size law has $b \approx 2$, craters that shrank with the surface would hide the crowding at any fixed threshold; the measured excess is itself the mark that the craters kept their size, and at the same threshold the polar median is the larger, 1.06 (1.01–1.12). Behind the scenes, an earlier reading assumed that the polar craters shrank by 9% and compared medians at a lowered threshold; that comparison in fact measured the threshold, and the paper follows the stiff shell.

The young craters, formed on the body after it rounded, must show no polar excess: the measurements give 0.92 on the Copernican craters of the LPI database, and 1.12–1.16 on the physically dated sample of Mazrouei et al. (2019) — values consistent with 1.0, but too few to exclude 1.2 (probabilities of 0.22–0.52); separating the two values needs about 150 dated young craters in the caps and 300 in the equatorial band. The young population is also uniform in longitude (3.11 against 2.84 craters per million km², far side against near side), so the mechanism that crowded the old craters had stopped before the young ones formed. *North and south:* after the mask, the south-to-north density ratio is 1.02 (0.78–1.32) in the cap of 15° and 1.02 (0.94–1.10) in the cap of 60° — equality inside every interval. *Five tests of size* ask whether the polar craters shrank by the factor of 0.79 that the deformation would give: the largest craters per equal area are larger at the poles, 1.16 (1.04–1.29); the hump of the size distribution stays at 80–113 km, with a scale factor of at least 1.09; old against fresh craters give a double ratio of 1.02 (0.89–1.16); the ellipticity at mid-latitudes exceeds that of the caps and belt by only about 0.01, a tenth of what the deformation would leave; and the simple-to-complex transition lies at 16.4 or 14.0 km in the caps, against 16.8–16.9 km in the belt, where shrinkage would predict about 13. Four tests find the size kept; none finds the shrink.

Part E — What remains. One process accounts for six observations, each with a number computed before the count: the polar excess and its rise with crater diameter, the larger polar median, the equality of the two poles, the order along the six ends of the axes, the absence of an

excess among the young craters, and the recovery of the flattest figure from the largest craters. Alongside these, five observations were measured but not derived from the model, and are stated as such: the trailing-side excess (1.96 in the raw count, 1.45 after the mask), whose link to the openings about 90°E awaits the ejection-and-return calculation; the crown of the far side, which shows a high density (221) where the geometry asks for less, a finding stated as a requirement that the crown of the dome did not contract as a free end; polar craters of 20–100 km shallower than equatorial ones (depth-to-diameter about 0.062 against 0.074 at 20–30 km); a meridional alignment of the most elongated craters at latitudes 30°–60° (45–51% against 20–25%), with the lighting of the images as the first suspect; and the crater-poor centre of South Pole–Aitken, passed to Paper 8. The further work centres on turning requirements into calculations: the figure of a body with strength; a verified and larger list of young craters (the table published by Mazrouei et al. and the 1,615 rayed craters of Werner & Medvedev); the two-layer relaxation solved along the attached stage; the pressure trapped at detachment; the tide's share of the crowding on a body held by its base; the spacing of polar craters as a direct measure of the contraction; the shortening and extensional structures the shell should have left; the yield of craters from the openings about 90°E and 90°W; and the innermost caps, below 20 km, with the next catalogue.

C.8 Paper 8 — South Pole–Aitken

> **THE CLAIM**
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> The concentrated crater deficit at the centre of South Pole–Aitken records a long-lived heat source beneath the far side — the burner — that worked opposite the openings of the near side, with two exits: Imbrium the main one and Serenitatis the secondary. During the attached period the heat came from the Earth, entering through Imbrium and Serenitatis; after release it was replaced by the Moon's own internal heat, together with the radiation of the nearby Earth, which faded as the Moon receded, and everything cooled slowly. The heat reached the surface where the crust was thin and raised it into a dome: the softened crust slid toward the margins, volatile and even refractory material was driven off from it, the layers of returned material were cooked and compacted, and iron-rich material from the Earth's interior came to the base. As the heat waned the dome cooled and collapsed into the basin seen today. The dome lies south of both exits, a direction set by the pear and the rotation, and the change from the flattened figure to the sphere enlarged the distance from 17–21° to the 24–29° of today.

Part A — Introduction: the basin and its enigma. South Pole–Aitken (SPA) is the largest basin on the Moon, about 2,400 by 2,050 km. Its floor lies 6–8 km below the surrounding highlands — the deepest ground on the Moon — and it is the oldest large basin that can still be recognised (about 4.25 Ga from its samples). Chang'e-4 landed on its floor, and Chang'e-6 returned from it the first far-side samples. Since craters serve as the Moon's clock, a surface this old should be full of them; yet the centre of the basin departs from this trend. Counting in rings outward from the centre, on ground cleaned of the lava plains and the maria, shows that the centre holds only about half the craters of

the ground just outside the rim; the place where they are fewest — *the minimum* — lies about 270 km north-north-east of the centre, and the craters that survive there are shallow, under high rims.

The basin is set apart in other records as well: thin crust ringed by thickened crust, an iron- and thorium-rich floor with little visible lava, a dense mass deep beneath it, a mantle dried and poor in oxygen, melts that rose over some two billion years, and strong crustal magnetism on its margins. The accepted reading attributes these features to a giant oblique impact about 4.25 Ga — from the north in most readings, toward the south in Andrews-Hanna et al. (2025) — with each feature given its own explanation. Six findings, however, remain unexplained or strained: a deficit concentrated in the centre that sharpens as the lava is removed, its off-centre position and its loss of the middle sizes, the scarcity of lava precisely in the thinnest crust, an east–west rather than north–south asymmetry of the ring, a span of melts of two billion years, and a weak magnetic field at the minimum. The model proposes one process in their place. The burner is heat from the Earth's interior that crossed the attached body along the line of the openings and gathered beneath the far side, warming the ground from below as a flame warms the base of a pot. While the burner was active, SPA lay about 27° from the far tip of the elongated body — the point opposite the centre of the contact — and on the sphere of today the same separation reads about 39.5° (southward 22° $\rightarrow$ 33°, eastward 15° $\rightarrow$ 22°). This change results from the passage from the flattened figure to the sphere, not from a motion over the surface (an earlier version, measured from the point opposite the old reference point 23°N 320°E, gave 34–36° $\rightarrow$ 48.6°). An antipodal relation survives both the turning of the body and its rounding, so a point and its opposite still face each other through the centre.

Part B — The anomalies of SPA (overview). This part examines the anomalies one by one, each in six steps: what is measured and by whom; why it is strange; what science says; where that falls short, closed by a verdict (explains, explains partly, explains with difficulty, does not explain); what the model says; and how the measurement fits it, closed by its own verdict (fits, fits partly). The two verdicts stand in separate boxes, and neither compares itself with the other. Where the paper made the measurement itself, a study in the seven steps of a research report is attached (Studies 1–5). All crater comparisons are like for like, on old ground: the lava plains of the LROC map, the 34 maria of Paper 5 together with the core of the feed dome beneath each, and a halo of 200 km around each mare are removed from both counts and areas on both sides, and every density is normalised by that of the ring just outside the rim (39.57–50° from the centre at 53°S, 191°E).

B1 — The crater record (Study 1). SPA as a whole holds 7% fewer craters of 20 km and more than the rest of the Moon on the same type of ground (213.9 against 230.2 per million km²), and 11% fewer than ground at the same latitudes, and the shortfall is concentrated in the centre: 0.51 of the outer density within 10°, 0.81 in the next ring, and about nine tenths from 20° outward. The cleaner the ground, the sharper the deficit (0.59 on all ground, 0.56 without lava plains and maria, 0.51 without the halos too); so young surfaces blur the deficit rather than cause it. The deficit also survives on the basin's ellipse (122–132 in the core). A continuous density map places the minimum at 45.4°S, 197.9°E, record strength 0.45, on the meridian of the point opposite Serenitatis (197.9° against 198.8°) and 212 km from the Chang'e-6 landing site; the poorest of eight sectors are the north-east and the north. The middle sizes are missing most (0.47 at 20–40 km, against 0.91 at 10–

20 km; 4.62 small craters per larger one in the centre, against 2.13 elsewhere), and the craters of 40–80 km in the central 20° are 18% (4–33%) shallower than those in the rim belt, with rims standing higher than in a matched control (1.35 against 0.92). Science: does not explain — the oblique-impact model gives no answer to the deficit concentrated in the centre, the loss of the middle sizes or the raised rims. Model: fits — heat from below erased and relaxed the craters, later craters landed on compacted ground and developed high rims, and a late spray of small fragments fell on top.

B2 — Shape and depth (Study 2). SPA is the lowest region of the Moon: the floor between its craters lies at -4.8 km, 6–9 km below its rim, depending on how the rim is defined. Measurement only on the ground between craters — every crater of 20 km and more removed with a margin, and with them the openings, the 36 original marks of the series — shows that the basin is deeper than its thin crust explains by isostasy by 1.1–1.8 km at the centre, and this gap falls off gradually to about 0.2 km at the rim, for every margin width tested. Science: explains. Model: fits — a dome that collapsed, with compaction strongest at the centre; a column 9–15 km thick that lost 12% of its voids sinks by 1.1–1.8 km, and the lowest ground lies above the dense mass.

B3 — Thin crust. In all four GRAIL crustal models the crust at the centre of the basin is 15–28 km, against 41–51 km outside: the basin has lost about half its crust or more. The thinnest crust (about 61°S, 184°E), the lowest intercrater floor (61°S, 199°E) and the dense mass (62°S, 200°E) all lie together in the south, within about 230 km of one another (an earlier version gave about 200 km). Science: explains (a southward-travelling impact; Apollo by a blow of its own). Model: fits — the softened crust slid off the heated dome, was pushed outward by the expanding hot material and melted from below, leaving small domes of thin crust over the two exits, Ingenii and Apollo. The question why the weakest crater record lies on the northern side of the thin zone, about 450 km from its thinnest point, is left open.

B4 — The outer ring (Study 3). With the regional far-side trend removed, the crust around SPA is thickened in all four models by 3.8–5.7 km. The thickening lies 50–55° from the centre, about 320–470 km beyond the rim (an earlier version gave 350–430 km), in a band about 450 km wide; the ring stands above the crust farther out in seven of eight directions, thick to the east (7.8 km toward 97°) and missing to the west, and it holds about 37% of the crust missing from the basin. Science: explains partly — an asymmetry along the east–west axis is predicted by no impact direction. Model: fits — the material that slid off the dome piled at the margins: narrow and close in the north-east, where it met the thick far-side crust, and wide in the western and south-western sectors.

B5 — The inner ring (Study 4). A second ring of thickened crust, found in 22 of 24 directions, is centred at 51.4°S, 196.8°E, about 118 km east-north-east of the ellipse’s centre; its inner wall is steepest to the north-east (30.7 m/km) and gentlest to the south-west (11.5 m/km). Science: explains partly. Model: fits — the ring is the edge of the dome, and its centre is the dome’s centre, which lies today 29° from the Imbrium exit and 24° from the Serenitatis exit. Both rings follow the thickness of the surrounding crust, and pile steeply against the firm crust of the north-east, as snow piles against a kerb.

B6 — Density and porosity (Study 5). At the surface SPA is as porous as the highlands (19.7% against 20.3%); but at 10 km depth its pores are nearly closed: 2.9–4.4% at the centre, against 6.2–7.9% in the far-side highlands and 9.6–12.8% in the rest of the far side, and they rise steadily toward the rim in both depth models; at the same time the grain density is higher at the centre (3,005 against 2,897–2,920 kg/m³). Science: explains partly — composition explains the density, but not a compaction at depth that weakens from the centre outward, since the weight of the overlying layers closes pores alike everywhere. Model: fits — heat held long beneath the centre cooked and compacted the porous layers of returned material, and the porosity difference over 0–20 km accounts for 0.5–1.2 km of the subsidence of B2.

B7 — An unusual floor. Lunar Prospector gamma-ray maps show iron falling steadily from the centre outward (9.5, 8.6, 6.6, 4.9, 4.4%), thorium low at the centre (1.7 ppm) with its maximum at 10–20° and hot spots of 4–5 ppm in a crescent facing north-west, and a potassium-to-thorium ratio of 451–455 across the whole interior of the basin, 16–20% below the comparison regions. Science: explains, each finding by its own cause. Model: fits — a melt lake above the burner, south of the two exits; heavy material raised with the heat; thorium carried with the melt toward the trailing side; and volatile loss across the whole heated zone.

B8 — Little visible lava. The maria make up only 4.1% of the area of SPA, against 30% of the near side (an earlier version gave 32%); the cover is 14.6% in the inner 300 km and only 0.7–1.3% at the margins. A central cap of near-side basin size holds 8.4%, less than any of the eight near-side basins with similar crust, limb basins excluded (22–74%, mean 51%; an earlier version gave a mean of 42%, which included the two limb basins). Science: explains partly. Model: fits — the basin was heated and melted from below rather than flooded with lava from above; only at the hot centre did melt reach the surface, and the vast heated area dilutes the overall percentage, which accounts for most of the gap.

B9 — Buried basalts and the central mare. The buried lava mapped by Wang et al. (2024) lies almost wholly inside a circle of radius 15° about the dome's centre (13.6–24.1% of its area), together with most of the visible lava, and thorium forms a one-sided crescent at 12–32° from the centre, a layer weakening with distance. Science: explains. Model: fits — the dome first held a large, ordinary mare, which was later closed and buried. The burner's exits lie at 37.6°S, 160.0°E and 27.3°S, 198.4°E, the antipodes of the centres of Imbrium (37.6°N, 340.0°E) and Serenitatis (27.3°N, 18.4°E) — the crater counts use the antipodes of the opening points, 37°S, 161.5°E and 25.4°S, 198.8°E — and the outflowing liquid was turned by the rotation toward the trailing side, as a simulation at the 5.77-hour day reproduces, and only while the body turns fast.

B10 — The dense mass. GRAIL gravity data point to a diffuse density excess of at least 2.18×10^{18} kg in the mantle beneath SPA, deeper than 300 km in the preferred bound (James et al. 2019), in the southern half of the basin. Science: explains partly — dense material held in a stiff mantle, but its source rests on a cored striking body, and its southern position is not derived. Model: fits — iron from the Earth brought by the burner, pushed up over the mare zone (radius about 12.4° against 15°), sinking while the mantle was hot and freezing in place as it stiffened, leaning toward the warm, soft south.

B11 — A mantle that lost its volatiles. The Chang'e-6 samples, collected about 11.6° from the dome's centre, carry isotopically heavy potassium and iron (δ^{56}Fe 0.13–0.21), and the potassium-to-thorium ratio is low and flat over the whole heated region, against 563 in the highlands and 537 in the rest of the far side. Science: explains partly — a brief impact flash can give heavy potassium, but not a uniform deficit over 2,400 km, nor heavy iron. Model: fits — strong, concentrated and sustained heating, a low flame under the pot rather than a blowtorch passed once, drove off moderately volatile elements and even part of the iron's light atoms.

B12 — An oxygen-poor mantle. Two independent methods on 23 Chang'e-6 basalt fragments give $\Delta\text{IW} = -1.93$ beneath SPA, against -0.80 beneath the near side: about thirteen times less available oxygen, and the most reduced mantle known. Science: explains partly — two possible mechanisms exist, not yet separated. Model: fits — the prolonged heat drove off the oxygen-bearing substances, and the metallic iron that came from the Earth consumed the rest, as steel wool rusts in water, and the iron sank with it (B10).

B13 — A depleted source. The Chang'e-6 samples from the same region include a KREEP-rich basalt at 4.2 Ga ($\mu \approx 1,620$), alongside an ultra-depleted source at 2.8 Ga ($\mu \approx 360$). Science: explains. Model: fits — the material arrived from the Earth rich in the escaping elements and was then washed out by repeated melting from below, like a tea bag used again and again; the elements that left are found in the thorium crescent, and the refractory residue explains the scarce late lava of B8.

B14 — Metal in the ancient norites. Two norite clasts of about 4.25 Ga carry iron–nickel metal with Ni/Co of 8.5–11: above the metal of lunar basalts (below 5), and below the Earth's mantle (18.7), the Earth's core (about 20) and primitive meteorites (21.0). Science: explains, as material of the striking body. Model: fits partly — all metal on the Moon is of terrestrial origin, and so is this; its ratio was lowered by re-melting in a reduced setting and by mixing about half and half with low-ratio metal that is itself terrestrial (giving about 10–12). The ratio cannot decide between a terrestrial and a foreign source; the isotopes of Cr, Ti, Mo, Ru and Os can, and this is the sharpest test of the paper.

B15 — Prolonged volcanism. The dated record in the basin spans a long sequence: it begins with the norites (4.25 Ga) and the rich basalt (4.20 Ga), passes through a reset at 3.873 ± 0.008 Ga, and runs to thin late deposits of 3.7–2.2 Ga, with the visible lava peaking at 3.6–3.2 Ga. Science: explains with difficulty, through two impacts at one point and two mantle sources. Model: fits — one sequence: the burner during the attached period, fed by the Earth's heat; the separation at 3.873 Ga, which closed the attached stage; and then the Moon's own heat, together with the fading radiation of the receding Earth, cooling slowly in shrinking volumes. By this sequence, the buried lava of the central circle is predicted to be 3.87–3.6 Ga old.

B16 — The magnetism of SPA. In the field model T2015_449 at 25 km, the centre is nearly bare (0.8% of the area above 5 nT within 10°, and 2.0% at 10–20°), the ring holds 14%, the north, west and north-west sectors 32–42%, and the east and south sectors none; at the minimum the field is 1.9 nT (0.85 nT at a test point about 20 km from it, the value an earlier version gave for the minimum itself). Science: explains partly — the weak centre and the strong rim each get an explanation, but the side, and its coincidence with the thorium, rest on chosen assumptions. Model:

fits — the heat erased the centre's magnetisation and much of its iron sank out of reach, while iron carried with the fluid to the trailing side sank there. Magnetism and thorium share the same side, but not the same place (the magnetic peaks lie some 8° north of the thorium centre). Who magnetised the iron is a question for future work.

B17 — The light swirls. The bright, reliefless swirls sit on magnetic anomalies. In Mare Ingenii, inside the crescent and near the Imbrium antipode, the mean field is 5.9 nT, and 48% of its area lies above 5 nT; by contrast, in Mare Moscoviense, a far-side mare with lava flows, the field is only 1.3 nT, and it has no such area at all. Science: explains partly — the brightening is explained, but the source and place of the iron are not. Model: fits — swirls form where a flow froze over metallic iron from the deep Earth, within the crescent, and not where there was flow without that iron.

B18 — The anomalies together. Of the seventeen anomalies, science explains six fully, nine in part, one with difficulty, and one — the crater record — not at all; the model fits sixteen, and one in part (B14, which the isotopes will decide). The difference lies in how the answers are built: science gives each anomaly its own cause, while the model orders them all about one long-lived centre, and in layers by depth: the exits with their small domes (Ingenii 12–16 km, Apollo 5–9 km of crust) in the north, the dome and the hot centre south of them, and the crescent of thorium, magnetism and swirls on the trailing side — as a flame kept under a pot leaves a pattern about the hottest point, a pattern one hard stir would not leave.

Part C — The antipodes: from a flattened figure to a sphere. The points opposite Imbrium (37°S, 161.5°E) and Serenitatis (25.4°S, 198.8°E) both fall in the northern part of SPA, 1,022 km apart, and their midpoint lies about 650 km from the basin's centre (an earlier version gave about 700 km). A rigid copy of the near-side basins, printed about their antipodes, gives envelopes of $2,144 \times 1,340$ km (axis ratio 1.60) from the rings and $2,034 \times 1,185$ km (1.72) from the 19 mare polygons, against $2,402 \times 2,056$ km (1.17) for SPA, and with orientations that do not agree. The negative result is kept, because the burner sets where craters were erased, not the outline of the basin, which belongs to the dome and its collapse.

When the centres of Part B are set on one map, they fall in three groups along one north–south line: in the north, the two exits with their small domes; in the middle, the dome's centre, the basin's centre, the magnetic peak nearest the minimum and the crater minimum, within 6° of the dome's centre (and within 9° of one another; an earlier version said within 6° of one another); and in the south, the lowest ground, the dense mass and the thinnest crust, within about 230 km. The line from the Earth's centre through the contact centre (24.1°N, 335.2°E) leaves the far side at 24.1°S, 155.2°E, 41.8° from the dome. Inside the rotating pear, however, the push is perpendicular to the Earth's spin axis, so the heat entered the contact disc north of its centre, where the two burners lie, and left the far side up to about 22.5° south of the point opposite: the pear stood at 10–25°N, and the outward push of rotation is level at any height, as on a carousel. The exits and the dome were originally laid out on the elongated body ($1 : 0.637 : 0.601$); when it became a sphere, distances from the end of its long axis opened without anything moving over the surface: the coordinates of the Imbrium exit changed by 5.5°, of the Serenitatis exit by 11.8°, of the dome's centre by 13.0° and of the minimum by 12.6°, and the distance between the dome and the exits grew from 17° and 21° to the 24° and 29° of today. The rotation decided that the dome lies south; the change from the

flattened figure to the sphere decided how far. A change of the Earth's axis after the separation may make SPA appear out of line with the burner; the axes are left for future work.

Part D — The dynamic version: column, rotation and angle. This part asks whether the numbers allow material and heat to cross from the Earth's interior to the far side. Conservation of angular momentum gives radial speeds of 1.8–2.6 km/s and transit times of 1.2–1.5 hours, a fraction of one turn; the kinetic energy, 1.6–3.4 MJ/kg, converts ideally to pressures of 5–10 GPa at a density of 3,000 kg/m³. The effective potential $\Phi_{\mathrm{eff}} = -GM_0/r - \frac{1}{2}\Omega^2 r^2$ rises from the surface to a hilltop at the corotation radius; from the local radius of the flattened Earth at the source latitude, 25°N (6,602 km), to 16,399 km at the 5.77-hour day, the barrier is 26.2 MJ/kg, and a parcel of material rising alone falls short of it by a factor of seven to sixteen. But the model's material is not a lone parcel; it is a continuous body under positive pressure: the part beyond the corotation radius lowers the pressure beneath it, and material from below is pushed up as in a siphon, while the engine remains the source pressure of about 150 GPa of Paper 3 — the relief acts only at the late stages, and the effective barrier and the reduced base pressure for a body of Paper 3's shape are a requirement. On the far side the rotation helps material out of the vents, stripping a late spray of small fragments that return as small fresh craters on the large ones; material leaving below the escape speed, about 2.3 km/s at the radius of then, 1,890 km, and rising less than about 700 km, stays within the Moon's Hill sphere (about 2,600 km from its centre at 16,400 km, 1.4 radii of the Moon of then) and lands back on it (an earlier version, with today's radius, gave 2.4 km/s and 900 km). A Coriolis deflection cannot place both exits relative to the basin (the antipode of Serenitatis lies 7.8° from the centre in the sense of that deflection, and that of Imbrium on the other side), so the geometry comes from the horizontal push of Part C: a small exit angle of 1–3°, kept over a path of 9,447–11,199 km, places the heat 165–590 km from the point straight opposite.

Part E — Challenges and responses. Seven published findings that seem to stand against the proposal are answered from the paper's own material. Iridium and gold in Chang'e-6 metal grains (Liu et al. 2026) lie in undated regolith fragments, where fine space dust later drawn onto the surface layer is allowed, and element ratios do not separate an asteroid from the Earth's metal-rich interior. The Ni/Co of 8.5–11 in the norites (Su et al. 2025) places the metal above the metal of lunar basalts, but both the Earth's mantle (18.7) and CI chondrites (21.0) lie above it, and a grain's ratio changes each time it re-melts; so the ratio cannot tell the two sources apart, and the decision passes to the isotopes. The dense mass lies about 500 km south of the minimum, because the minimum marks where heat erased craters at the surface, and the mass where heavy material sank. The nearest strong magnetic peak lies about 230 km from the minimum, where the field is weak (1.9 nT), as heat that erases magnetisation at its core and leaves it at the cooler margins requires. The crust at the minimum (19–28 km) is not the thinnest (15–24 km at the nominal centre), because the claim is that the deficit lies inside the zone of thin crust, its place set by the direction of the exits. The buried basalts of Wang et al. (2024), over at least 8.7×10^4 km², are a product of the burner, not a rival explanation; the test is whether the shallowing of craters persists outside them. Finally, the rule "except in the regolith" concerns the source of a sample, not where it was collected: foreign material in an ancient sample, if found at all, is expected only in tiny amounts.

Part F — Predictions and tests. The sharpest prediction is that the Moon's ancient samples hold no material of non-terrestrial origin: the metal in the two dated norites must carry a terrestrial signature in Cr, Ti, Mo, Ru or Os. If foreign material is found at all, it is expected only in tiny amounts — space dust, or one or two meteorites landed over four billion years. Alongside it stand further predictions: volatile loss and reduction strongest near the minimum and weaker toward the rim; shallow craters persisting outside the buried basalts; floor-fractured craters concentrated in the deficit zone; magnetisation strongest at the margins of the heated core and sourced at depth; a deficit that follows the antipodes of the near-side openings; a large early mare, closed and buried, with a gravity signature of its fill; a late spray of small fragments, in which the small craters on large hosts are fresh while the hosts themselves are degraded (a frozen sample of 20 hosts inside SPA and 20 in a control, with 2,634 candidate small craters, awaits inspection); two crater populations of different age; and young craters with higher rims inside SPA than outside. Two predictions are already measured in the paper: the rings of crust that slid outward are confirmed, and the compaction at depth is supported. The tests that need new data are listed: the spatial minimum of depth after the mare mask; a count of floor-fractured craters; thorium, iron oxide and hydroxyl sampled along the crater profile; the depth and direction of the magnetisation; a null test for the choice of Serenitatis among all the openings; and an out-of-sample test.

Part G — Conclusions and further work. The five studies were each stated with a threshold fixed before the measurement, and all five came out green: the crater record (0.51 at the centre, the minimum on the Serenitatis meridian, the middle sizes missing, shallow craters with high rims; limit: small central counts); the depth remainder of 1.1–1.8 km (limit: the dense mass can fill part of it); the outer ring in seven of eight directions (limit: it holds only about 37% of the missing crust); the inner ring in 22 of 24 directions; and the closure of pores at depth (limit: extrapolated from published inversions).

The conclusions are given in the order of the argument. The centre of the basin is poor in craters and holds shallow ones, and the deficit leans toward the point opposite Serenitatis. This finding is read as a heat source beneath the far side, fed through Imbrium and Serenitatis during the attached stage — at least about 10^7 years, and in the model about 600 Myr from the start of the bulge to release at 3.873 Ga. The feed from the Earth ended at release, after which the Moon's own heat and the fading radiation of the receding Earth acted, and everything cooled slowly. The depth, the rings, the closed pores, the chemistry of Chang'e-6, the reduced mantle, the prolonged volcanism and the magnetism fit the same reading. A rigid mirror of the near side does not make the basin's shape, and this negative result is kept. The centres lie on one north–south line, in three groups; the southward direction is set by the pear and the rotation, and enlarged by the rounding; the Earth's axis is left for future work; and the sharpest prediction can be decided by the isotopes. The further work includes: a physical model of the heat path and its energy budget; a redox budget to explain the reduction; the dynamic tests recomputed with the series' constants, giving the effective barrier and the base pressure of Part D; the tests of Part F; and, as future work outside the series' papers, the Moon's and the Earth's axes at the time of the rounding.

C.9 Paper 9 — The Stripes

> **THE CLAIM**
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> The long, straight, parallel stripes on the floor of the North Pacific are the leading side of the pear on which the Moon was built: they were stretched in the direction of the Earth's rotation while the pear stood, and folded into stripes when it was gathered back into the Earth. The pear's tip — the place where the Moon was formed — lies west of the base, on the trailing side, and left its outline in two bends of the stripes, in a mirror imprint of the near side, and in two chains of ridges drawn up as the tip was lifted away.

Part A — Introduction: the stripes. Six stripes — Mendocino, Pioneer, Murray, Molokai, Clarion and Clipperton — cross the floor of the North Pacific for thousands of kilometres; Clipperton alone runs for some 7,240 km, more than the radius of the Earth. These structures are seen best on maps of the vertical gravity gradient, which highlight where the pull of gravity changes sharply over a short distance, and so pick out narrow folds and troughs, whether they lie exposed on the floor or buried beneath younger material. All six are *parallels of one equator*: each keeps a constant distance from one tilted great circle, as the lines of latitude keep a constant distance from today's equator. Lines of this kind are what one turning motion about one axis leaves behind, and six such lines sharing one equator, from 33.6° north of it (Mendocino) to only 1.8° (Clipperton), place a strong constraint on how they were made.

Part A then links the stripes to the pear of Paper 3: one bulge of the young Earth, broad at its base near Kiritimati (1.9°N 157.5°W), narrowing to the tip on which the Moon sat, and leaning back against the rotation, as a child walking outward on a turning roundabout is pushed backward. Its front, facing the rotation — the *leading side*, on the east — was stretched, while its back, the *trailing side*, lay beneath the leaning tip. The Moon's release did not make the pear collapse: the sorting, which acts on the pear whether or not a Moon sits on it, gathered it back into the Earth over tens to hundreds of millions of years after release. The mass did not leave the Earth, but moved from high outside to deep inside; in the course of this the leading wall was laid along the floor and folded back on itself into the stripes.

Part B — The observations. This part examines eight observations, each stated first as what the map shows and then as what the model expects. The stripes form one belt about one equator (B.1), with two further stripes about 6° and 14° south of it (B.4), so that the base lies inside the system, not at its edge. The base of the pear, placed in Paper 3 from the deep mantle before the stripes were considered at all, lies only 1.6° from that equator, beside Clipperton at 1.8° (B.2): this is a finding, not a choice, since a body rising from a spinning planet rises where the outward pull of the rotation is greatest — on the equator of its time. The stripes run from the base eastward to America, and do not continue as straight parallels to its west (B.3).

At four crossings — near 36°N 178°W, in the Line Islands near 6°N 159°W, near the Phoenix Islands at 2°S 175°W, and near 13°S 152°W — the stripes pass beneath the volcanic ridges, and

the ridges stand over them whole (B.5). A groove cut from above would cut the ridge, as a knife drawn across a tablecloth meets the book lying on it; by contrast, a fold made from the sides and from below passes beneath it, as a rucked cloth lifts the book whole. West of the ridges the outer stripes leave their parallels and turn toward each other — about 14° and 8° in oblique views with vertical exaggeration of 30 and 200, a first reading to be measured on a flat map (B.6). Beneath the base the lowermost part of the mantle is slow in all five models of the seismic vote maps, and the centre of the Pacific low-velocity province lies 1,480–1,660 km south-south-west of it (B.7). West of the tip the floor and its margins are crumpled in every direction — the Philippine and Mariana arcs, the arcs of Indonesia, and the hooked Banda arc — at 8,100 to 10,300 km from the base, about one pear-length (B.8). This is the back of the pear, which had no single pull of its own and was gathered back in no order, as a rug pushed into a heap crumples rather than lying flat.

Part C — The pear tip: the place where the Moon was formed. West of the base the northern stripe bends down at 23.8°N 164.9°E, and the southern stripe bends up at 5.3°S 173.2°E. The distance between the two bends is 3,360 km (3,040–3,700 km when each mark is moved by 15 pixels), between the 3,000-km contact on which the Moon sat at release and the 3,650-km dome above it. Their centre, 9.3°N 169.3°E, lies about 3,760 km west of the base (3,762 km by the coordinates; an earlier version gave 3,770 km), on the trailing side, where a pear that leaned back would bring its tip down. The base and the southern edge of the tip lie on the equator of the stripes, and the centre of the tip lies about 15° north of it (14.4°–16.4° for inclinations of 10°–14°). That is not the release latitude of 22.5°, which is the direction of the Moon's centre from the Earth's centre: these are two different quantities in different frames, and relating them belongs to the future work on the axes (C.1–C.2).

Inside this region the imprint of the near side is preserved as a *mirror image* — two faces pressed together exchange east and west and keep north, as a rubber stamp prints the reverse of its face — enlarged by $(3344/1300)^{1/3} = 1.37$, the factor by which every length on the Moon has shrunk since its density rose from 1,300 kg m^{-3} on the tip to 3,344 today (a second route, through the stages of the series, gives 1.31). With Imbrium kept at the author's mark, 13.19°N 179.04°E, the mirror rule places Serenitatis on the Marshall chain, 410 km from the centre of the tip, and the contact centre 1,260 km from it, both inside the footprint of radius about 1,680 km; Nectaris, far south on the Moon as well, falls just outside it, at 1,970 km (C.3). A first reading by eye had marked Imbrium, Serenitatis and Nectaris as a direct image; the requirement that two faces in contact leave a mirror image showed that the three marks could not all be right, and the rule of the mirror, applied from Imbrium alone, replaced it. Finally, the stripe system, from the place of the tip to the eastern end of Clipperton near America, is 9,360 km long, against a pear that reached at release 9,680–9,800 km from the Earth's surface to the corotation line (16,399 km, a day of 5.77 h): a wall laid flat covers about as much floor as it once stood tall, as a felled tree does, and the two lengths agree to within 4%, with nothing fitted between them (C.4).

One width, measured four ways	km
The contact zone on which the Moon sat at release (Paper 3)	3,000
The spacing of the spatula ridges, island pair to island pair (Part D)	2,800–3,500

One width, measured four ways	km
The width between the two bends of the stripes (C.1)	3,360
The dome above the contact zone (Paper 3)	3,650
The offset from the centre of the tip to the base (C.2)	3,760

How to read the table: the rows are ordered by size, and the shade deepens with it. Two rows come from the pear of Paper 3, and three from the measurements on the floor of the Pacific; all five converge within about three and a half thousand kilometres — the width of the place where the Moon sat.

Part D — The spatula ridges. On either side of the mirror imprint rise the large ridge systems of the central Pacific: the Marshall–Gilbert–Tuvalu chains on the western flank — with the Mid-Pacific Mountains and Wake in the more distant north-western sector — and the Line Islands chain on the eastern flank, which includes Palmyra and Kiritimati. The paper reads these structures as ridges drawn up from above at the moment of separation, by a mechanism like a plasterer pulling a flat spatula straight off a layer of fresh plaster: the soft medium clings to both surfaces, is stretched as the gap opens, and is left in ridges along the edges of the spatula. In the scenario discussed, the "spatula" is the tip of the extension on which the Moon sat, the "wall" is the floor, and the "plaster" is a viscous layer that separated them. The distance between the two chains lies in the range 2,800–3,500 km (Majuro–Kiritimati 3,490 km, Tarawa–Palmyra 2,810 km, and middle to middle 3,290 km); the western chain passes through the centre of the tip — Kwajalein lies 180 km from it — while the eastern chain carries the base itself within it. A ridge drawn up in a single moment should not show the steady age progression typical of a *plume track* — the line of volcanoes a fixed column of hot material leaves as the floor moves over it, like the scorch marks a candle leaves beneath a sheet of paper drawn across it. The data support this distinction: the Line Islands erupted in two episodes, 81–86 and 68–73 Ma, each at once over 1,200 to more than 4,000 km, and the literature attributes them to stretching of the outer shell, not to a source rising from below; the Marshall Islands chain spans 66–140 Ma with no simple chronological order; only Hawaii–Emperor shows a clean age progression, from 0 to about 81 Ma. A comparison between these chains and ridges outside the Pacific read as built from below — Ninetyeast (5,000 km long, from 77 Ma in the north to 43 Ma in the south), Walvis and Chagos–Laccadive — in terms of age, shape, flexural moat (the trough into which the shell bends under a load), crustal root (the thickened keel beneath a ridge) and orientation, is set out within Study 7.

Part E — The equator of then. Every stripe measured shows a rise northward toward the east: Mendocino +2.4°, Murray +12.7°, Clarion +9.0°, Clipperton +11.0°, and the southern stripe +11.7° (E.1). Lines of this kind are how the parallels of a tilted equator are laid out on today's map. When the stripes are fitted together, they emerge as parallels of a single equator inclined to today's by at least 10°: a line lying near a tilted equator rises at most at about the angle of the tilt, so a smaller inclination cannot produce Clipperton and the southern stripe; inclinations up to about 20° fit almost equally well, because the measurement points span only some 55° of longitude (E.2). For inclinations of 10°–14° the base lies on the equator of the system; the first fit, at 11°, crosses today's

equator at longitude 159°W, only 1.5° from the base. The stripes were laid down as the pear was gathered back, after the release (E.3). Until that stage a mass of nine to twelve lunar masses (derived in Paper 3 from the width of the base) stood high above one side of the Earth. When this mass was moved from high outside to deep inside, it changed the distribution of the Earth's mass about its axis, as a lump of putty pressed into a spinning ball forces the ball to roll slowly to a new balance. The stripes therefore record the equator at the time the pear was gathered back, and perhaps also the beginning of that turn. The question of how far and how fast the Earth turned is left to future work on the axes.

Part F — Challenges. This part answers three objections. First, the floor between the stripes is dated to at most about 180 million years. In the model the floor is a reservoir that gathers layers over time: a new layer is laid on top while the layer beneath is taken back and renewed from below, so dating methods measure the age of the last layer, not of the stripes — as dating fresh snow covering a ploughed field gives the date of the snowfall, not the age of the furrows that still show through it. The maps agree with this: on the topography the stripes are faint, while on the gravity gradient — which sees density contrasts beneath the surface — they are sharp, as remnants buried beneath new material would be. The requirement is that where drilling or seismic profiles cross a stripe, the fold continues below the youngest layers, and the layers above it are thinner over its crests (Study 9). Second, the fact that the stripes pass beneath the ridges does not by itself fix which structure came first; a ridge that already stood above a fold as it formed should show shortening where the fold passes (Study 6). Third, the Hawaiian–Emperor chain shows a clean age progression; if it is a spatula ridge, the dated lavas must record a later sequence of eruptions on a ridge that already stood, and Hawaii remains the open case among the three chains.

Part G — Predictions, tests and the studies to come. This part sets out nine studies, each with the result it expects: (1) the equator refitted on the full traces of the stripes — a common equator that passes through the base; (2) a test of chance for the position of the base within $\pm 2°$ of the equator; (3) one side only — stripes that do not continue west of the base; (4) steps in depth and age across each stripe, ordered about the base; (5) the east–west asymmetry of the Pacific about the base; (6) continuity beneath the ridges on a flat map without vertical exaggeration — stripes unbroken by the ridges, and ridges disturbed where a fold passed beneath them; (7) the spatula ridges against the three ridges built from below — a distinction between them and plume tracks; (8) circular features at the sites that mirror the positions of Serenitatis, Nectaris, the contact centre and the other near-side maria, at the scale 1.37; (9) the stripes beneath the dated layers in seismic profiles and drill holes — continuing below the youngest layer, with layers whose thickness changes across the fold.

Part H — Conclusions. The traffic light of the paper shows five green rows — the base on the equator of the stripes, the width of the tip, the total length of the stripes, the near side inside the bounds of the tip as a mirror image, and the continuity of the stripes beneath the ridges at four crossings — alongside four orange rows waiting on the studies: the trailing side, the convergence west of the ridges, the spatula ridges and the age of the sea floor. When these components are combined into one picture, it emerges that a pear of the Earth's own material stood on the Pacific, its base near Kiritimati and its tip some 3,760 km to the west of it, with the Moon on that tip; after the

Moon's detachment, as the pear was gathered back, its leading wall was laid eastward toward America as some 9,360 km of stripes, its back crumpled into the island arcs of the western Pacific, the tip drew up the ridge chains on either side of it, and the face of the Moon left its mirror image on the floor at the scale of 1.37. The stripes record the equator of that period, tilted by at least 10° relative to today's equator.

C.10 Paper 10 — Mars

> **THE CLAIM**
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> The moons of Mars were made of Mars's own material, not gathered from debris. A young Mars, 1.5 times its present radius and turning on its axis once in about five hours, fired two plugs of its own crust from a single volcanic conduit at Pavonis Mons, pushed by trapped gas. A ring that Mars had built in its early stages from its own steam gave each plug the sideways speed it lacked, rounded its orbit and drew it into the equatorial plane; the same friction also ground it. The tide of the soft young Mars then pushed both moons outward. Deimos, fired first and long in the ring, passed the final corotation line and keeps receding, smooth; Phobos, the second, braked, met only the thin edge of a waning ring, kept its grooves, was caught by the moving line, and has been falling since.

Part I — What happened. Phobos and Deimos are small bodies (27 and 15 km across), dark and porous, on near-circular orbits above the equator, on the two sides of the corotation radius of Mars, 20,428 km. Deimos shows a smooth, rounded surface, while Phobos is scored by long grooves. Capture does not give equatorial circular orbits; an impact disc, the leading account, requires a mixing of Mars with the impactor; and in both accounts the surface features need separate causes. The paper applies the founding idea of the series — a body made of its parent's interior material, given its speed by the parent's rotation — but follows it on Mars by another route. Mars built no pear and no closed dome. A young Mars, of radius 5,084 km and turning in five hours, gave all material leaving its equator a free sideways speed of 1.78 km s^{-1}, and held it down with only a quarter of today's gravity, 1.04 m s^{-2}; its corotation line stood at a height of 7,057 km, less than 2,000 km above the surface. Beneath a sealed crust the planet held water and gas. A soft, warm plug sealed a volcanic conduit a few kilometres below the vent, and gas trapped in a deep chamber at 0.25–0.5 GPa pushed it along a path of 200–300 km, the walls supported by the weight of the material around them. A free volcanic blast is limited to about 1 km s^{-1}; a confined push of this kind gives 2.4–2.9 km s^{-1}. This single conduit fired twice: the first plug, Deimos, small, light and wet, left cleanly; the second plug, Phobos, larger and harder, dragged along the walls and was braked.

Shot	Top height, km	Outward speed, km s^{-1}	Sideways speed at top	Orbital speed there
Phobos, braked, lower	10,000	2.44	0.90	2.07

Shot	Top height, km	Outward speed, km s^{-1}	Sideways speed at top	Orbital speed there
Phobos, braked	11,000	2.57	0.82	1.97
Deimos, to the ring	15,000	2.89	0.60	1.69
Deimos, straight to its orbit	23,463	3.19	0.38	1.35

Read from top to bottom: the higher the shot, the greater the push needed from the gas (shaded, deeper with speed), and the further the sideways speed at the top (km s^{-1}) falls short of the speed needed for an orbit there. Each plug needs only to reach the ring; the ring and the tide do the rest.

A body fired from the surface reaches the height of the ring but not its orbit: without further intervention it falls back within one revolution. The missing speed was supplied by a ring that Mars had built in its early stages from the steam released from every fissure. Near the hot surface the steam stayed a clear gas; higher up it froze into ice crystals and dust particles circling at orbital speed, and beyond about 13,000 km — the Roche limit for ice — they gathered into clumps. A body crossing the ring is dragged forward toward its flow speed: it gains its orbit, rounded and equatorial, and the same friction grinds its surface. The ring must give at least about 0.6 of each body's mass in perfect coupling, and in practice of order 10^{16} kg; the account stands or falls with it. Beyond the line, the tide of the soft, fast young Mars, dissipating 8 to 20 times more energy than today, pushed each moon outward. Deimos was carried from about 15,000 to 23,463 km over half a billion to a billion years, passed 20,428 km before the line arrived there, and was ground smooth, mostly in the clumpy outer ring at the end of its climb. Phobos, by contrast, reached only the thin, gaseous edge of a waning ring: it gained enough speed not to fall, was etched by gas and not smoothed by water, and receded more slowly than the line, which moved outward as the spin of Mars slowed. The line caught it at about 19,000 km, when the day was about 22 hours long, and it has been falling since; running its fall backward at today's dissipation rate places it at 19,100–19,500 km 3.9–4.5 Gyr ago. The inner ring fell back as dust; the rest of the material evaporated. The mass that left through the line, of order 5–20% of the starting mass and mostly volatiles, took about 92% of the spin with it, and Mars slowed and contracted to its present state. The water that did not leave stayed sealed beneath the crust, where InSight detected it.

Part II — The evidence. Pavonis Mons, located at 1.48°N and the closest of the four Tharsis volcanoes to the equator, has at its summit a single circular collapse pit 47 km wide and nearly 5 km deep — a steep-walled barrel with a depth-to-width ratio of about 0.10. Its volume, 8,300–8,700 km^3, is of the scale of the two moons together, 6,740 km^3 — the roof of a chamber that collapsed after the shots. For comparison, Olympus, located at 18.65°N, would have produced a moon on an initial orbit tilted by about 19°. Deimos is marked by a smooth, rounded surface and shallow hollows (median depth about 2% of width, against 5.5% on Phobos); it left larger than it is now, having lost 30–60% of its mass, but not larger than Phobos. Phobos lost only about 1%: its grooves are the scrapes of the conduit, its wrinkles reflect the contraction of a cooling body, and Stickney is the mark of a solid body that stood in the ring and struck Phobos while it was warm. Both moons show a

sharp end and a blunt end, and the blunt end faces Mars: the tip was formed by the flow from behind, and tidal locking turned them at a later stage. The hollows on both bodies are folded inward, evidence that they were struck while the bodies were warm and plastic; the moons are therefore samples of a young Mars soaked with water beneath its surface. The Earth's Moon, by contrast, never passed through a ring, and so kept its craters.

Part III — The predictions. Seven predictions give a value, an instrument and a date, against what the accepted account expects at the same place: Phobos material on the Martian line in oxygen, chromium and titanium, with no impactor component, no evaporation shift in zinc and potassium, and with closed vesicles of a cooled foam; pits that never cross grooves, and sharp groove rims; Deimos's shallow hollows and its composition, identical to that of Phobos; a mass deficit beneath the Pavonis caldera; the centre of mass of Phobos displaced toward Mars, toward its blunt end; a water-saturated mid-crust beneath Tharsis; and in the Martian dust, an ancient component, older than 4 Gyr, with grains carrying solar-wind helium and neon, whose abundance peaks in the equatorial band. The Martian Moons eXploration mission of JAXA, launched in October 2026, is expected to land on Phobos in 2027 and to return its samples in 2031; a dedicated appendix lists fourteen proposed measurements, mission by mission. The account requires a fast, expanded young Mars, a soft Mars, the confined gas push — its critical step — and the ring.

C.11 Paper 11 — The Requirements

> **THE CLAIM**
>
> A Moon built this way must show particular properties, and it does show them. Paper 11 gathers eighteen requirements — seven classical properties, the three exposure zones, seven requirements that follow from the pear and three added in the wake of the later papers — each with the quantity that tests it, and reports them one by one rather than adding them into a combined score. The strongest of them, the ordering of the crust and the old craters about the axis of the old contact zone, was run against the data with thresholds fixed in advance: two of its tests passed; in the third, the crust is thinnest in the contact zone and thickest on the far side in all four crustal models, as predicted, and the contrast, about 10 km, is a number for calibration.

Part A — The seven requirements and their assumptions. A *requirement* is a property the Moon or the Earth must have if the Moon formed by this route; to each requirement the paper attaches the physical assumption needed to derive it, and so links each test to the part of the model it concerns. R1, affinity to terrestrial material: the Moon's isotope ratios, a label of origin that later processes barely move, must be reachable as one mixture of the two terrestrial sources defined in Paper 3 — the deep feed and the rising sialic shell. R2, a small metallic core: all the Moon's metal is terrestrial; significant volumes of iron entered the pear, the concentration decreased toward the tip, and the feed material that reached the Moon was mainly silicate with a small metallic component. The core is small because most of the core material returned to the Earth: at detachment the heavy core

material lay in the last third of the body, did not cross the corotation line, and went back down the pear. For a core of 1.4% of the mass and a deep share of 0.95, the metal that reached the centre must have been about 1.47% of the fed material, so that “little metal” becomes a number Paper 3 must explain. R3, asymmetry linked to the aperture: the far hemisphere carries a thick crust made of many layers of compacted sial, which migrated to the tip during the bulge under the forces of the spin, together with material that returned to it again and again; the near hemisphere served as the base. R4, orientation: the connection side faces the Earth, as two dancers holding hands face each other while both turn relative to the room. R5, an order of preservation: the far hemisphere kept its record before the fed side quieted, and its maria fell still when cooling ended their internal pressure. R6, volatiles in the amount and phase the feed needed, at each depth. R7, the centre of mass: a thickening of light crust on one side moves the balance point toward the opposite side, as a ball carrying a thick patch of cork on one side balances off-centre. Paper 10 serves as a separate test; its outcome does not change the parameters of the Earth–Moon system.

Part B — The requirements against the measurements. Each requirement is set directly against its measurement, and the terms “consistent”, “open” and “failed” are used in their strict sense. The Earth–Moon oxygen isotope difference is $\Delta^{17}\mathrm{O} = -1 \pm 5$ ppm: for LEF this is an expected result, not a finding that must be explained; the overall multi-system balance is set as an open question (B.1). The seismic core reaches a radius of only about 330 km, about 1.4% of the mass (B.2). GRAIL data show that the far-side crust is thicker; the mass of the shell on the dome of Paper 3, 2.85×10^{21} kg (3.9% of the lunar mass), matches the mass of the far-side crust in a far hemisphere 55 km thick, 2.6×10^{21} kg, to within about 10%, with no parameter adjusted between them; while the near-side crust — the sial of the base zone peeled away with the body, about 1.5×10^{21} kg — is a separate term (B.3). Present synchronism is a weak test (B.4). Four clocks — crystallisation, the literature’s impact age, basaltic fill and exposure — may be compared only within one kind; just as the firing of the bricks, a later fire, a coat of paint and the weathering of the facade are four true dates of one house (B.5). Melt inclusions hold 615–1,410 ppm of water, sealed in when they formed: the Moon was wet then, when enormous amounts of volatiles were in and around its material; these have since escaped to space, and the interior measured today is dry. The amount the feed needed is still to be derived (B.6). A crust of the modelled shape displaces the centre of mass toward the near side by 1.24–3.11 km, for density contrasts of 300–750 kg m^{-3}, against about 2 km observed — a sensitivity range, not a prediction (B.7). Then R8, the exposure zones (B.8): the contact was exposed last, the ring of the base during the rounding, and the dome’s surface from the start, so the Moon is ordered in three zones about the near side. Five tests were fixed before the data were opened, and the maria are not masked, since they are part of the prediction. The tests were run in September 2026 on the four GRAIL crustal-thickness models and on the Robbins crater catalogue, about the sub-Earth axis and with the boundaries defined at that time: the contact disc (0–41°), the limb belt (41–90°) and the far side (90–180°). T5, the thickness trend in twelve belts of 15°: a rank correlation of $+0.87$ on all four models, from 26.8 km at 30–45° to 45.6 km at 165–180°, monotonic over ten consecutive belts, against -0.43 and $+0.43$ about the poles — the predicted axis orders the crust better than any control axis; *passes*. T2, craters of 20 km and more: 99, 164 and 219 per 10^6 km^2 in the three zones, with ratios of 1.34, 2.21 and 1.65, each above its pass threshold; *passes*, and the order survives the conventional mare mask. T1, the contrast between the far side and the contact disc: 9.65, 10.0, 10.3 and 10.6 km on Models 1–4, and the ordering disc

$<$ belt $<$ far side holds in all four; the prediction is the ordering, and the size of the contrast, about 10 km, is a number for calibration. The other conditions fixed in advance — far/limb at least 1.1, a monotonic order, and no control axis with a sharper trend — were met. Recomputed on the current geometry — the contact centre at 24.1°N 335.2°E, the contact cap 0–31°, the belt 31–70° and the far part 70–180°, as the translation rule of Paper 3 places them — the crater densities, with the maria and South Pole–Aitken removed on both sides, are 53, 187 and 220 per 10^6 km^2 (far/belt 1.17, far/cap 4.15, belt/cap 3.55), and the crust of Model 1 is 26.3, 30.3 and 36.2 km with the maria kept (far minus cap 9.9 km), and 31.3, 33.0 and 38.9 km with the maria and South Pole–Aitken removed (7.5 km): the ordering holds on both counts. Because a thick far side, an offset centre of mass and maria on the thin side are different expressions of one distribution of material, the rows are not added into a score (B.9).

Part C — Further signatures of the rise and the connection. This part gathers signatures that only this route predicts. The rigid shell carried its craters whole and kept their size — the median crater diameter at the poles is 1.06 times that of the equatorial belt — while the rounding crowded them, by a ratio that rises with size, from 1.09 at 20–30 km to 1.53 at 60–100 km (C.1). On the Earth, the source region is the base of the pear near Kiritimati, and the tip lies at 9.3°N 169.3°E (Paper 9); on the Moon, the hill from which the body was built is centred near 16.6°N 13.1°E, and the last contact at 24.1°N 335.2°E, with Imbrium the main exit and Serenitatis the secondary — close to the latitude of 22.5° at which, according to Paper 4, the Moon left (C.2). The openings fall into three groups — on the crown, on the flank and under the base — by how long each of them was open to space (C.3). The body completed its rounding while still attached, and left as a round body with a rigid shell, by a process of peeling: over the 6,000-km base the sial slid up the pear toward the tip, in the direction of outward sorting; this motion was eased under shear by the lubricated layer between sial and sima, and the sial parted from the sima as tape lifted from a painted wall takes the paint with it and leaves the plaster; over the 3,000-km contact the heat melted the sial away; and slightly after the body's centre of mass reached the corotation line, about 210 km beyond it, the last of the contact was pulled free with force. There was no neck (C.4). A link between the feeding lobe and the present deep structure remains a research direction (C.5). The pear adds seven requirements (C.6): R9, a pear about 9,000 km wide lying inside the corotation line, which is part of the Earth and held by the Earth's gravity like any other part of it, and is not thrown out because its speed is not a throwing speed and because it has not crossed the corotation line; R10, nine to twelve lunar masses, derived in Paper 3 from the width of the base, gathered back whole into the Earth, with nothing left in orbit; R11, the base located above the Pacific low-velocity province, which we propose is largely the returned pear — part of the sorted material that went back into the Earth; R12, the thorium of the contact, brought from the Earth's depths; R13, the core as a gauge of how close the feed came to the descending iron; R14, no foreign material beyond tiny amounts; R15, polar crowding by size — measured and passed. The papers closed after the pear add three requirements (C.7): R16, the 34 maria as the openings and the source of the returning material, supported by a density of 0.8 sharp craters of 20 km and more per million km^2 on mare floors, against 18.3 on the plains and 207 in the highlands (Paper 6); R17, two exits beneath South Pole–Aitken, with the dome located south of both, at 17–21° on the flattened figure and 24–29° on today's sphere (Paper 8); R18, the Tharsis source lay on the Martian equator when Deimos left (Paper 10).

Part D — Consistency among the papers and conclusions. A summary table lists what each paper hands to this one and the condition it still leaves open: corotation does not by itself prove release (Paper 2); the radial balance holds only for a passage of about 1,000 km radius holding 0.46 lunar masses under a sideways load of order 10^{22} N, so the connection zone is the broad base of the dome and not a stem (Paper 3); the shell's viscosity, $\gtrsim 10^{23}$ Pa s, is required, not measured (D.1). The chronology is a central constraint: the hafnium–tungsten clock, with a half-life of 8.9 million years, leaves about 5.1×10^{-21} of its parent isotope after the 600 Myr of the series' time line — some 400 for the bulge and the pear, about 100 for the flattening stage and about 100 more to release at 3.873 Ga — so the tungsten signatures were set early, and they must be carried through the whole window without resetting; a consistent reservoir history is set to be computed (D.2). A combined map brings together every claim with its test and with the branch of the model it rests on (D.3). The novelty lies not in any single process but in the combination: an interior fed from below while its skin was the Earth's own shell lifted on the dome (D.4). Seven conclusions follow, each with its meaning; R8's T5 and T2, as well as R15, R16 and R17, have passed or are supported (D.5).

Appendix A — Reproduction of the centre-of-mass calculation. The appendix sets out the short integration behind the offset: the symmetric part of the Moon contributes nothing, and only the denser mantle, which reaches higher beneath the thin crust, moves the balance point. For $M_L = 7.35 \times 10^{22}$ kg, $R_L = 1,737.4$ km, a mean crust of 45 km and an amplitude of 15 km, the offset is the amplitude multiplied by the ratio of the density contrast to the mean lunar density. The appendix also shows that the contact centre, 24.1°N 335.2°E, lies 34.0° from the sub-Earth point, so the computed offset need not point straight along the mean Earth line, and the test compares a direction and not only a length.

Appendix B — Map of prior research and novelty. This appendix classes each component of the work as standard physics, prior observation, prior art, model hypothesis, this work or open prediction. The measurements of oxygen, core, crust, basalt and water are assigned to the cited studies; the idea that a hot Earth heated the near side is attributed to Roy and colleagues, and the reading of Procellarum as a magmatic system to Andrews-Hanna and colleagues; the contribution of the series is the simultaneous internal feed and rise of the outer shell, and the joint test of thickness, mass and offset in one model.

What Part C establishes. Read in order, the ten papers set out one process that leaves a measured trace at each step. Papers 2 and 3 show that a body built from the Earth's own material on the tip of a pear was released at the point where the motion it already had was the motion of an orbit, and Paper 4 carries it from 22.5° and 17,496 km to today's orbit. Papers 5 to 8 read the Moon as the record of that construction: the maria as openings, the craters as returning material that landed and was carried on a rigid shell, the poles as crowded by the rounding, and South Pole–Aitken as the mark of the burner on the far side. Paper 9 finds the other end of the same event on the Earth — the pear's leading wall folded into the stripes, and the place of the tip between their bends — and Paper 10 follows the same idea on Mars by another route — two plugs of crust fired from a single conduit, their orbits completed by a ring — and explains two small moons and the reason one of them falls. Paper 11 gathers what all of this requires and states the test of each

requirement; the key test, the ordering of the crust and the old craters about the old contact zone, has been run: two of its tests have passed, the ordering holds in every model, and the size of the crustal-thickness contrast, about 10 km, is left to calibration. Part D now gathers the series' traffic light.

PART D

The traffic light of the series

THE PROBLEM AND THE APPROACH

Ten papers have each measured or computed, separately, their own part of the process, and each of them has judged its results independently. A reader who wants to know where the model stands as a whole should not have to open ten papers. **The approach:** every study and observation that has been given a status in any paper of the series is gathered in one table, with its number and its light; then the data the series used are listed, so that the reader can see what was measured by the series itself, and on what data.

While Part C presented the papers one by one, this Part looks at them all together. It does not judge anything anew: each light below is the light the paper itself gave its result, or — in Papers 2–4, which compute rather than measure — the status the paper itself states for each claim. The purpose is a single view — the whole series in one table — and an honest count of what has been measured and fits, what fits in part or waits for a study, and what was cancelled.

D.1 One table: every study and observation, with its light

The three colours. *Green* means that the hypothesis of the study, stated before the measurement, held in full on the measured sample — or, in Papers 2–4, that the calculation closes on the stated inputs. A green row may still name a limit (such as a small sample, or a sub-test not yet beyond chance), provided that this limit does not contradict the hypothesis. *Orange* means that one part of the result held while another has not yet separated, or that the result is stated as a requirement and waits for the calculation or the measurement that will decide it. *Grey* marks a study that was cancelled; its number is kept so that the continuity of the paper's numbering is not disturbed. No study of the series is red. T1 of Paper 11 — in which the predicted ordering holds in all four GRAIL models, while the size of the crustal contrast, about 10 km, is left to calibration — is marked orange, as a result in which one part held and another waits for calibration.

How the rows were taken. Papers 5, 6, 8 and 9 each close with a traffic-light table of their own (Paper 5, §M.2; Paper 6, §O.2; Paper 8, §G.1 and the verdicts of §B18; Paper 9, Part H), and their

lights were copied unchanged. Paper 7 gives a result for each of its five tests (§D); “passes” is copied as green, and “consistent; not yet decisive” as orange. Paper 10 grades the links of its account on a feasibility map (§G.5); links that were computed and hold are marked green, and links computed on a stated input, stated as a requirement or waiting for data are marked orange. Paper 11 gives the state of each requirement (§B.9) and the results of the exposure-zone tests (§B.8). Papers 2, 3 and 4 state the status of each claim in their summary chapters and test tables (Paper 2, §C.1 and §D.1; Paper 3, §D.1 and §E.4; Paper 4, Part E). The number in the third column is the value the paper reports; the section where it is set out in detail is named in that paper’s own table.

fits measured or computed, and fits partial / awaiting not yet decided cancelled

Paper	Subject of the study or observation	Finding	Status
Paper 2 — LEF Mechanism in Principle `7 green · 1 orange`			
2	Closing the velocity gap	At the corotation radius the tangential speed of rotation equals the orbital speed: 15,818 km, 5,020 m/s for a parent body with a 5.5-h day (at surface level the gap is 2,133 against 7,701 m/s)	fits
2	The bridging by the pear structure	A continuous pear structure bridges the Earth to the corotation line; the Moon is released whole, in one piece	fits
2	Length of the day at the moment of release	Combined effect of the pear and of the sinking iron: a day of 5.77 h at release (had the pear acted alone, 6.44 h)	fits
2	Corotation at release	The Moon’s centre of mass lies on the line: 16,399 km (two-body calculation; 16,331 km in the test-particle approximation)	fits
2	The angular-momentum budget	The system’s angular momentum closes at 1.007 times today’s value (an earlier version gave 1.008); for comparison, the classical route needs 2.74 times	fits
2	The Roche limit	A rigid, round figure: axis ratio ≤ 1.04 for $\mu \geq 10$ GPa; the release radius (16,400 km) lies outside the rigid Roche limit (9,484–12,994 km)	fits
2	The start of tidal evolution	The Moon recedes from the parent body from the moment of release; a margin of $\Delta r \simeq 2d_0$ beyond the line	fits
2	Core centralisation	$D \approx 2$ during the mass transfer — a system requirement: the Earth differentiates in parallel with the transfer, not after release	awaiting

Paper	Subject of the study or observation	Finding	Status
Paper 3 — LEF Mechanism in Practice `4 green · 3 orange`			
3	The hydrostatic pressure test	A pressure of 150.26 GPa at the source; a deep melt of density below about 3,864 kg/m³ reaches the corotation line (a density of 3,500 kg/m³ leaves an excess of 10.49 GPa)	fits
3	The geometry of the body on the tip	A hemisphere on a 6,000-km base, holding the lunar mass, has a density of 1,300 kg/m³ without adjustment; a dome height of 3,650 km; a surface three times that of the far hemisphere	fits
3	The release distance from the line	A viscous contact releases the body at about $2d_0$ beyond the line: about 210 km for a contact layer about 105 km thick (Paper 3, §C.5a)	fits
3	Anchoring the base over the Pacific province	In all 5 tomographic models examined the mantle is slow beneath Kiritimati; the site lies inside the -1% contour of SMEAN (-2.25%) and of SMEAN2 (-2.36%)	fits
3	The joint test of pressures, flows and loads	The balance of radial forces is reached only above a passage of about 1,000 km, already holding 0.46 lunar masses under a lateral load of order 10^{22} N: a thin-pipe model fails, and the broad base is required	partial
3	The two deep mantle provinces	Comparison of the composition, shape and plume isotopic signatures of the Pacific province against the African — set as a measurement programme	awaiting
3	The requirements handed on	Three-dimensional stability, an equation of state for the deep material, energy balance and the time of failure of the contact layer — set as a calculation programme	awaiting
Paper 4 — The Orbit `3 green · 4 orange`			
4	Source of the orbital inclination	$i_0 = \lvert\lambda\rvert$: release at latitude 22.5° gives an orbit inclined 22.5° to the equatorial plane of that time	fits
4	The initial orbit	Entry into a nearly circular orbit on release at latitude 22.5° and at 17,496 km, about 210 km beyond the line (equations 6–7)	fits
4	The initial recession	$\langle a \rangle > 0$ from the first revolution	fits

Paper	Subject of the study or observation	Finding	Status
4	Survivability	About 7,200 km outside the rigid Roche limit of 10,314 km — the Roche threshold test passed; the deformation check is set as an open question	partial
4	The present inclination	The reduced decay law leads to 5.145° at today's distance, equal to the observed value — full validation requires the full integration	partial
4	Distance and the orbital clock	About 384,400 km and a rate of 3.8 cm/yr today; no constant value of Q reconciles both measures together ($Q_{\oplus} \simeq 12$ recovers the rate, $Q_{\oplus} \simeq 29$ the 3.87 Gyr since release) — requires integration	awaiting
4	The geographic link	Independent reconstruction of the ancient latitude of the centre of mass, separate from the orbit	awaiting
Paper 5 — The Maria `13 green · 2 orange · 1 cancelled`			
5	1. Anchoring of the maria over the feed domes	33 of 34 maria lie over a dome; none of 20,000 random sets of 34 points reached 33	fits
5	2. The material-source test	The craters carry the iron signature of their ground, falling with distance: from 6.9 to 3.1 wt% FeO within 150 km, against 2.3 in the control regions	fits
5	3. Basin or crater: the crustal jump	The crustal jump is identified under 29 of 36 marked openings, against only 4 of 169 ordinary craters	fits
5	4. The buried domes	44 of 73 domes buried; the iron and thorium concentrations above them equal those of the surrounding ground; the crust above them thicker by 8–13 km	fits
5	5. Subsidence of the centre over the dome	The centre is sunken in 10 of 12 open maria (median -267 m; $\rho = -0.59$); the exceptions are Serenitatis ($+60$ m) and Imbrium ($+43$ m), the two maria of the contact zone: Serenitatis froze first, very close to the Earth, and was drawn outward more; Imbrium froze after detachment, as the Moon receded	fits
5	6. One floor level	All 16 floors at one uniform level, sunk below the rim (median 3.4 km) and below the surrounding ground (median 1.6 km); chance probability 1 in 65,000	fits

Paper	Subject of the study or observation	Finding	Status
5	7. Cleanness of the floors	The density of 2–5 km craters on the floors is only about a twelfth of that of the surrounding ground; every floor is cleaner in every size range	fits
5	8. Size and state	Craters erased and melted into the floors; the share of melted craters rises with size; small craters keep a sharp outline in the contact zone (shared with Paper 6, Study 5)	fits
5	9. The rings of iron and titanium	The concentration excess of the maria extends to 600 km	fits
5	10. Thorium and the contact zone	The contact zone, 17.5% of the surface, gathers within it 95% of the thorium above 3 ppm; its centre, fixed in advance, ranks third of 16,200 centres, 47 km from the leader	fits
5	11. The foreign-mass balance	Reading the maria as basins of outside bodies requires about 1×10^{19} kg of foreign material; the melts carry no known foreign material	fits
5	12. The order of freezing	A consistent order at every size from 2 km; the maria of the contact zone are the cleanest	fits
5	13. Thorium and titanium	The thorium component held: the floors of the contact zone are a third poorer than their plains; the titanium component did not separate the zones	partial
5	14. —	Cancelled; identification number kept	cancelled
5	15. The plains of the contact zone	Rim lost, and the surviving crest blunted, on the side facing the hot plain — only in the contact zone; matched crest comparison on four pairs	fits
5	16. The collar of the far maria	Elevation above the control around the far maria, and most of all around the buried ones; around the 10 open far maria alone (median 207 m) the signal did not separate	partial
Paper 6 — The Craters 11 green · 3 orange · 1 cancelled			
6	1. Crater density by longitude	249–252 craters per 10^6 km² at 90°E, against 139–175 at 90°W; ratio 1.79 (1.56–2.05)	fits

Paper	Subject of the study or observation	Finding	Status
6	2. The material-source test	FeO of 5.96 wt% within 50 km of the six maria, against 2.30 in 5,922 control regions; the excess persists to 600 km	fits
6	3. The buried domes and the large craters	No local excess over the buried domes; the large craters come from the visible maria and the buried domes alike	fits
6	4. Craters melted into the floors	49 of 79 craters of 10 km and more melted, the share rising toward the contact zone (chance about 3 in a million); not one crater of 20 km and more survived on the 703,000 km² of the contact floors	fits
6	5. Size and state on the maria	0.8 sharp craters ≥ 20 km per 10^6 km² on the floors, against 18.3 on the plains and 207 in the highlands	fits
6	6. Rim, floor and peak by longitude	Evidence of hard landing at 90°W (toothed rims 1.14–1.25 times those at 90°E); some intervals of the larger rims include zero	partial
6	7. The inside of the crater	One material from the centre to the rim; in 1,214–1,235 pairs, every confidence interval between the halves includes zero	fits
6	8. The foreign-mass balance	The visible craters require 4.4–5.0×10^{19} kg of foreign mass; the crust actually holds 0.9–2% foreign material, against 2–6% required (an earlier version gave 2–5%); the ruthenium is terrestrial in its signature	fits
6	9. The Earth as the last scatterer	An excess of the last small craters on the near side, 1.12–1.27, which persists after the secondaries are removed	fits
6	10. The raised floors	About a third of the large floors are raised above the basalt outside them; 84 of 86 lie on the plains	fits
6	11. Iron in the high and low floors	The low floors are richer in iron; against the basalt just outside them, -0.44 against -0.64 wt%, the gap does not go beyond chance	partial
6	12. The material after release	The ejecta are poorer in iron than the basalt; among craters within the basalt, the values fall inside the scatter of the maps	partial

Paper	Subject of the study or observation	Finding	Status
6	13. —	Cancelled; identification number kept	cancelled
6	14. The mass under the melted craters	A mass excess relative to their own floor beneath 13 of 16 (Wilcoxon, $p = 0.04$); no such excess is identified beneath the plains	fits
6	15. The rim: worn by heat or flooded	The rim lost on the side facing the hot plain — only in the contact zone; crest comparison on four pairs	fits
Paper 7 — The Poles `4 green · 1 orange`			
7	D.1 The crater-density ratio	1.20 ± 0.05 on the clean count (maria and SPA removed on both sides); the ratio rises with diameter, from 1.09 at 20–30 km to 1.53 at 60–100 km	fits
7	D.2 Preservation of diameters	Polar median of 1.06 (1.01–1.12) under the same threshold: the craters kept their size	fits
7	D.3 The young craters	0.92 (0.39–2.04) and 1.12 (0.60–2.03): the results are consistent, but have not yet crossed the decision threshold	partial
7	D.4 South against north	South/north ratio of 1.02 (0.78–1.32) at 15°	fits
7	D.5 Crater size	Largest per unit area 1.16 (1.04–1.29); old/fresh 1.02 (0.89–1.16); 4 of 5 size tests passed	fits
Paper 8 — South Pole–Aitken `6 green · 1 orange`			
8	1. The crater record	Density in the central ring only 0.51 of the density outside, in a like-for-like comparison; the minimum at 45.4°S 197.9°E, on the meridian opposite Serenitatis	fits
8	2. The depth against the thin crust	A depth remainder of 1.1–1.8 km at the centre, falling to 0.2–0.3 km at the basin's rim	fits
8	3. The outer ring	Thickened crust in 7 of 8 directions, thickest to the east; the ring holds about 37% of the missing crust	fits
8	4. The inner ring	Present in 22 of 24 directions; its centre at 51.4°S 196.8°E, 118 km east-north-east of the basin's centre	fits

Paper	Subject of the study or observation	Finding	Status
8	5. The porosity at depth	At a depth of 10 km: 2.9–4.4% at the centre, 7.1–8.8% at the rim, and 6.2–12.8% in the surroundings	fits
8	The anomalies B1–B17	The model offers a fit to 16 of the 17 anomalies (accepted science: 6 fully, 9 in part, 1 with difficulty, 1 not at all)	fits
8	B14 Metal in the norites	Ni/Co ratios and iron–nickel grains; ages of 4.247 and 3.873 Ga — partial fit; the decision is passed to the isotopes	partial
Paper 9 — The Stripes `5 green · 4 orange`			
9	Position of the base on the equator of the stripes	An offset of only 1.6° from it, for an inclination of 10°–14°	fits
9	The width of the tip	3,360 km, against the 3,000-km contact zone and the 3,650-km dome	fits
9	The total length of the stripes	9,360 km from the tip to the eastern end, against a pear length of 9,680–9,800 km	fits
9	The imprint of the near side on the Earth	Mirror image enlarged by $\times 1.37$ with Imbrium in place; Serenitatis and the centre of the contact zone lie inside the tip	fits
9	The stripes beneath the ridges	The stripes pass beneath the ridges at all four crossings observed	fits
9	The trailing side	Crumpled arcs west of the tip, 8,100–10,300 km from the base	partial
9	Convergence of the stripes west of the ridges	First reading on oblique views; the quantitative measurement is yet to be made	awaiting
9	The spatula ridges	Line and Marshall without plume age progression; the comparison with plume ridges is yet to be made	awaiting
9	The age of the sea floor	The dated surface is the last layer; the continuity of the fold beneath it is to be tested in seismic profiles	awaiting

Paper	Subject of the study or observation	Finding	Status
Paper 10 — Mars `4 green · 6 orange`			
10	The velocity budget	The equations of energy and angular momentum: 2.44–2.89 km/s outward to reach the ring, from a speed of 1.78 km/s at the equator	fits
10	The equatorial site	Pavonis at 1.48°N; both moons within 2° of the equatorial plane, whereas Olympus would have given an inclination of about 19°	fits
10	The scale of the pit	Volume of the Pavonis caldera 8,300–8,700 km³, against the volume of the two moons together, 6,740 km³	fits
10	The two faces	Ratio of hollow depth to width in the shape models: median of 1.9% on Deimos, against 5.5% on Phobos	fits
10	The trapped-gas push	A push at pressures of 0.25–0.5 GPa along 200–300 km, against a confining pressure of 0.26–0.39 GPa; the critical step, passed on to modelling	partial
10	The mass of the ring	At least 0.6 of each body's mass in perfect coupling, and of order 10^{16} kg in practice; a requirement	partial
10	The climb of Deimos	From 15,000 to 23,463 km over 0.5–1 Gyr, for a Mars dissipating 8–20 times today's energy	partial
10	The race of Phobos	Caught by the moving corotation line at about 19,000 km (day about 22 h); running backward at today's dissipation places it at 19,100–19,500 km 3.9–4.5 Gyr ago	partial
10	The spin and mass lost	About 92% of the spin was lost, together with 5–20% of the starting mass, through the line; a requirement	awaiting
10	Material, centre of mass and dust	Seven predictions; awaiting the samples of the MMX mission (2031) and the returned Martian regolith	awaiting
Paper 11 — The Requirements `3 green · 5 orange`			
11	R8, T5: axis specificity	Spearman coefficient $+0.87$ ($p = 2 \times 10^{-4}$) about the sub-Earth axis in all four GRAIL models; the control axes give -0.43 to $+0.76$	fits
11	R8, T2: oldest craters	Density ratios: far/limb 1.34, far/disc 2.21, limb/disc 1.65 (Robbins, 6,973 craters ≥ 20 km)	fits

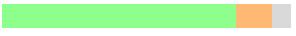
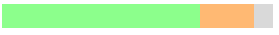
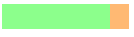
Paper	Subject of the study or observation	Finding	Status
11	The crust balance	Shell mass 2.85×10^{21} kg against far-side crust mass 2.6×10^{21} kg at a thickness of 55 km: about 10%	fits
11	R8, T1: crustal thickness	Far-minus-disc gap of 9.65 / 10.0 / 10.3 / 10.6 km in Models 1–4; the ordering holds in all four; the size of the contrast is a number for calibration. On the current geometry (centre of the contact zone 24.1°N 335.2°E; ranges 0–31°, 31–70°, 70–180°), Model 1 gives 26.3 / 30.3 / 36.2 km, a far-minus-contact gap of 9.9 km, and the ordering holds	partial
11	R8, T3 and T4: mascon basins	Not yet run	awaiting
11	R7: the centre-of-mass offset	An offset of 1.24–3.11 km for density contrasts of 300–750 kg/m³, against about 2 km observed; the joint density model is yet to be built	partial
11	R1–R6	Oxygen -1 ± 5 ppm; core mass 1–2% of the mass; water 615–1,410 ppm in melt inclusions — each still lacks its closing calculation	partial
11	R9–R14, R18	Stated alongside their threshold tests (§C.6–C.7); the position of R11 measured in Paper 3	awaiting

How to read the table. The reading is done in two stages: first by colour, and then by the order of the papers. The green rows in Papers 5–9 and 11 are measurements the series made itself on public data, each measurement set against a hypothesis and a pass line defined in advance, before the count. The green rows in Papers 2–4 are calculations that close consistently for stated input values. The green rows in Paper 10 are links of the Martian account that close by calculation. The orange rows fall into two kinds, and the table keeps them apart in the wording of the status cell. “Partial” marks a result in which one component passed the test while another has not yet separated — the titanium analysis in Paper 5 (Study 13); the larger rims in Paper 6 (Study 6); and the crustal contrast of test T1, in which the ordering holds in all four models while the size of the contrast, about 10 km, is left to calibration. “Awaiting” marks a requirement or a prediction not yet decided by measurement or calculation — the plume isotopes in Paper 3, the thermal-orbital integration in Paper 4, the seismic profiles in Paper 9, and the samples of the MMX mission. The table contains no red row.

Three rows need a note of bookkeeping, to avoid double counting. Study 5 of Paper 6 and Study 8 of Paper 5 are one measurement — the state of the craters on the mare floors — which each paper counts in its own table; the series therefore contains 88 distinct studies and observations (an earlier

version gave 89), and two cancelled numbers. Paper 11's requirements R15, R16 and R17 have already passed, but their measurements were made in Papers 7, 6 and 8, and they are counted there (Paper 7, D.1–D.2; Paper 6, Study 5; Paper 8, the anomalies and Figure C.1). Finally, the crater count of test T2 in Paper 11 deliberately keeps the maria in all three belts: the maria are the thinnest and youngest ground of the contact disc, and that is the prediction itself. When the conventional mare mask is applied alike to all three belts, the density ratios are 1.13, 1.37 and 1.21 — the same order (Paper 11, §B.8).

The count, paper by paper. The table below gathers the distribution of lights in each paper. The bar in the last column draws each paper's rows to the same scale, so that the reader sees at once where the series has made most of its measurements, and where it still waits.

Paper	Subject	Green	Orange	Cancelled	Rows	
2	LEF Mechanism in Principle	7	1	—	8	
3	LEF Mechanism in Practice	4	3	—	7	
4	The Orbit	3	4	—	7	
5	The Maria	13	2	1	16	
6	The Craters	11	3	1	15	
7	The Poles	4	1	—	5	
8	South Pole–Aitken	6	1	—	7	
9	The Stripes	5	4	—	9	
10	Mars	4	6	—	10	
11	The Requirements	3	5	—	8	
	The series	**60**	**30**	**2**	**92**	

The distribution of the lights reflects the structure of the series as a whole. The papers that focus on reading the Moon's surface — the maria, the craters, the poles and South Pole–Aitken — are mostly green: they were built as measurement sets, in which the pass thresholds were fixed in advance, before the data were opened, and the measurements were indeed run. By contrast, the papers that build the mechanism (2–4) are green in the sectors where the calculation closes, and orange in the sectors where they hand a requirement to a calculation not yet made — the stability and the equation of state of the feed, and the integration of the orbit. The Mars paper is more orange than green (6 orange rows against 4 green), because its links wait for a model of the gas push and for one mission. Of the 92 rows, 60 are marked green, 30 orange, and none red. The overall count is not a score: as explained in Paper 11 (§B.9), linked results must not be added as

though they were independent experiments, and the table is to be read row by row, by the test of each row.

D.2 What the series measured itself

Most of the green rows above are measurements the series made with its own scripts on public databases, not results borrowed from the literature. This distinction matters for the credibility of a new model: an account that merely collects published findings that support it proves little; an account that states a hypothesis, fixes a threshold in advance, and then runs the count itself puts its hypotheses to a real test, and that is what was done here. The list below records what was measured, and on what data. Every comparison was made like for like — the same terrain removed on both sides, in the same size bins and at the same latitudes — and each paper reports this in its study.

Data set	What the series measured on it	Papers
Robbins global crater catalogue [14]	Crater densities by zone, longitude, latitude and diameter range; the oldest-crater test T2 (6,973 craters ≥ 20 km); the polar crowding profile; the floors of the maria; the crater deficit of SPA	5, 6, 7, 8, 11
Wang et al. crater catalogue [15]	Depth-to-diameter ratios, rim heights and crater state; the melted craters	6, 7
LPI crater database; lists of young craters and of rayed craters [16, 17]	The young crater population of the polar regions (Test D.3)	7
GRAIL gravity field: Bouguer field and gradients [18, 19]	Mapping of the 73 feed domes; the 34 maria on the gradients; ringed and unringed mascons; the dense mass under the melted craters	5, 6, 8, 11
GRAIL crustal-thickness models [10]	The crustal jump; the buried domes; tests T1 and T5 on all four models; the rings of SPA; the crust balance	5, 8, 11
Combined LOLA and SELENE topography (SLDEM2015) [20]	Floor levels, rims, collars, central peaks and floor roughness; the depth of SPA	5, 6, 8
Mare outlines from the LROC mission [21]	The basalt boundaries, removed alike on both sides of every crater comparison	5, 6, 7
Kaguya Multiband Imager, FeO and TiO_2 [22]	The iron and titanium rings around the maria; the composition of the ground around and inside each crater	5, 6

Data set	What the series measured on it	Papers
Lunar Prospector thorium maps [23]	The thorium of the contact disc; the thorium of the maria against their surrounding ground	5, 8
GRAIL basin list [24]	The basins of 500 km and more, removed from the polar counts	7
Lowermost-mantle tomography: vote maps, SMEAN, SMEAN2 [25, 26, 27]	Position of the base of the pear over the Pacific province	3, 9
Marine gravity of the Pacific floor [28]	The geometry of the stripes, their bend points, the tip and the ridges	9
Published laboratory analyses of Chang’e-6 and Apollo samples	The chemical composition, ages, metal and magnetism of SPA; the siderophile budget	5, 6, 8
Orbit, shape and density data of Phobos and Deimos	The launch speeds, the orbital inclinations, the depth of the hollows and the centre of mass of Phobos	10

The table is read by its shading. The blue-shaded rows are the crater catalogues and the gravity field, on which most of the green rows rest; the red-tinted rows are the maps of the surface — the basalt outlines and the chemistry — used to remove the maria from both sides of each comparison and to trace the distribution of the material thrown from them; the white rows are the data of the Earth and of Mars. The studies themselves, in summary:

- *The Moon’s surface, by zones.* The exposure-zone tests T1, T2 and T5 (Paper 11); the polar crowding profile as a function of crater size, the crater diameters, the young population and the two poles (Paper 7); the crater deficit, depth, rings and porosity of the South Pole–Aitken basin (Paper 8, Studies 1–5).
- *The maria as openings.* The link between the maria and the gravity domes, the crustal jump, the buried domes, the sinking centres, the one-level floors, the clean floors and their order of freezing, the iron and titanium rings, the thorium of the contact, and the collar of the far maria (Paper 5, Studies 1–16).
- *The craters as landings.* The count and shape of craters by longitude, the source test, the melted and raised floors, the mass under the melted craters, the inside of the crater, the foreign mass and the last scatterer (Paper 6, Studies 1–15).
- *The Earth.* The position of the base of the pear against five tomographic models and two averaged models (Paper 3, §E.4); the geometry of the stripes, the tip and the ridges on the marine gravity field of the Pacific (Paper 9).

The scripts, the means by latitude belt and the input data are deposited with the reproduction package (Code availability).

PART E

Requirements and the key observations

> **THE PROBLEM AND THE APPROACH**
>
> what must hold if the model is right, and what do its key observations show? **The answer:** each step is stated as a requirement alongside its test, and the two key observations of the model have been tested against the data.

Part D presented what has been measured. This Part turns to what the model must still deliver, and to its two key observations. The requirements are those stated in Paper 11, where each requirement is presented together with its test; here each requirement appears in one line alongside its status, so that the reader can see the whole list at a glance and go to Paper 11 for the full argument.

E.1 The requirements

The conditions of the process. The process is admissible under standard physics only if the following conditions hold. Each of them is a condition the papers compute and state alongside its test. The table presents the conditions of the first edition of this paper; the rows sharpened in later papers are given in their updated form.

Condition	Value	Paper	What tests it
A fast, expanded parent body, its core still gathering	A day of 5.5 h (window 4.90–5.54 h); radial expansion of at least 98 km; $k \gtrsim 0.397$ at the start of the mass transfer	2	The spin rate of the early Earth before the Moon formed, within the bounds of the window
A deep, light melt	Density below about 3,864 kg/m³ along the whole ascent path	3	A volatile route that keeps the density of the deep melt low under the pressures of the lower mantle

Condition	Value	Paper	What tests it
A broad, massive, viscous connection between the bodies	A passage radius of about 1,000 km at least (0.46 lunar masses); contact 6,000 km, and 3,000 km at release; a layer that flows rather than fractures	3	A viscous layer that flows without fracturing, and carries the body beyond corotation
Release beyond corotation	Margin of about 210 km: $\Delta r \simeq 2d_0$, $d_0 \approx 105$ km (Paper 3, §C.5a)	2, 3	The final detachment state, beyond the corotation radius
A rigid, round body at the time of release	$\mu \geq 10$ GPa, axis ratio ≤ 1.04; 16,400 km lies inside the fluid Roche limit for 2,600 kg/m³ (19,971 km) and beyond the rigid one (10,314 km)	2, 4, 7	Rigidity of the body at the moment of release
A stiff shell	Surface viscosity $\gtrsim 10^{23}$ Pa s throughout all stages of the ascent	3, 6, 7	A crust that held its craters through the rise; a crust thicker on the far side than on the near side
The release latitude	22.5°, the latitude of the centre of the contact and the upper bound of the release latitude and of i_0 (25° kept as a sensitivity case)	4	A source region placed at a latitude that leads to an inclination of about 5° today

Two rows have changed since the first edition, and both changes made the condition sharper, not looser. The bound on the passage radius was first set at a value above 854 km; the joint test of Paper 3 (§A.13) now places the threshold at a passage radius of about 1,000 km, already holding about half the Moon's mass — which is why the connection zone is the broad base of the dome. The survival requirement originally rested on an interior viscosity of order 10^{18}–10^{20} Pa s for the time the body spent inside the fluid Roche limit; it now requires a body rigid and round at release, because the flattening and the rounding happened while the Moon was attached (Paper 7), and what left the line was a body with a rigid shell (Paper 2, §B.7). The release latitude, first about 25°, is now the 22.5° of the contact centre.

The eighteen requirements. Paper 11 gathers what a Moon built in this way must show into a list of eighteen requirements: the seven classical properties of the Moon (R1–R7), the exposure zones (R8), seven requirements that follow from the pear (R9–R15), and three added from the papers that closed after it (R16–R18). Each row in the table below names the requirement, its key number and its status. The colours follow the scheme of Part D.

	Requirement, in one line	Key number	Status
R1	The Moon is made of terrestrial material, in several isotope systems at once	The oxygen isotope gap: -1 ± 5 ppm between the Earth and the Moon	supported
R2	A metal deficit: a small core of terrestrial iron; most of the core material that entered the pear sank back down it to the Earth	Core mass 1–2% of the mass (seismic value about 1.4%)	supported
R3	An asymmetric crustal profile: a thicker crust on the side away from the contact	A thickness excess of about 8–10 km on the far side; shell 2.85 against crust 2.6×10^{21} kg	supported
R4	One face toward the Earth, kept from the attached stage	The present synchronous state of the Moon	supported
R5	A defined order of closure: the far side closed and kept before the fed side quieted	Comparable units in both hemispheres	supported
R6	Volatiles and water in the amount and state the feed needs	Water concentrations of 615–1,410 ppm in melt inclusions	supported
R7	The centre of mass displaced toward the Earth, by the same crust	A computed range of 1.24–3.11 km, against about 2 km observed	supported
R8	Three concentric exposure zones about the near-side centre	T5 $+0.87$ and T2 1.34 passed; T1 — the ordering holds, contrast of 9.65–10.6 km (its size left to calibration); T3, T4 not yet run	partly passed
R9	A pear that is part of the Earth, held by the Earth's gravity, inside the line	A base width of about 9,000 km, a contact of 3,000 km	awaiting
R10	The pear is gathered back whole into the Earth, leaving no mass in orbit	Nine to twelve lunar masses, from a base of 9,000–11,000 km (Paper 3)	awaiting
R11	The base of the pear above the Pacific province	5 of 5 tomographic models slow beneath Kiritimati; composition and shape to come	position passed
R12	The thorium of the contact disc, brought from the Earth's depths	95% of the thorium with concentrations above 3 ppm lies inside the contact (Paper 5)	awaiting

	Requirement, in one line	Key number	Status
R13	The core as a gauge of how deep the feed reached	1–2% of the mass	awaiting
R14	Little or no foreign material, in the Moon and in the Pacific plumes	Multi-system isotope balance	awaiting
R15	Polar crater crowding rising with crater size	1.09 → 1.53 (Paper 7)	passed
R16	The 34 maria are the openings; craters landing on active floors melted	0.8 against 18.3 and 207 (Paper 6)	passed
R17	Two exits, Imbrium and Serenitatis, and a dome south of both	17–21° on the flattened figure, 24–29° today (Paper 8)	passed
R18	On Mars, both moons fired from one equatorial conduit	Pavonis 1.48°N; both moons within 2° of the equator; MMX samples	awaiting

The table of requirements is read from top to bottom, as the chronological sequence of the model's testing. Each of the seven classical requirements has observational support, and each lacks one calculation that would turn the match into a test — a multi-system isotope balance, a transport model of the metal, or a joint density model for the crust and the centre of mass. The exposure-zone requirement (R8) is the only requirement of the first edition that has been run as a statistical test with its thresholds fixed in advance, and it is the first key observation of the model (E.2). Of the ten requirements added later, three have already passed on the measurements of Papers 6, 7 and 8, and the rest stand alongside the test that will decide them. The rows are not to be added into a single index: R3 (crustal thickness) and R7 (centre-of-mass offset) are two views of one distribution of crust, and the maria are a third view; counting them as three independent successes would mean weighing one parcel on three scales (Paper 11, §B.9).

E.2 The two key observations

The first key observation: the exposure zones. If the Moon rounded while on the tip of the pear — a ball squashed on the tip, its contact diameter narrowing from 6,000 km to 3,000 km before detachment — then the surface of the Moon carries the exposure sequence of that peeling process, in the form of three concentric zones about the centre of the near side. The contact disc, at the centre, was exposed last, at the moment of release; the base annulus, at the limb, was exposed gradually as the contact shrank during the rounding — the way a suction cup peeled from glass lets go first at its edge; and the surface of the dome, on the far side, was exposed first. The share rule of Paper 3 places the boundaries: a contact cap of about 31° about the contact centre — the tip of the pear, at latitudes of about 22–24°N. The zones are computed about 24.1°N 335.2°E, the centre of the contact as the author outlined it, which lies 34.0° from the sub-Earth point; the limb belt extends to 70°, and the far side lies beyond it. For comparison, the tests of September 2026 were run on the

earlier boundaries, 41° and 90° about the sub-Earth point (Paper 11, §B.8). Three data sets must therefore order themselves by angular distance from the near-side centre, and by no other axis: *the crustal thickness*, since the far side is built of many layers of compacted sial that migrated up the pear to its tip, and of material that returned to it again and again; *the oldest crater population*, carried up on the shell, and densest where it was exposed longest; and *the character of the basins* — which of them ejected material, and which only conducted it. No Moon assembled from pieces predicts a concentric ordering about the near-side centre.

The tests are stated as numbers fixed in advance, before the count (Paper 11, §B.8): the crust thickens monotonically from the contact disc, through the limb belt, to the far side; the oldest crater population of the far side is denser than that of the limb belt by a factor of at least 1.1; the ringed mascon basins are concentrated in the limb belt (at least 40% of them at 41–105°); and no control axis orders the data as well as the sub-Earth axis. Together they read the peeling of the body from the tip of the pear at stage 4, the rounding stage — the geometry on which the whole construction rests.

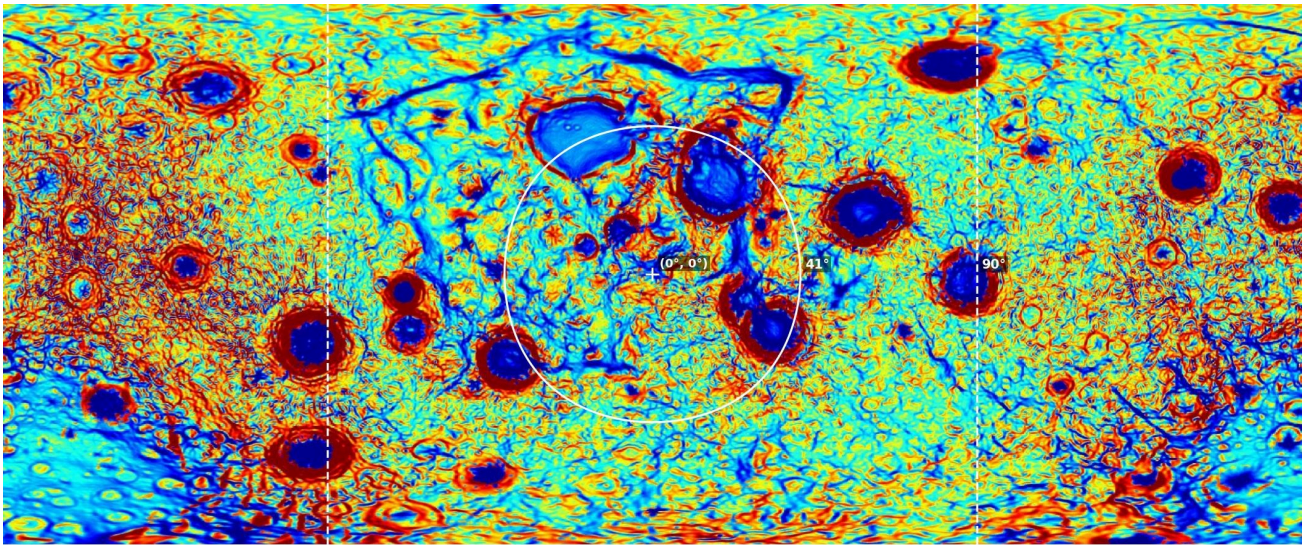


Figure E.1. The three exposure zones on the GRAIL Bouguer-gradient map. Circles at 41° (solid) and 90° (dashed) from the sub-Earth point (0°, 0°) mark the contact disc, the limb belt and the far side as used for the September tests (Paper 3, §B.4; current boundaries in Paper 11, §B.8). Blue: mass concentrations under thin crust; red: their rims. The strongest unringed mascons lie inside the disc, the ringed basins in the belt, and the basins on thick crust beyond it (Paper 11, §B.8, tests T1–T5). Cylindrical projection centred on 0°E, ±75° latitude. Credit: LROC QuickMap (NASA/GSFC/Arizona State University); GRAIL Bouguer gravity gradients (NASA/JPL-Caltech/MIT).

The figure is read from the centre outward. Inside the solid circle (the contact disc) are concentrated the strong mascons that have no rings — Imbrium, Serenitatis and Procellarum; in the band between the circles (the limb belt) are arrayed the ringed basins — Orientale, Crisium, Humorum, Nectaris, Smythii, Humboldtianum and Moscoviense; beyond the dashed circle (the far side) lie basins on thick, high crust, with neither strong mascons nor rings. That is the sequence the dome predicts. The map alone, however, is a reading, not a test; the test is the belt statistics.

The result. When the test was run in September 2026 on the four GRAIL crustal-thickness models and on the Robbins crater catalogue (Paper 11, §B.8), the ordering was found. The crust thickens from the contact disc to the limb belt and to the far side in all four models, in a consistent order over twelve belts of 15° about the predicted axis ($+0.87$ in every model; on Model 1 the rise is monotonic over ten consecutive belts), and about no control axis (the poles -0.43 and $+0.43$, the east limb $+0.76$). The oldest crater population orders itself as predicted: far/limb 1.34, far/disc 2.21, limb/disc 1.65, and each ratio crosses the pass line set for it. The thickness contrast — far side minus contact disc — is 9.65, 10.0, 10.3 and 10.6 km in Models 1–4, respectively, and the ordering holds in all four; the size of the contrast is a parameter to be calibrated. A recomputation on the current geometry (contact cap 0–31°, belt 31–70°, far part 70–180°) shows that the ordering holds in both quantities: for craters ≥ 20 km, after removal of the maria and the SPA basin, the densities are 53, 187 and 220 per 10^6 km² (far/belt 1.17, far/cap 4.15, belt/cap 3.55); the crustal thickness of Model 1, with the maria included, is 26.3, 30.3 and 36.2 km, and the far side minus the cap is 9.9 km. The ordering test has therefore passed on the craters and on the axis; as for the thickness, the ordering is present in every model, and the size of the contrast, about 10 km, is left to calibration. The basin-class tests, T3 and T4, have not yet been run.

The second key observation: the material of the craters. According to the model, nearly all the Moon's craters, some 99%, are made of returned terrestrial material (Paper 6); foreign material, if it is found at all, will be in small amounts only, and fine space dust is not counted. The craters therefore record the material that left the Earth and came back to it, not a bombardment from outside. The decisive quantity for testing the claim is the foreign mass. **The result:** read as impacts, the visible craters require 4.4–5.0×10^{19} kg of foreign material, which is 2–6% of the upper 5 km of the crust; whereas in fact the crust holds 0.9–2% at most, and the ruthenium of the crater melts is terrestrial (Paper 6, Study 8).

E.3 The status of the model

The LEF model is a process model: it proposes a sequence, computes each step quantitatively, and states the test of each step. At this stage it does not present a simulation of the ascent; the three-dimensional stability of the connected body, the energy that drives the upwelling, the equation of state of the melt at depth, and the time at which the contact fails are the calculations the model requires and has not yet performed (Paper 3, §D.4). The model claims that the Moon formed this way. In support of this it points to where the classical objection to fission went wrong: it computed the release at the Earth's surface, where it would need enormous speeds that would break the Earth apart. In the LEF mechanism, the pear raises itself to the corotation region and releases the material at its tip — in our case, the Moon; the pear grows until the material at its tip detaches, and then the sorting draws the pear back in. The classical objection computed at the Earth's surface; the model computes at the tip of the pear. The model further shows that the route is open at the corotation radius, with an angular-momentum budget of 1.007 times today's; that a hemisphere of the right size has the right density and the right surface, without adjustment; that the same process, applied to Mars, gives what Mars has; that the Moon's surface carries the ordering the process

leaves, on the axis it predicts; and that the maria, the craters, the poles and South Pole–Aitken each pass the tests their papers set for them — 58 green rows and not a single red one.

The model does not claim proof. Most requirements have now moved from the status of open questions to that of calibration targets — the numbers are fixed, the tests are written, and the data exist or are coming. In this sense the series is a milestone: each requirement stands alongside its test, row by row, and Section E.2 presents the two key observations with their results; the requirements and their tests are set out in full in Paper 11.

PART F

The series

THE PROBLEM AND THE APPROACH

twelve papers carry one model. How are they divided, what does each answer, and what is left? **The approach:** one table of the papers with the question and the number of each; then the open work, stated as calculations to be made and as topics the model opens for others.

F.1 The twelve papers

The papers were built to be read in order, but each of them stands on its own: each paper carries the environment table, its own predictions and its own tests. The table gives each paper the question it answers and the number that carries the answer. The shading follows the structural division of the series: navy for the papers that state the model and build and release the body (1–4); red for the papers that read the Moon's surface (5–8); white for the papers that test the model outside the Moon and gather its requirements (9–11); and cream for the paper of research topics.

No.	Title	The question it answers	Its central number
1	The Model	What is the whole process, from the hot early Earth to today's Moon?	Release at 16,399 km, 3.873 Ga
2	LEF Mechanism in Principle	Can the Moon leave the Earth at orbital speed without being thrown, and does the angular-momentum balance close?	16,399 km at a 5.77-h day; budget 1.007, against 2.74 for the classical route

No.	Title	The question it answers	Its central number
3	LEF Mechanism in Practice	How was the body built — fed from within, risen from without — and how did it let go?	150 GPa at the source; 1,300 kg/m³ at the tip
4	The Orbit	Where does the inclination come from, and does the Moon recede from the very start?	$i_0 = 22.5° \to 5.145°$ today
5	The Maria	What is a mare?	33 of 34 on a feed dome; 44 of 73 domes buried
6	The Craters	What made the craters, and what do they record?	About 99% from returning material; 249–252 against 139–175 per 10^6 km²
7	The Poles	Why are the poles crowded with craters?	1.20 ± 0.05, rising 1.09 $\to$ 1.53 with crater size
8	South Pole–Aitken	What is the great basin of the far side?	Central deficit 0.51; the model fits 16 of 17 anomalies
9	The Stripes	Where did the pear go, and where did the Moon sit?	Stripes 9,360 km against a pear of 9,680–9,800; tip 3,360 km
10	Mars	Can a fast young planet fire its own moons?	Gas push 2.4–2.9 km/s; a ring completes the orbit; Phobos caught by the line near 19,000 km
11	The Requirements	What must a Moon built this way look like, and does it indeed look so?	18 requirements; T5 $+0.87$, T2 1.34
12	Research topics	What should be measured and computed next?	The open work of F.2–F.3

Reading the third column from top to bottom shows that the questions follow the process itself: how the Moon could leave (2), how it was built (3), how it moved in its orbit (4), what it carries on its surface (5–8), what the Earth kept (9), whether the same happened elsewhere (10), and what the whole must satisfy (11).

F.2 What remains open — future work

Two central subjects lie deliberately outside the bounds of the series, and are left for future work.

The system of axes. The axes of the Earth and the Moon, from the moment of release to today, are not solved in this series. The problem involves four reference planes — the Earth's equator, the

Moon’s orbit, the Moon’s equator and the ecliptic; the axes move in two different ways and at very different rates; the memory of the frozen bulge is unknown; and the components affect one another. The series therefore gives every position in the coordinate frame of the present, and says so where the past orientation matters: the release latitude, 22.5°; the equator of the stripes, inclined at least 10° to today’s equator; and the southward direction of the dome of South Pole–Aitken. A change of the Earth’s axis after the detachment may make SPA appear out of line with the burner. The solution of the axis system may change parts of the model, and for that reason it is listed here first.

The magnetism of the Moon. The source of the Moon’s ancient magnetism is not treated in the series. Its pattern over South Pole–Aitken — erased at the centre, strong on the trailing side, and the swirls of Ingenii — is read as a record of heat and of iron that came from the Earth’s interior; the source of the field itself is passed to future work.

The open computations. Within the series, each paper sets out the calculations it requires and has not yet made. They are gathered in the table below, each alongside what it will close. None of them changes a number fixed in the papers; each of them turns a requirement into a result.

Computation	Named in	What it would close
A three-dimensional numerical simulation of the ascent	3	A body of the lunar mass, foamed to a density of 1,300 kg/m³, fed under pressure through a 6,000-km base on a spinning, expanded parent: the first three-dimensional test of the mechanism. The initial state is fully specified in the series
The equation of state of volatile-rich melt at 50–150 GPa	3, 10	Whether the density of the feed falls below 3,864 kg/m³ (shock and diamond-anvil experiments on hydrous and carbonated melts)
Energy balance and the time of failure of the contact	3	The mechanical work of the lift, and the moment the viscous layer lets go
The stratified Maclaurin problem	2	A Clairaut–Darwin–Radau parent with sinking iron: the corotation radius within 15,382–17,569 km (from $D = 3$ to $D = 1$), and Paper 4’s starting orbit
The thermal-orbital integration	4	No constant Q fits both today’s rate of recession and the age; with the Laplace and Cassini transitions resolved, $i = \lambda$ becomes a prediction of 5.145°
The trapped pressure at the moment of detachment	7	Its contribution to the Moon’s initial speed, and whether the release velocity of Paper 4 needs a correction
The Hf–W reservoir history	11	One chronology for signatures, metal transfer and angular momentum through the 600-Myr window

Computation	Named in	What it would close
Metal transport and a joint density model	11	R2 and R13 (the core fraction); R3 and R7 together (crustal thickness and centre of mass)
The figure with strength and the two-layer relaxation	7	The polar crowding as a result of a stiff shell over a softer body; the stresses in the shell
The tide on a body held by its base	7	The six-axis order as a number; the far-side crown; the largest craters (1.84)
The engine of the openings	5, 7	The critical flux and the sealing time of each mare; the yield of craters from the openings about 90°E
The heat axis, its energy and its redox budget	8	Whether the burner's heat compacts the column, closes the pores and erases the craters of SPA; the dynamic tests recomputed at a day of 5.77 h
The gas push in the conduit, and the ring	10	The critical step of the Martian account: a soft plug pushed 200–300 km by trapped gas; the ring that completes the orbits

F.3 Research topics the model opens

A process model earns its place by the work it makes possible. The research topics below follow directly from the sequence of the model; each is a measurement or a study that would advance a particular step, and each is stated alongside the result the model predicts. The topics are gathered in Paper 12, *Research topics* — a companion paper that stands outside the papers of the model itself: it adds no step to the process, but collects, from all the papers and from the working notes of the series, 118 topics that the model opens to other researchers, each topic setting out the question, the model's prediction, the data and the method.

1. **The belt statistics, completed (R8).** Running the basin-class tests T3 and T4, and running T2 on an age-selected crater population. The axis, the belts and the pass thresholds were fixed in advance (Paper 11, §B.8), and the GRAIL, LOLA and crater-catalogue data suffice for this.

2. **The samples of MMX.** Material from Phobos is expected to reach the Earth in 2031. The Mars paper predicts a series of distinct results: Martian isotopes with no impactor component, closed vesicles of a cooled foam, and a centre of mass displaced toward Mars, to the blunt end (Paper 10, Part III; mission schedule [29]). That pits never cross grooves can be tested now, on existing images.

3. **The far-side crust as the Earth's sial.** If the far-side crust is built of many layers of compacted terrestrial sial that migrated up the pear to its tip, and of material that returned to it again and again, its bulk composition should be sialic in the sense of the early Earth's surface. It should

grow denser and richer in iron with depth (central peaks against their depth of origin, Paper 3), and it may reach deeper than the crustal models now assume. Chang'e-6 material and future far-side samples test the first claim; central peaks from different depths of origin test the second.

4. **The metal of the ancient norites.** Measurement of Cr, Ti, Mo, Ru and Os isotopes in the iron–nickel metal grains of the two dated Chang'e-6 norites, and of Ru and Os isotopes in the melts of each of the mare basins: the model predicts that this metal is terrestrial. Foreign material, if identified at all, should appear only in tiny amounts — space dust, and one or two meteorites landed over four billion years (E.2; Papers 5, 8).

5. **The Pacific lower mantle.** The position of the base over the Pacific province has already passed the test; what remains to be tested is its composition, its shape against the African province, and the isotopes of the plumes rising from it. We propose — with reservation, since the nature of the deep provinces is still open (the Pacific province may be core-related material that did not complete its sorting) — that the Pacific province is largely the returned pear, part of the sorted material that went back into the Earth; the model predicts that it is denser and richer in iron than the African province (Paper 3, Part E).

6. **The stripes.** The nine studies of Paper 9, Part G: refitting the equator on the full traces, the chance test of the position of the base on it, the one-sided order of the stripes, the steps in depth and age, the trailing side, continuity beneath the ridges, the spatula ridges against plume tracks, the mirror imprint at the scale 1.37, and seismic profiles across the fold beneath the dated layers.

7. **The young craters of the poles.** A dated sample of some 150 young craters in the polar caps will decide Test D.3 of Paper 7; the model predicts that it will show no density excess, because the young craters formed on the body after it rounded.

8. **The maria at higher resolution.** A thorium map at a resolution of 26 km over the feed domes; and the age of the youngest basalt of each mare against the distance from the contact centre — the basalt should be younger closer to it, if the contact maria sealed last (Paper 5).

9. **The other bodies.** The process, run on Venus (one turn in 243 days) and on Mercury (59 days), lifts nothing; on a fast-spinning super-Earth, it lifts more. A parameter study in spin rate, mass and volatile content would turn the family argument of Paper 10 into a prediction for the moons of other planets.

None of these is exotic: each is a calculation or a measurement within reach of an existing research group, and each would either strengthen a step of the process or remove it.

Code availability

Paper 1 computes nothing new: every number in it is taken from Papers 2–11, and each of them is reproduced by the scripts of the LEF reproduction package. The code is written in Python 3 (with the numpy and scipy libraries), and prints the values of each section from the input data stated in the text, together with the counts, the means by latitude belt and the catalogue selections behind

the traffic light of Part D. The package is attached to this preprint as supplementary material (LEF_code_and_data.zip). An earlier version of the series and of the package is deposited at doi:10.5281/zenodo.22743615.

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The series: Duran, Z. Papers 2–12 of this series (2026): 2 *LEF Mechanism in Principle*; 3 *LEF Mechanism in Practice*; 4 *The Orbit*; 5 *The Maria*; 6 *The Craters*; 7 *The Poles*; 8 *South Pole–Aitken*; 9 *The Stripes*; 10 *Mars*; 11 *The Requirements*; 12 *Research topics*. Author's manuscripts, posted in sequence as preprints on Preprints.org, with earlier versions at doi:10.5281/zenodo.22743615; they are the source of the model's calculations and are not external validation of it.

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